

Notes on

Many-Body Theory in Condensed Matter

Christos Kourris

`christos.kourris@tu-dresden.de`

These notes are mainly based on a course at TU-Dresden given by Dr. Hong-Hao Tu in the Summer Semester of 2021-2022. The formalism developed and used throughout the different parts of the course is the path-integral formalism, for spins, bosons and fermions. More material will be added on the way. For more resources you can check out:

- ...

Contents

1	Lecture 1: Review and QM Path Integral	3
2	Lecture 2: Classical XY model	9
3	Lecture 3: Quantum XY model	14
4	Lecture 4: Goldstone Boson	17
5	Lecture 5: Mermin-Wagner Theorem	20
6	Lecture 6: Kosterlitz-Thouless (KT) Phase	25
7	Lecture 7: KT Transition (I)	29
8	Lecture 8: KT Transition (II)	33
9	Lecture 9: Bosonic Systems	38
10	Lecture 10: BEC and Superfluidity (I)	40
11	Lecture 11: BEC and Superfluidity (II)	44
12	Lecture 12: Fermionic Systems	48
13	Lecture 13: BCS Theory	52
14	Lecture 14: Majorana zero mode & Ising anyon	57
15	Lecture 15: Majorana field theory & Jackiw-Rebbi mechanism	61
16	Lecture 16: Schwinger boson & spin coherent state	65
17	Lecture 17: Haldane's conjecture	71
18	Lecture 18: AKLT Model	77
19	Lecture 19: Parton Construction	83
20	Lecture 20: Kitaev's Honeycomb Model	89

1 Lecture 1: Review and QM Path Integral

1.1 Brief Review of QM

In Quantum Mechanics we learned how to solve one-body Hamiltonians of the form,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(\vec{r})$$

where $\hat{p}$ denotes the (3-dimensional) momentum operator of the particle and $\hat{V}(\vec{r})$ is the potential. Given a wavefunction at time $t = 0$, the solution is given at all times by solving the time-dependent Schrodinger's equation,

$$i\hbar \frac{\partial}{\partial t} \psi(\vec{r}, t) = \hat{H} \psi(\vec{r}, t) \rightarrow \psi(\vec{r}, t) = e^{-\frac{i}{\hbar} \hat{H} t} \psi(\vec{r}, 0)$$

The position and momentum operators are given by,

$$\hat{p} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$$

$$\hat{r} = (\hat{x}, \hat{y}, \hat{z})$$

and satisfy the commutation relations,

$$[x_i, p_j] = i\hbar \delta_{ij}$$

The eigenbases of the two operators provide Hilbert spaces in which we can write the states, such that,

- Position Eigenbasis:

$$\hat{r} |\vec{r}\rangle = \vec{r} |\vec{r}\rangle \quad \langle \vec{r} | \vec{r}' \rangle = \delta(\vec{r} - \vec{r}') \quad \int d^3 r |\vec{r}\rangle \langle \vec{r}| = \mathbf{1}$$

- Momentum Eigenbasis:

$$\hat{p} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle \quad \langle \vec{p} | \vec{p}' \rangle = \delta(\vec{p} - \vec{p}') \quad \int d^3 p |\vec{p}\rangle \langle \vec{p}| = \mathbf{1}$$

$$|\vec{p}\rangle = \int d^3 r |\vec{r}\rangle \langle \vec{r} | \vec{p} \rangle = \int d^3 r \frac{1}{(2\pi\hbar)^{3/2}} e^{i/\hbar \vec{p} \cdot \vec{r}} |\vec{r}\rangle$$

As usual, the interpretation of a wavefunction ψ , is that for a particle in state $|\psi(t)\rangle$ will be found at position $\vec{r}$ with probability,

$$P = |\langle \vec{r} | \psi(t) \rangle|^2$$

1.2 Feynman's Path Integral

Consider a general Hamiltonian for a particle of mass m of the form,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}(\vec{r}).$$

Suppose that the particle at $t = 0$ is in some initial state $|\vec{r}'\rangle$. The probability of finding the particle at $|\vec{r}\rangle$ after time t is given by the amplitude $|\langle \vec{r}' | e^{-i/\hbar \hat{H}t} | \vec{r} \rangle|^2$. The idea of the construction of the path integral is to view this amplitude as a propagator of the state from $|\vec{r}'\rangle$ at $t' = 0$ to $|\vec{r}\rangle$ at $t' = t$, and split into many pieces of propagation that take place in the infinitesimal time interval Δt , such that $N\Delta t = t$. To this end, we can expand the *Propagator* in the following way,

$$\begin{aligned} \langle \vec{r}' | e^{-i/\hbar \hat{H}t} | \vec{r} \rangle &= \langle \vec{r}' | \underbrace{e^{-i/\hbar \hat{H}\Delta t} e^{-i/\hbar \hat{H}\Delta t} \dots e^{-i/\hbar \hat{H}\Delta t}}_{N \text{ pieces}} | \vec{r} \rangle \\ &= \langle \vec{r}' | e^{-i/\hbar \hat{H}\Delta t} \mathbf{I} e^{-i/\hbar \hat{H}\Delta t} \mathbf{I} \dots \mathbf{I} e^{-i/\hbar \hat{H}\Delta t} | \vec{r} \rangle \\ &= \int d^3r_1 d^3r_2 \dots d^3r_{N-1} \langle \vec{r}' | e^{-i/\hbar \hat{H}\Delta t} | \vec{r}_{N-1} \rangle \langle \vec{r}_{N-1} | \dots \langle \vec{r}_2 | e^{-i/\hbar \hat{H}\Delta t} | \vec{r}_1 \rangle \langle \vec{r}_1 | e^{-i/\hbar \hat{H}\Delta t} | \vec{r} \rangle \end{aligned}$$

We can now isolate one of these pieces that make up the propagator, $\langle \vec{r}_1 | e^{-i/\hbar \hat{H}\Delta t} | \vec{r} \rangle$, and simplify it in the limit that $\Delta t \rightarrow 0$. For that, we first compute the matrix elements of $\hat{H}$, such that,

$$\langle \vec{r}_1 | \hat{H} | \vec{p}_0 \rangle = \left[\frac{\vec{p}_0^2}{2m} + V(\vec{r}_1) \right] \langle \vec{r}_1 | \vec{p}_0 \rangle \equiv H(\vec{p}_0, \vec{r}_0) \langle \vec{r}_1 | \vec{p}_0 \rangle,$$

where $\langle \vec{r}_1 | \vec{p}_0 \rangle = \frac{1}{(2\pi\hbar)^{3/2} e^{+i/\hbar \vec{p}_0 \cdot \vec{r}_1}}$. This can now be used by expanding $\langle \vec{r}_1 | e^{-i/\hbar \hat{H}\Delta t} | \vec{r} \rangle$ in a momentum set eigenbasis,

$$\begin{aligned} \langle \vec{r}_1 | e^{-i/\hbar \hat{H}\Delta t} | \vec{r} \rangle &= \langle \vec{r}_1 | \left[\mathbf{I} - i/\hbar \hat{H}\Delta t + \mathcal{O}(\Delta t^2) \right] | \vec{r} \rangle \\ &= \int d^3p_0 \langle \vec{r}_1 | \left[\mathbf{I} - i/\hbar \hat{H}\Delta t + \mathcal{O}(\Delta t^2) \right] | \vec{p}_0 \rangle \langle \vec{p}_0 | \vec{r} \rangle \\ &= \int \frac{d^3p_0}{(2\pi\hbar)^3} e^{i/\hbar \vec{p}_0 \cdot (\vec{r}_1 - \vec{r})} e^{-i/\hbar H(\vec{p}_0, \vec{r}_0)\Delta t} \end{aligned}$$

We can substitute the infinitesimal result in the expression for the propagator to get,

$$\begin{aligned} \langle \vec{r}' | e^{-i/\hbar \hat{H}t} | \vec{r} \rangle &= \int \prod_{m=1}^{N-1} d^3r_m \prod_{n=0}^{N-1} \frac{d^3p_n}{(2\pi\hbar)^3} e^{i/\hbar \vec{p}_0 \cdot (\vec{r}_1 - \vec{r}_0) + i/\hbar \vec{p}_1 \cdot (\vec{r}_2 - \vec{r}_1) + \dots} \\ &\quad \times e^{-i/\hbar H(\vec{p}_0, \vec{r}_1)\Delta t - i/\hbar H(\vec{p}_1, \vec{r}_2)\Delta t + \dots} \\ &= \int \prod_{m=1}^{N-1} d^3r_m \prod_{n=0}^{N-1} \frac{d^3p_n}{(2\pi\hbar)^3} \exp \left[\frac{i}{\hbar} \sum_{n=0}^{N-1} \vec{p}_n \cdot (\vec{r}_{n+1} - \vec{r}_n) - \frac{i}{\hbar} \sum_{n=0}^{N-1} H(\vec{p}_n, \vec{r}_{n+1})\Delta t \right] \end{aligned}$$

In the continuous time limit, $\Delta t \rightarrow 0$, which must be taken with care, we can think of the big list of intermediate position vectors $(\vec{r}_0, \vec{r}_1, \dots, \vec{r}_N)$ as a function $\vec{r}(t')$ where t' is now a continuous variable in $[0, t]$. Taking,

$$\vec{r}_{n+1} \approx \vec{r}_n + \dot{\vec{r}}_n \Delta t$$

so that, to first order in Δt , we have $V(\vec{r}_{n+1}) = V(\vec{r}_n)$, allows us to turn the sums in the exponentials to integrals as in,

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{i}{\hbar} \sum_{n=0}^{N-1} \vec{p}_n \cdot (\vec{r}_{n+1} - \vec{r}_n) &= \frac{i}{\hbar} \int_0^t dt' \vec{p}(t') \cdot \dot{\vec{r}}(t') \\ \lim_{N \rightarrow \infty} \frac{i}{\hbar} \sum_{n=0}^{N-1} H(\vec{p}_n, \vec{r}_{n+1})\Delta t &= \frac{i}{\hbar} \int_0^t dt' H(\vec{p}(t'), \vec{r}(t')) \end{aligned}$$

The measure of the integration also becomes the measure of the *path integral*,

$$\lim_{N \rightarrow \infty} \int \prod_{m=1}^{N-1} d^3 r_m \prod_{n=0}^{N-1} \frac{d^3 p_n}{(2\pi\hbar)^3} = \int_{\substack{\vec{r}(t'=0)=\vec{r} \\ \vec{r}(t'=t)=\vec{r}'}} \mathcal{D}\vec{r}(t') \mathcal{D}\vec{p}(t')$$

where the integral limits denote the boundary conditions which we require by construction.

Finally we can collect everything together into the formal result for propagator,

$$\langle \vec{r}' | e^{-i/\hbar \hat{H}t} | \vec{r} \rangle = \int_{\substack{\vec{r}(t'=0)=\vec{r} \\ \vec{r}(t'=t)=\vec{r}'}} \mathcal{D}\vec{r}(t') \mathcal{D}\vec{p}(t') \exp \left[\frac{i}{\hbar} \int_0^t dt' (\vec{p} \cdot \dot{\vec{r}} - H(\vec{p}, \vec{r})) \right] \quad (1)$$

The discretisation of the path into infinitely many small pieces allows us to interpret this as a *sum over trajectories*, each weighted by the exponential in the expression. From classical mechanics, we can already recognise the expression in the exponential as the *action* of the system. The path integral is in fact a *functional integral*, and the measure should be thought as the infinite subdivision limit of a discretised expression. Finally this is exactly the reasoning that led Feynman (and others) to the construction of the path integral. The intuition is that the single postulate we need to explain quantum mechanics is that the probability amplitude for a particle propagating from a position $\vec{r}_0$ to a position $\vec{r}_N$ can be obtained by summing over all possible paths connecting $\vec{r}_0$ and $\vec{r}_N$, and each path is weighted by a phase factor $\exp[iS/\hbar]$, where S is the **classical action** of the path.

Let us elaborate a bit on the last point, regarding the classical action of the path. The action is given by,

$$S = \int_0^t dt' (\vec{p} \cdot \dot{\vec{r}} - H(\vec{p}, \vec{r}))$$

For the classical path, the action is an extremum, thus $\delta S = 0$. Let us take as an example a 1D system so that $\vec{r} \mapsto x$, and start by integrating out the momentum, by means of completing the square, by writing $p\dot{x} - \frac{p^2}{2m} = -\frac{1}{2m}(p - m\dot{x})^2 + \frac{m\dot{x}^2}{2}$,

$$\begin{aligned} \langle x' | e^{-\frac{i}{\hbar} \hat{H}t} | x \rangle &= \int_{\substack{x(t'=0)=x \\ x(t'=t)=x'}} \mathcal{D}x(t') \mathcal{D}p(t') \exp \left[\frac{i}{\hbar} \int_0^t dt' (p\dot{x} - \frac{p^2}{2m} - V(x)) \right] \\ &= (\text{const.}) \int_{\substack{x(t'=0)=x \\ x(t'=t)=x'}} \mathcal{D}x(t') \exp \left[\frac{i}{\hbar} \int_0^t dt' (\frac{1}{2}m\dot{x}^2 - V(x)) \right] \end{aligned}$$

where we have used the Gaussian integral $\int_{-\infty}^{\infty} dp e^{-\alpha p^2} = \sqrt{\frac{\pi}{\alpha}}$. We recognise the term in the exponential as the usual form of the classical action,

$$S[x] = \int_0^t dt' \left[\frac{1}{2}m\dot{x}^2 - V(x) \right]$$

In the limit where $\hbar \rightarrow 0$, we can use the stationary phase approximation to infer that the path integral will be dominated by the stationary points of the action $S[x]$,

$$\begin{aligned} \delta S = S[x + \delta x] - S[x] &= \int_0^t \left[\frac{1}{2m}(\dot{x} + \delta\dot{x})^2 - \frac{1}{2m}\dot{x}^2 - V(x + \delta x) + V(x) \right] \\ &= [m\dot{x}\delta x]_0^t - \int_0^t dt' m\ddot{x}\delta x - \int_0^t dt' V'(x)\delta x \\ &= \int_0^t dt' [-m\ddot{x} - V'(x)]\delta x = 0 \end{aligned}$$

where we used that for the variation δx , the boundary conditions are $\delta x(t' = 0) = \delta x(t' = t) = 0$. Therefore, at the stationary points, $\delta S = 0$, we recover the classical EOM,

$$m\ddot{x} = -V'(x)$$

1.3 Relation to Classical Statistical Mechanics

We can use the path integral method to compute the partition function of a system at a finite temperature $T = 1/\beta$,

$$Z = \text{tr } e^{-\beta \hat{H}} = \int dx \langle x | e^{-\beta \hat{H}} | x \rangle$$

By introducing the *imaginary time* $\tau = it$, we can rewrite the partition function in terms of a path integral for a particle propagating in a periodic path from time $\tau = 0$ to time $\tau = \beta$. To this end, we start from the imaginary time propagator,

$$\begin{aligned} \langle x | e^{-\beta \hat{H}} | x \rangle &= \langle x | \underbrace{e^{-\Delta\tau \hat{H}} e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}}}_{N\Delta\tau=\beta} | x \rangle \\ &= \int dx_1 \dots dx_{N-1} \langle x | e^{-\Delta\tau \hat{H}} | x_{N-1} \rangle \langle x_{N-1} | e^{-\Delta\tau \hat{H}} | x_{N-2} \rangle \dots | x_1 \rangle \mathbf{I} \langle x_1 | e^{-\Delta\tau \hat{H}} | x \rangle \end{aligned}$$

Upon inserting $\mathbf{I} = \int dp_0 |p_0\rangle \langle p_0|$, similar manipulations to the previous case lead us to,

$$\langle x | e^{-\beta \hat{H}} | x \rangle = \int_{\substack{x(0)=x \\ x(\beta)=x}} \mathcal{D}x(\tau') \mathcal{D}p(\tau') \exp \left[- \int_0^\beta d\tau' (-ip(\tau')\dot{x}(\tau') + H(p(\tau'), x(\tau'))) \right]$$

For Hamiltonians of the form, $H = \frac{p^2}{2m} + V(x)$, we can integrate out the momentum by writing,

$$\frac{p^2}{2m} - ip\dot{x} = \frac{1}{2m}(p - im\dot{x})^2 + \frac{m\dot{x}^2}{2},$$

so that the partition function is,

$$Z = \int_{x(0)=x(\beta)} \mathcal{D}x(\tau') \exp \left[- \int_0^\beta d\tau' \left(\frac{1}{2m} \dot{x}(\tau')^2 + V(x) \right) \right]$$

The restriction to periodic paths arises because $Z(\beta)$ is a trace. Note also that the notation $\mathcal{D}x(\tau)$ means very different things in the two path integrals developed, as a result of involving very different classes of paths that are “integrated over”.

Quantum to Classical Correspondence (Following Parameswaran)

Let us review what we have derived in this section in order to uncover a deeper relationship between the two path integrals. The Feynman propagator for a particle moving in **real time**, from $x(0) = 0$ to $x(t_R) = x_1$, is,

$$F(x_0, x_1; t_R) = \langle x_1 | e^{-\frac{i}{\hbar} \hat{H} t_R} | x_0 \rangle = \int_{\substack{x(0)=x_0 \\ x(t_R)=x_1}} \mathcal{D}x(t) \exp \left[\frac{i}{\hbar} \int_0^{t_R} dt \left(\frac{1}{2} m \dot{x}^2 - V(x) \right) \right]$$

We can imagine propagating the particle in imaginary time, by replacing $t_R \rightarrow -it_I$, which corresponds to the propagator,

$$\begin{aligned} Z(x_0, x_1; t_I) &= \langle x_1 | e^{-\hbar \hat{H} t_I} | x_0 \rangle = F(x_0, x_1; -it_I) \\ &= \int_{\substack{x(0)=x_0 \\ x(-it_I)=x_1}} \mathcal{D}x(t) \exp \left[\frac{i}{\hbar} \int_0^{-it_I} dt \left(\frac{1}{2} m \dot{x}^2 - V(x) \right) \right] \\ &= \int_{\substack{x(0)=x_0 \\ x(t_I)=x_1}} \mathcal{D}x(\tau) \exp \left[-\frac{1}{\hbar} \int_0^{t_I} d\tau \left\{ \frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x(\tau)) \right\} \right] \end{aligned}$$

where on the last step we make a change of variable $\tau = it$, and the boundary conditions are now for the new variable τ .

So far we have a propagator for a *quantum mechanical problem*, in one dimension, but this can be generalised to D spatial dimensions. To see how this can be linked to a problem in classical mechanics, let us consider a one dimensional string of length T_I and tension m . We can define the classical energy functional,

$$E_{cl}[x(t)] = \frac{1}{\delta} \int_0^{t_I} d\tau \left\{ \frac{m}{2} \left(\frac{dx}{d\tau} \right)^2 + V(x(\tau)) \right\}$$

with boundary conditions such that the endpoints are fixed, $x(0) = x_0$, $x(T_I) = x_1$. The constant δ fixes the units of the energy since $[\text{action}] = [\text{energy}] \times [\text{time}]$, and therefore its units are the units of time. The corresponding propagator is then,

$$F(x_0, x_1; -it_I) = Z(x_0, x_1; t_I) = \int_{\substack{x(0)=x_0 \\ x(t_I)=x_1}} \mathcal{D}x(\tau) e^{-\frac{\delta}{\hbar} E_{cl}[x(\tau)]}$$

Following our development of the path integral formulation of the partition function, we can interpret the integral of the string as a partition function of a classical mechanical problem at temperature $T_{cl} = \frac{\hbar}{\delta k_B}$. The conclusion is that, the propagator of a quantum mechanical particle over the imaginary time interval $-it_I$ is equal to the classical partition function of a string of length T_I with its endpoints fixed, at a classical temperature T_{cl} . The classical problem has $D + 1$ dimensions (here $1 + 1$) because we have promoted the time direction of the quantum problem to a classical spatial coordinate.

We can develop this a step further, which is something we will do later, but let us introduce it here already. The partition function of a *quantum statistical mechanical system* is given by,

$$Z_{QM}(\beta) = \int dx \langle x | e^{-\beta \hat{H}} | x \rangle = \int dx F(x, x; -i\beta \hbar)$$

which can be interpreted as a relationship between the partition function of a quantum mechanical partition function at temperature $1/\beta$ and the propagator of a quantum particle to return to its original position in imaginary time $\beta \hbar$. $F(x_0, x_1; -it_I)$ then describes two different statistical mechanical problems, based on our interpretation of the imaginary time parameter T .

- T_I is the length of a classical system with fixed end points at an **effective** classical temperature T_{cl} , which corresponds to one extra dimension.

- $T = \beta\hbar$ and upon taking the trace, F corresponds to the quantum partition function.

The relation between the two can be seen by writing

$$Z_{QM}(\beta) = \int dx F(x, x; -i\beta\hbar) = \int dx Z(x, x; \beta\hbar) = Z_{QM}^{\beta\hbar}(T_{cl})$$

2 Lecture 2: Classical XY model

We start by studying the classical XY model which is a canonical model of magnetism. It consists of 2D spins, $\vec{I}_j = (I_j^x, I_j^y) = I(\cos \theta_j, \sin \theta_j)$ with $\theta_j \in [0, 2\pi]$ a continuous variable that represents the angle of the spin relative to a fixed axis, interacting via a nearest neighbour interaction,

$$\vec{I}_j \cdot \vec{I}_l = I^2 \cos(\theta_j - \theta_l)$$

The Hamiltonian of the model is,

$$H = -J \sum_{\langle ij \rangle} \vec{I}_j \cdot \vec{I}_l = -JI^2 \sum_{\langle ij \rangle} \cos(\theta_j - \theta_l)$$

and the corresponding partition function,

$$Z = \int_0^{2\pi} d\theta_1 \dots d\theta_N e^{-\beta H(\{\theta\})}$$

The model possesses a continuous $O(2)$ symmetry: $\vec{I}_j = M\vec{I}_j$, with $\det M = \pm 1$. There are two proper subgroups of the symmetry group,

$$\begin{aligned} SO(2): \quad \theta_j &\rightarrow \theta_j + \alpha \quad \forall \alpha \\ \mathbb{Z}_2: \quad \vec{I}_j &\rightarrow -\vec{I}_j \end{aligned}$$

We will later see that the existence of this continuous symmetry in two dimensions does not allow the system to have long range order, even at zero temperature. However, we will find a phase transition between a phase with short-range correlations at high temperatures, to a phase of *quasi-long-range order* at low temperatures.

For $J > 0$ the system is ferromagnetic and at low temperatures the spins are fully aligned with each other. For $J < 0$, the system is antiferromagnetic, and at low temperatures the spins are fully anti-aligned with their nearest neighbours. The two phases are equivalent on a bipartite lattice by taking $\vec{I}_{j \in B_{\text{subl}}} = -\vec{I}_{j \in \text{subl}}$, or $\theta_{j \in \text{subl}} = -\theta_{j \in \text{subl}}$.

2.1 Large Temperature Analysis

We take $J > 0$ and consider the correlation function $C(\vec{R}_i, \vec{R}_j) = \langle \vec{I}_i \cdot \vec{I}_j \rangle_\beta$ at a temperature $T = \frac{1}{\beta}$. This is computed in the usual statistical mechanical sense as an average,

$$\begin{aligned} C(\vec{R}_i, \vec{R}_j) &= \frac{1}{Z} \int_0^{2\pi} d\theta_1 \dots d\theta_N \vec{I}_i \cdot \vec{I}_j e^{-\beta H(\{\theta\})} \\ &= \frac{1}{Z} \int_0^{2\pi} d\theta_1 \dots d\theta_N I^2 \cos(\theta_i - \theta_j) e^{-\beta H(\{\theta\})} \end{aligned}$$

At $T = 0$, the state is fully ordered, so that $\vec{I}_i \cdot \vec{I}_j = I^2$ for all i, j and therefore the correlation function simply reduces to $C(\vec{R}_i, \vec{R}_j) = I^2$ for all i, j . This is our first example of *long-range order*.

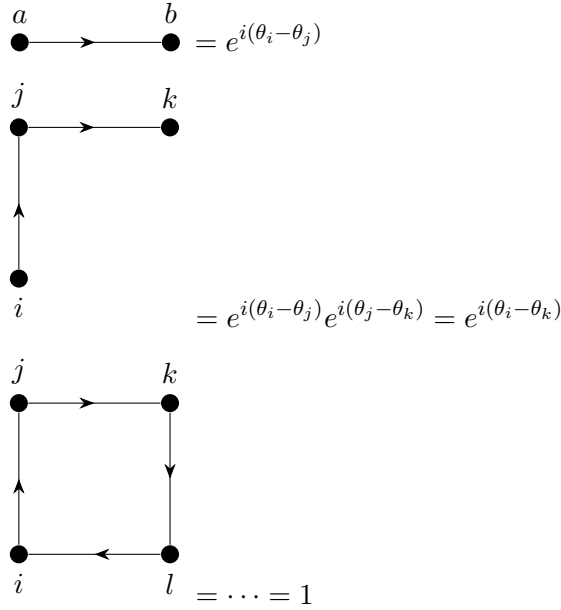
We can also compute the partition function by first Taylor expanding the exponential in powers of βH ,

$$\begin{aligned} e^{\beta J I^2 \sum_{\langle ij \rangle} \cos(\theta_j - \theta_i)} &= \sum_{m=0}^{\infty} \frac{1}{m!} (\beta J I^2)^m \left[\sum_{\langle ij \rangle} \cos(\theta_j - \theta_i) \right]^m \\ &= \sum_{m=0}^{\infty} \frac{1}{m!} (\beta J I^2)^m \left[\frac{1}{2} \sum_{\langle ij \rangle} e^{i(\theta_j - \theta_i)} + e^{-i(\theta_j - \theta_i)} \right]^m \end{aligned}$$

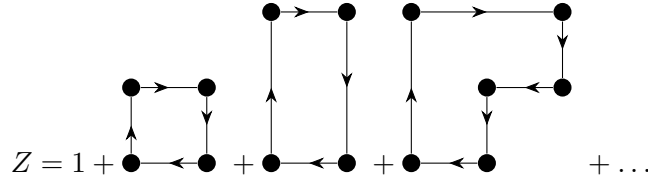
so that the partition function can be written as,

$$Z = \int d\{\theta\} \sum_{m=0}^{\infty} \frac{1}{m!} (\beta J I^2)^m \left[\frac{1}{2} \sum_{\langle ij \rangle} e^{i(\theta_j - \theta_i)} + e^{-i(\theta_j - \theta_i)} \right]^m$$

There is a graphical way to represent this partition function which makes it computation order by order easier to tackle. We represented with directed lines,



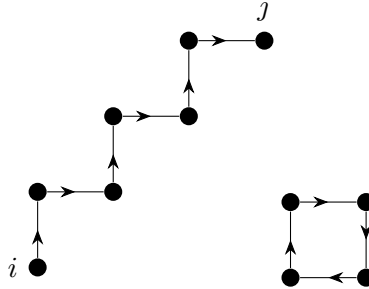
As we can see, all the closed loops will give 1 and all pieces which are open will have a leftover phase $\sim e^{im\delta\theta}$ which integrates to 0 for all integer $m \neq 0$. Therefore, the partition function is a sum of all closed oriented loops, which come with a degeneracy on an $N \times N$ lattice,



Similarly, we can find a graphical representation for the correlation function,

$$C(\vec{R}_i, \vec{R}_j) = \frac{1}{Z} \int d\{\theta\} \frac{I^2}{2} \left[e^{i(\theta_j - \theta_i)} + e^{-i(\theta_j - \theta_i)} \right] \sum_{m=0}^{\infty} \frac{1}{m!} (\beta J I^2)^m \left[\frac{1}{2} \sum_{\langle kl \rangle} e^{i(\theta_k - \theta_l)} + e^{-i(\theta_k - \theta_l)} \right]^m$$

The appearance of the extra factor of the exponentials means that the contributing graphs are now all the open oriented graphs from site i to site j . Due to the suppression factor of $\frac{\beta J I^2}{2}$, shortest paths are favoured. Note that graphs which are allowed include also open paths from i to j multiplied by any closed loop of the form above, eg



Carrying out the sums, we find that to leading order,

$$C(\vec{R}_i, \vec{R}_j) \approx \left(\frac{\beta J I^2}{2} \right)^{|\vec{R}_i - \vec{R}_j|} \equiv e^{-|\vec{R}_i - \vec{R}_j|/\xi}$$

where we $|\vec{R}_i - \vec{R}_j|$ is the length of the shortest open string, and $\xi = \frac{1}{\ln\left(\frac{2T}{J I^2}\right)}$ is the correlation length. Its meaning is that within a region of size (of radius) ξ , most spins will be pointing in the same direction. Note that as T decreases, the correlation length ξ increases, which is precisely the meaning of long-range order!

2.2 Phase Diagrams and Phenomenology

Let us conclude this short study by giving the phase diagrams of the classical XY model in various dimensions.

$d = 1$

In one spatial dimension, there is no phase transition. At exactly $T = 0$, the system sits in a ferromagnetic state in which all the spins point in the same direction. At *any finite temperature* T , the system is disordered. The correlation function in the disordered phase is given by,

$$C(\vec{R}_i, \vec{R}_j) \sim e^{-|\vec{R}_i - \vec{R}_j|/\xi(T)}$$

and the phase diagram is,

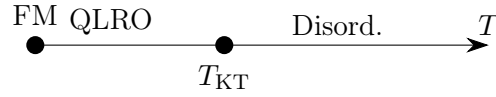


$d = 2$

The case of two dimensions is also special. There is no long-range order at any finite T and at exactly $T = 0$, the system again sits in a ferromagnetic state in which all the spins point in the same direction. The difference now is that there is some temperature T_{KT} , in which the system exhibits *quasi-long-range ordered* with algebraic correlations, and above T_{KT} , the system is disordered.

$$C(\vec{R}_i, \vec{R}_j)_{T < T_{KT}} \sim \frac{1}{|\vec{R}_i - \vec{R}_j|^{y(T)}}$$

and the correlation function in the disordered phase is as above. The phase diagram now includes a finite T phase transition, and the phase diagram is,

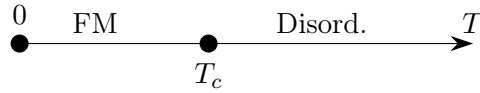


$d \geq 3$

For $d \geq 3$, the continuous symmetry of the system is not enough to prevent long range ordering at finite T . In some sense, each spin interacts with “enough” spins and this interaction ultimately leads to some ordering at small enough temperatures. Thus, there exists a T_c below which the system is in a ferromagnetic state, and above which the system is disordered. The correlation function in the ferromagnetic case, and in the thermodynamic limit, is,

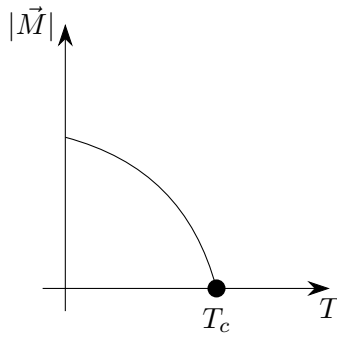
$$C(\vec{R}_i, \vec{R}_j) \rightarrow |\vec{M}|^2 \text{ as } |\vec{R}_i - \vec{R}_j| \rightarrow \infty$$

where $\vec{M}$ is a constant vector. The phase diagram looks is now,



2.3 Order Parameter and Symmetry Breaking

To quantify the difference between two phases (excluding topologically ordered phases), we need the concept of an *order parameter*. The order parameter is a quantity which takes a non-zero value only below T_c (equivalently, above J_c). In the case of the XY model, the order parameter is the magnetization vector $\vec{M}$ whose magnitude $|\vec{M}|$ is non-zero when the spins are ordered around it. The behaviour of $|\vec{M}|$ as a function of T is shown below,



Using the order parameter, we can obtain an effective theory of a coarse-grained system by writing a functional for the free energy in terms of the order parameter field,

$$F[\vec{M}(\vec{r})] = \int d^d \vec{r} \left[|\nabla_\mu \vec{M}(\vec{r})|^2 + \alpha \frac{T - T_c}{T_c} |\vec{M}(\vec{r})|^2 + \lambda |\vec{M}(\vec{r})|^4 + \dots \right]$$

where we included terms that respect the symmetries of the system ($O(2)$ symmetry here so that terms like $\vec{M}_x^2 \vec{M}_y$ is ruled out). The other implicit assumption that went into constructing the

functional $F[\vec{M}(\vec{r})]$ is that it is a slowly varying function and analytic in $\vec{M}(\vec{r})$ (which rules out terms like $|\vec{M}(\vec{r})|^3$). For stability we also assume that $\lambda > 0$ but the sign of the quadratic term is changing. For $T < T_c$ there is a unique minimum at $\vec{M} = 0$ but for $T > T_c$ the minima lie across a circle (mexican hat potential).

3 Lecture 3: Quantum XY model

Having studied the classical XY model and understood its ordered and disordered phases, we now turn our attention to a different but related model, the quantum XY model, also known as the $O(2)$ rotor model. We recall first the quantum mechanics of a single rotor described by the Hamiltonian, but even before doing that let us very briefly recall the properties of classical particle on a ring of radius r . This is described by the Lagrangian,

$$L = \frac{m}{2} r^2 \dot{\phi}^2 + X \dot{\phi}$$

where X in the potential term is a constant parameter and the whole term respects the translational symmetry of the problem. For X constant, the classical equations of motion are still,

$$r^2 \ddot{\phi} = 0$$

The reason is that the potential term is actually a total derivative and thus contributes only in the boundaries. The equation of motion is however derived by the bulk properties of the Hamiltonian, and thus the action, $\delta S = 0$.

We now move to the quantum mechanical case and include the X term to see its effect,

$$\hat{H} = \frac{(\hat{L}_z - X)^2}{2C}, \quad \hat{L}_z = -i \frac{\partial}{\partial \theta}$$

with $C > 0$. Physically, this Hamiltonian describes a quantum mechanical particle on a ring. Recalling that $\hat{L}_z \psi_m(\theta) = m \psi_m(\theta)$ are the eigenstates of the z-angular momentum operator, we can immediately write down the eigenstates of the quantum rotor, which we identify as 2π periodic waves,

$$\psi_m(\theta) = \frac{1}{2\pi} e^{im\theta}, \quad \theta \in [0, 2\pi]$$

$$E_m = \frac{(m - X)^2}{2C}$$

where m is an integer with $m = 0$ excluded because of boundary conditions (if we enforce vanishing of the wavefunction at the boundaries, then the constant wavefunction cannot vanish there). We first note that the eigenenergies now *do* depend on X . The spectra of the rotor at X and $X + \mathbf{Z}$ are the same, and therefore we can identify inequivalent X 's by rewriting X in the following form,

$$X = \frac{\theta}{2\pi}, \quad \theta \cong \theta + 2\pi$$

and we call this a θ term.

For $X \neq 0$, the ground state is the $m = 0$ eigenstate and describes left and right movers with the same energy. For $X = 0$, the ground state is the $n = 1$ eigenstate. Oddly enough, the ground state is degenerate at $\theta = \pi$ ($X = 1/2$) with $n = 0, n = 1$ both being ground states. The system is gapless at $\theta = \pi$ only. This absence of the gap usually means travelling waves with no energy cost and thus long-range correlations.

3.1 Path Integral for a Single Rotor

Let us write the hamiltonian of the single quantum rotor for simplicity,

$$H = \frac{(\hat{L}^z)^2}{2C} + f(\hat{\theta})$$

The finite temperature partition function can be worked using a Trotter decomposition as before, and writing $N\Delta\tau = \beta$ we have,

$$\begin{aligned} Z &= \text{tr} e^{-\beta \hat{H}} = \int_0^{2\pi} d\theta_0 \langle \theta_0 | e^{-\beta \hat{H}} | \theta_0 \rangle \\ &= \int_0^{2\pi} d\theta_0 \langle \theta_0 | e^{-\Delta\tau \hat{H}} e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}} | \theta_0 \rangle \\ &= \int_0^{2\pi} d\theta_0 \dots d\theta_{N-1} \langle \theta_0 | e^{-\Delta\tau \hat{H}} | \theta_{N-1} \rangle \langle \theta_{N-1} | e^{-\Delta\tau \hat{H}} | \theta_{N-2} \rangle \dots e^{-\Delta\tau \hat{H}} | \theta_1 \rangle \langle \theta_1 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle \end{aligned}$$

At this stage we need to workout the matrix element, $\langle \theta_1 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle$, by using the $\{m\}$ eigenbasis of $\hat{H}$,

$$\begin{aligned} \langle \theta_1 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle &= \sum_{m_0 \in \mathbb{Z}} \langle \theta_1 | m_0 \rangle \langle m_0 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle \\ &= \sum_{m_0 \in \mathbb{Z}} \langle \theta_1 | m_0 \rangle \langle m_0 | \theta_0 \rangle \left[1 - \Delta\tau \left(\frac{m_0^2}{2C} + f(\theta_0) + \mathcal{O}(\Delta\tau^2) \right) \right] \\ &= \sum_{m_0 \in \mathbb{Z}} \frac{e^{im_0(\theta_1 - \theta_0)}}{2\pi} e^{-\Delta\tau \left(\frac{m_0^2}{2C} + f(\theta_0) \right)} \end{aligned}$$

At this point we also introduce an auxiliary integration variable x and use the Posson summation formula make the integral over x Gaussian, the matrix element is,

$$\begin{aligned} \langle \theta_1 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle &= \int_{-\infty}^{\infty} dx \sum_{m_0} \delta(x - m_0) \frac{1}{2\pi} e^{ix(\theta_1 - \theta_0)} e^{-\Delta\tau \left(\frac{m_0^2}{2C} + f(\theta_0) \right)} \\ &= \sum_n \int_{-\infty}^{\infty} dx \frac{1}{2\pi} e^{i(2\pi n + \theta_1 - \theta_0) - \frac{\Delta\tau}{2C} x^2} e^{-\Delta\tau f(\theta_0)} \\ &= \sum_{n=-\infty}^{\infty} \sqrt{\frac{2\pi C}{\Delta\tau}} e^{-\frac{C}{2\Delta\tau} (2\pi n + \theta_1 - \theta_0)^2} \frac{1}{2\pi} e^{-\Delta\tau f(\theta_0)} \end{aligned}$$

In the continuum limit, we replace, $\theta_1 - \theta_0 \rightarrow \dot{\theta}(\tau=0)\Delta\tau$, and furthermore, as $\Delta\tau \rightarrow 0$, we only need to consider the $n=0$ term. All together,

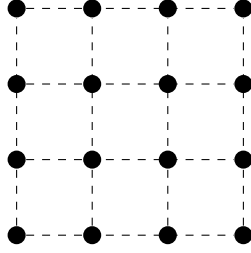
$$\langle \theta_1 | e^{-\Delta\tau \hat{H}} | \theta_0 \rangle = \sqrt{\frac{C}{2\pi\Delta\tau}} e^{-\frac{C}{2\Delta\tau} (2\pi n + \theta_1 - \theta_0)^2 - \Delta\tau f(\theta_0)}$$

The partition function thus admits the functional integral form,

$$\begin{aligned} Z &= \int d\theta_0 \dots d\theta_{N-1} \left(\frac{C}{2\pi\Delta\tau} \right)^{N/2} \exp \left\{ - \sum_{n=0}^{N-1} \frac{C}{2\Delta\tau} (\theta_{n+1} - \theta_n)^2 - \sum_{n=0}^{N-1} \Delta\tau f(\theta_n) \right\} \\ &= \int_{\substack{\theta(0)=\theta(\tau) \\ \theta(\tau) \in [0, 2\pi]}} \mathcal{D}\theta(\tau) e^{-\int_0^\beta d\tau \left[\frac{C}{2} \dot{\theta}^2(\tau) + f(\theta) \right]} \end{aligned}$$

3.2 Lattice $O(2)$ rotor model

We now turn our attention to a lattice of quantum rotors, and for simplicity we consider a square lattice,



Each site is now described by two non-commuting operators, $\hat{\theta}_j, \hat{L}_j^z$, $j = 1, \dots, N$, with $[\hat{\theta}_j, \hat{L}_k^z] = i\delta_{jk}$ and,

$$\begin{aligned}\hat{\theta}_j |\theta_j\rangle &= \theta_j |\theta_j\rangle, & \theta_j &\in [0, 2\pi) \\ \hat{L}_j^z |m_j\rangle &= m_j |m_j\rangle, & m &\in \mathbb{Z}\end{aligned}$$

and $\langle \theta_j | m_j \rangle = \psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}$. The rotors are coupled to each other via an interaction term, and the Hamiltonian takes the form,

$$H = -J \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) + \sum_j \frac{(\hat{L}_j^z)^2}{2C}$$

The phases of the system can be understood at least quantitatively by thinking the extreme limits of the two couplings.

- XY coupling dominant, $J \gg \frac{1}{C}$:

$$|GS\rangle = \prod_{j=1}^N |\theta_j = 0\rangle$$

This is a ferromagnetically ordered state and thus breaks a symmetry.

- Dynamical term dominant, $J \ll \frac{1}{C}$:

$$|GS\rangle = \prod_{j=1}^N |m_j = 0\rangle = \int_0^{2\pi} d\theta \frac{1}{\sqrt{2\pi}} |\theta\rangle$$

As can be seen, since θ is arbitrary at each site, this is the disorder phase. The ground state in the $\{m\}$ basis is unique.

At this point, we can trivially extend our single rotor path integral to the case of many rotors described by the Hamiltonian above,

$$Z = \int_{\substack{\theta_j(0)=\theta_j(\tau) \\ \theta_j(\tau) \in [0, 2\pi)}} \mathcal{D}\theta_j(\tau) \exp \left\{ - \int_0^\beta d\tau \left[\sum_{j=1}^L \frac{C}{2} \dot{\theta}_j^2(\tau) - J \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) \right] \right\}$$

4 Lecture 4: Goldstone Boson

As before, the classical “path” is given by the extremum of the action, $\delta S = 0$, which is solved trivially by $\theta_j(\tau) = 0$. To get an understanding of the excitations of the system, we analyze the quantum and thermal fluctuations around this classical path. This is convenient in the path integral formalism because the “classical path” which minimizes the free energy of the system is the most important contribution to the integral. Thus,

$$\begin{aligned}\theta_j(\tau) &= \langle \theta \rangle && \text{“Classical path”} \\ \rightarrow \theta_j(\tau) &= \langle \theta \rangle + \delta\theta_j(\tau) && \text{“Classical path”} + \text{fluctuations}\end{aligned}$$

By writing $\cos(\delta\theta_i - \delta\theta_j) \approx 1 - \frac{1}{2}(\delta\theta_i - \delta\theta_j)^2$, we can rewrite the path integral in terms of the fluctuation fields, noting also that the fluctuations do satisfy the PBC $\delta\theta(0) = \delta\theta(\beta)$,

$$\begin{aligned}Z &= \int_{\delta\theta(0)=\delta\theta(\beta)} \mathcal{D}\delta\theta_j(\tau) \exp \left\{ - \int_0^\beta d\tau \left[\sum_{j=1}^L \frac{C}{2} \dot{\delta\theta}_j^2(\tau) + \frac{J}{2} \sum_{\langle jl \rangle} (\theta_j - \theta_l)^2 \right] \right\} \\ &= \int \mathcal{D}\delta\theta_j(\tau) e^{-S_0[\delta\theta]}\end{aligned}$$

Note that this form of the path integral is now comparable to the harmonic approximation of lattice vibrations that gives rise to phonons. To calculate the path integral, we will make a Fourier transformation in both real space $\vec{r}$ and imaginary time direction τ ,

$$\delta\theta_j(\tau) = \frac{1}{\sqrt{\beta N}} \sum_{\vec{k}, i\omega_n} e^{i\vec{k} \cdot \vec{R}_j - i\omega_n \tau} \delta\theta(\vec{k}, i\omega_n)$$

where $\vec{k}$ is a vector in the first Brillouin zone which in our case is a d -dimensional cubic lattice, i.e. $\vec{k} = (k_1, \dots, k_d)$ with $k_i \in (-\frac{\pi}{a}, \frac{\pi}{a})$ and a the lattice spacing. The imaginary frequencies $i\omega_n$ are called the bosonic *Matsubara frequencies* and to satisfy the PBC they have to be $\omega_n = \frac{2\pi}{\beta} n$, $n \in \mathbb{Z}$ (because $\delta\theta_j(\tau) = \delta\theta_j(\tau + \beta) \iff e^{i\omega_n \beta} = +1$). The reality of the fluctuation fields enforces a condition on the Fourier transformed fields,

$$\theta(\vec{k}, i\omega_n) = \theta^*(-\vec{k}, -i\omega_n)$$

and we also recall the completeness relations in the convention we are using,

$$\frac{1}{N} \sum_{j=1}^N e^{i(\vec{k} - \vec{k}') \cdot \vec{R}_j} = \delta_{\vec{k}, \vec{k}'} \quad \frac{1}{\beta} \int_0^\beta d\tau e^{i(\omega_n - \omega_m)\tau} = \delta_{n,m}$$

In the Fourier transformed field, the partition function takes the form,

$$Z = \int \prod_{\vec{k}, i\omega_n} d\theta(\vec{k}, i\omega_n) e^{-S_0[\delta\theta]}$$

Let us work out the various terms in the transformed action by carrying out the algebra explicitly,

$$\begin{aligned}
& \int_0^\beta d\tau \sum_{j=1}^L \frac{C}{2} \dot{\theta}_j^2(\tau) \\
&= \frac{1}{\beta N} \int_0^\beta d\tau \sum_{\vec{k}, i\omega_n} \sum_{\vec{k}', i\omega_m} \sum_{j=1}^L \frac{C}{2} (-i\omega_n)(-i\omega_m) \delta\theta(\vec{k}, i\omega_n) \delta\theta(\vec{k}', i\omega_m) e^{i(\vec{k}+\vec{k}') \cdot \vec{R}_j} e^{-i(\omega_n+\omega_m)\tau} \\
&= \sum_{\vec{k}, i\omega_n} \frac{C}{2} \omega_n^2 \theta^*(\vec{k}, i\omega_n) \theta(\vec{k}, i\omega_n) \\
& \int_0^\beta d\tau \frac{J}{2} \sum_{\langle jl \rangle} (\delta\theta_j - \delta\theta_l)^2 = \int_0^\beta d\tau \frac{J}{2} \sum_{\langle jl \rangle} \delta\theta_j^2 + \delta\theta_l^2 - 2\delta\theta_j \delta\theta_l \\
&= \int_0^\beta d\tau \left(Jd \sum_j \delta\theta_j^2 - J \sum_j \sum_{\vec{\delta}} \delta\theta_j \delta\theta_{j+\vec{\delta}} \right) \\
&= \sum_{\vec{k}, i\omega_n} \left(Jd - J \sum_{\vec{\delta}} e^{i\vec{k} \cdot \vec{\delta}} \right) \theta^*(\vec{k}, i\omega_n) \theta(\vec{k}, i\omega_n)
\end{aligned}$$

Altogether, the action in terms of the fluctuation fields is,

$$S[\delta\theta] = \sum_{\vec{k}, i\omega_n} \left[\frac{C}{2} \omega_n^2 + \left(Jd - J \sum_{l=1}^d \cos(k_l a) \right) \right] \theta^*(\vec{k}, i\omega_n) \theta(\vec{k}, i\omega_n)$$

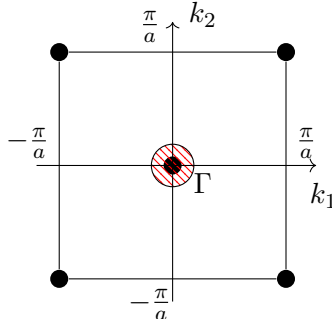
Note that this is directly comparable to the field theory of a harmonic oscillator, for which $x(\tau) = \frac{1}{\sqrt{\beta}} \sum_{i\omega_n} e^{-i\omega_n \tau} x(i\omega_n)$, with excitation energy (frequency) ω ,

$$\begin{aligned}
Z_{\text{h.o.}} &= \int \mathcal{D}x(\tau) \exp \left\{ - \int_0^\beta d\tau \left[\frac{1}{2} \dot{x}^2(\tau) + \frac{1}{2} m\omega^2 x^2(\tau) \right] \right\} \\
&= \int \prod_{i\omega_n} dx(i\omega_n) \exp \left\{ - \sum_{i\omega_n} \frac{1}{2} m(\omega_n^2 + \omega^2) x(i\omega_n) x^*(i\omega_n) \right\}
\end{aligned}$$

What we have therefore derived is that the partition function of N quantum rotors is precisely the partition function of N decoupled harmonic oscillators. The dispersion relation of the rotors are,

$$\omega_{\vec{k}} = \sqrt{\frac{2J}{C} \sum_{l=1}^d (1 - \cos(k_l a))}$$

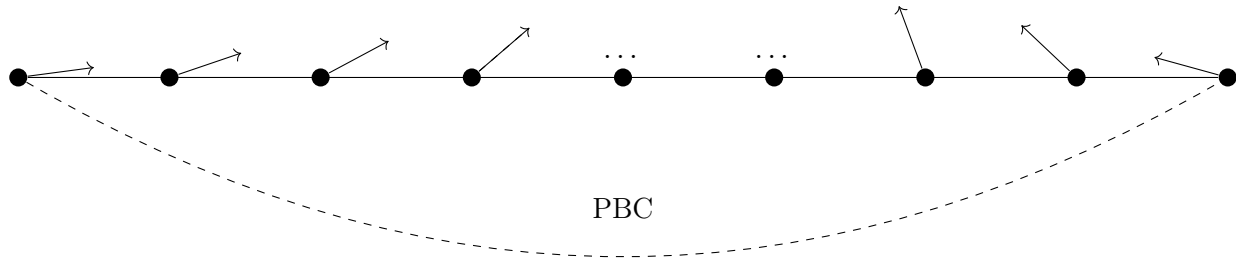
defined in the cubic d-dimensional Brillouin zone,



For small momenta $\vec{k}$, near the Γ point (within the shaded sphere in the diagram above), corresponding to the long wavelength excitations, the dispersion relation takes the form,

$$\omega_{\vec{k}} \approx \sqrt{\frac{Ja^2}{C}} |\vec{k}| \rightarrow \text{Linear} \rightarrow \text{Gapless!}$$

This is our first example of a *Goldstone boson*. These are excitations which arise due to the spontaneous breaking of a continuous symmetry and they are associated with slow long-wavelength fluctuations of the field or order parameter. In the present case and more generally in magnetism, the goldstone bosons are often called spin-waves which are wave-like modulations in the spin configuration. An example is shown below,



The goldstone mode here is massless and propagates at the constant velocity $v_s = \sqrt{\frac{Ja^2}{C}}$. The breaking of the continuous rotational symmetry implies that the system is enforced to choose one of the infinitely degenerate ground states, and an intuitive way to think about the goldstone mode is that it is trying to restore this symmetry. The symmetry is locally reorientated. Other examples of Goldstone modes include phonons (associated with slow density fluctuations of the crystal's atoms) and magnons.

As a final comment, we note that if we explicitly break the symmetry by adding, eg, a magnetic field, the spins now will pick a particular direction and the Goldstone bosons become “massive” or “gapped”. There is a more general mathematical statement to make, which is useful for relativistic theories. If a Hamiltonian admits a symmetry G , and the ground state of the Hamiltonian has a reduced symmetry H (so that G is spontaneously broken to H), then the number of goldstone bosons is given by,

$$(\# \text{ Goldstone bosons}) = (\dim G) - (\dim H)$$

This makes sense mathematically by noting that the manifold of ground states is understood to be the set of cosets G/H .

5 Lecture 5: Mermin-Wagner Theorem

In this lecture we will prove an important theorem that gives predictions on the kind of ordering that can happen at low temperatures. We will see that at “small enough” dimensions, fluctuations are always strong and destroy any long-range order, at any finite temperature. Recall then the partition function of the rotor model in the Gaussian approximation,

$$Z = \int \prod_{\vec{k}, i\omega_n} d\theta(\vec{k}, i\omega_n) \exp \left\{ - \sum_{\vec{k}, i\omega_n} \frac{C}{2} (\omega_n^2 + \omega^2) \theta(\vec{k}, i\omega_n) \theta^*(\vec{k}, i\omega_n) \right\}$$

We identify an order parameter $\vec{I} = (\cos \hat{\theta}, \sin \hat{\theta})$, and we can compute via path integral representation the expectation value of $\cos \hat{\theta}$,

$$\begin{aligned} \langle \cos \hat{\theta}_j \rangle &= \frac{1}{Z} \text{tr} \left\{ \cos \hat{\theta}_j \right\} \\ &= \frac{1}{Z} \int \prod_{j=1}^N d\theta_{\tau=0,j} \langle \{\theta_0\} | e^{-\Delta\tau \hat{H}} e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}} | \{\theta_0\} \rangle \end{aligned}$$

where we denoted $|\{\theta_0\}\rangle = |\theta_{0,l=1}, \theta_{0,l=2}, \dots, \theta_{0,l=N}\rangle$ and $\theta_0 \equiv \theta_{\tau=0}$. In carrying out the usual procedure performing a Trotter decomposition, we get in the limit $\Delta\tau \rightarrow 0$,

$$\langle \cos \hat{\theta}_j \rangle = \frac{1}{Z} \int \mathcal{D}\theta(\tau) \cos \theta_j(\tau=0) e^{-S[\theta]}$$

The “classical path” is such that $\theta_{cl} = 0$ so that $\cos(\theta_j(\tau)) = 1$ for all j, τ . Therefore,

$$\langle \cos \hat{\theta}_j \rangle_{cl} = 1$$

We consider the fluctuations around this classical solution by writing $\theta_j(\tau) = \theta_{cl} + \delta\theta_j(\tau)$, and we rewrite the order parameter in the form, $\cos \delta\theta_j(\tau) = \frac{1}{2} (e^{i\delta\theta_j(\tau=0)} + e^{-i\delta\theta_j(\tau=0)})$.

$$\begin{aligned} \langle \cos \hat{\theta}_j \rangle_0 &= \frac{1}{Z} \int \mathcal{D}\theta(\tau) \cos \theta_j(\tau=0) e^{-S[\theta]} \\ &= \frac{1}{Z_0} \int \mathcal{D}\delta\theta \cos \delta\theta_j(\tau=0) e^{-S_0[\delta\theta]} \end{aligned}$$

By translational symmetry, we only need to compute $\langle e^{i\delta\theta_{j=0}(\tau=0)} \rangle$, and we use the Fourier decomposition $\delta\theta_j(\tau) = \frac{1}{\sqrt{\beta N}} \sum_{\vec{k}, i\omega_n} e^{i\vec{k} \cdot \vec{R}_j - i\omega_n \tau} \delta\theta(\vec{k}, i\omega_n)$.

$$\begin{aligned} \langle e^{i\delta\theta_{j=0}(\tau=0)} \rangle &\approx \frac{1}{Z_0} \int \mathcal{D}\delta\theta e^{-S_0[\delta\theta] + i\delta\theta_{j=0}(\tau=0)} \\ &= \frac{1}{Z_0} \int \mathcal{D}\delta\theta \exp \left\{ - \sum_{\vec{k}, i\omega_n} \frac{C}{2} (\omega_n^2 + \omega^2) \theta(\vec{k}, i\omega_n) \theta^*(\vec{k}, i\omega_n) + \frac{i}{\sqrt{\beta N}} \sum_{\vec{k}, i\omega_n} \theta(\vec{k}, i\omega_n) \right\} \end{aligned}$$

To perform the integral, we complete the square by rewriting θ as $\theta(\vec{k}, i\omega_n) = \frac{1}{2} \left(\theta(\vec{k}, i\omega_n) + \theta^*(\vec{k}, i\omega_n) \right)$,

$$\begin{aligned} \langle e^{i\delta\theta_{j=0}(\tau=0)} \rangle &= \frac{1}{Z_0} \int \mathcal{D}\delta\theta \exp \left(- \sum_{\vec{k}, i\omega_n} \left[\theta^*(\vec{k}, i\omega_n) - \frac{1}{\sqrt{\beta N}} \frac{1}{C} (\omega_n^2 + \omega^2)^{-1} \right] \frac{C}{2} (\omega_n^2 + \omega^2) \right. \\ &\quad \left. \left[\theta(\vec{k}, i\omega_n) - \frac{1}{\sqrt{\beta N}} \frac{1}{C} (\omega_n^2 + \omega^2)^{-1} \right] \right) \\ &\quad \exp \left\{ \sum_{\vec{k}, i\omega_n} \frac{i^2}{\beta N} \frac{1}{2C} \frac{1}{\omega_n^2 + \omega^2} \right\} \end{aligned}$$

In its latter form, it is easier to see that the expression can be rewritten under a change of variables in $\theta(\vec{k}, i\omega_n)$ and perform a Gaussian integral, which will exactly cancel out the Z_0 in the denominator. Thus, the expectation value we are after is,

$$\langle e^{i\delta\theta_{j=0}(\tau=0)} \rangle = \exp \left\{ -\frac{1}{2C} \frac{1}{\beta N} \sum_{\vec{k}, i\omega_n} \frac{1}{\omega_n^2 + \omega^2} \right\} \langle e^{-i\delta\theta_{j=0}(\tau=0)} \rangle = \exp \left\{ -\frac{1}{2C} \frac{1}{\beta N} \sum_{\vec{k}, i\omega_n} \frac{1}{\omega_n^2 + \omega^2} \right\}$$

and therefore,

$$\langle \cos \delta\theta_j \rangle = \exp \left\{ -\frac{1}{2C} \frac{1}{\beta N} \sum_{\vec{k}, i\omega_n} \frac{1}{\omega_n^2 + \omega^2} \right\}$$

5.1 Summation over the Matsubara Frequencies

What is left now is to perform the sum over the Matsubara frequencies, $\{\omega_n | \omega_n = \frac{2\pi}{\beta} n, n \in \mathbb{Z}\}$. Let's then define the sum under question,

$$S = \frac{1}{\beta} \sum_{i\omega_n} \frac{1}{\omega_n^2 + \omega^2} = -\frac{1}{\beta} \sum_{i\omega_n} \frac{1}{(z - \omega_{\vec{k}})(z + \omega_{\vec{k}})}$$

The idea is to find a function $f(z)$ which is such that,

$$\oint \frac{dz}{2\pi i} f(z) \frac{1}{(z - \omega_{\vec{k}})(z + \omega_{\vec{k}})}$$

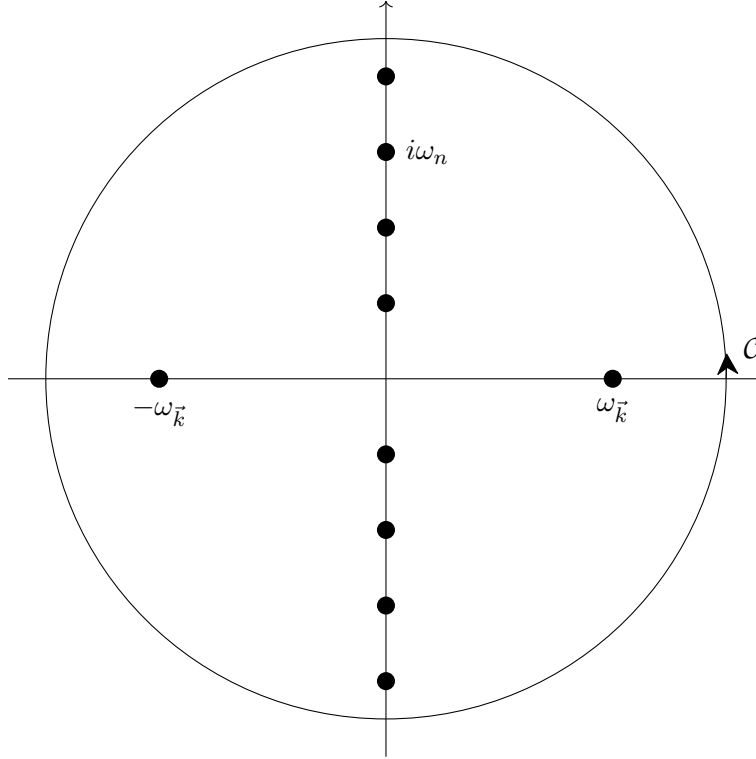
is precisely the sum we are looking for. A function $f(z)$ which has this property will have poles at all $z = i\omega_n$ and two extra poles at $z = \pm\omega_{\vec{k}}$. Additionally, we need $g(z) = f(z) \frac{1}{(z - \omega_{\vec{k}})(z + \omega_{\vec{k}})}$ to approach 0 not slower than $1/|z|^2$ as $|z| \rightarrow 0$. A function which satisfies these properties is the bosonic occupation number function,

$$n_B(z) = f(z) = \frac{1}{e^{\beta z} - 1}$$

To see why, we note that near $z = 0$, $e^{\beta z} - 1 = \beta z \left(1 + \frac{1}{2}\beta z + \frac{1}{3!}(\beta z)^2 + \dots \right)$, so that,

$$\lim_{z \rightarrow 0} \left[(z - 0) \frac{1}{e^{\beta z} - 1} \right] = \lim_{z \rightarrow 0} \left[\frac{1}{\beta e^{\beta z}} \right] = \frac{1}{\beta}$$

Consider now the contour with poles at all $z = i\omega_n$ and $z = \pm\omega_{\vec{k}}$ shown below,



where $\mathcal{C}$ is the contour which encircles all the poles of the function $n_B(z)$. By Cauchy's theorem, we know that if $g(z)$ decays not slower than $1/|z|^2$ as $|z| \rightarrow \infty$, we have,

$$\begin{aligned} 0 &= \oint \frac{dz}{2\pi i} g(z) = \oint \frac{dz}{2\pi i} f(z) \frac{1}{(z - \omega_{\vec{k}})(z + \omega_{\vec{k}})} \\ &= \sum_{i\omega_n} \text{Res}(z = i\omega_n) + \frac{n_B(\omega_{\vec{k}})}{2\omega_{\vec{k}}} + \frac{n_B(-\omega_{\vec{k}})}{(-2\omega_{\vec{k}})} \\ &= \sum_{i\omega_n} \frac{1}{\beta} \frac{1}{(i\omega_n - \omega_{\vec{k}})(i\omega_n + \omega_{\vec{k}})} + \frac{n_B(\omega_{\vec{k}})}{2\omega_{\vec{k}}} + \frac{(-1 - n_B(\omega_{\vec{k}}))}{(-2\omega_{\vec{k}})} \end{aligned}$$

and therefore, the sum is rewritten in terms of the bosonic occupation numbers,

$$\sum_{i\omega_n} \frac{1}{\beta} \frac{1}{(i\omega_n - \omega_{\vec{k}})(i\omega_n + \omega_{\vec{k}})} = - \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \frac{1}{\omega_{\vec{k}}}$$

The first term is temperature dependent and is therefore the thermal fluctuation contribution. The second term, $\frac{1}{2\omega_{\vec{k}}}$, is the quantum fluctuation (zero point motion) contribution. Altogether, the order parameter expectation value takes the form,

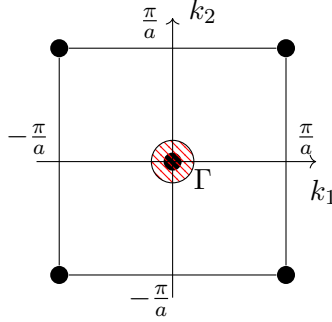
$$\langle \cos \theta_j \rangle = \exp \left\{ -\frac{1}{2C} \frac{1}{N} \sum_{\vec{k}} \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \frac{1}{\omega_{\vec{k}}} \right\}$$

Before moving forward with the calculation, we list here other useful results on different Matsubara summations for various summands $g(i\omega_n)$.

$$g(i\omega_n) = \frac{1}{i\omega_n - \omega} \rightarrow$$

5.2 Sum over the 1BZ

To complete the calculation we need to perform the last sum over the Brillouin zone. These sums are typically very difficult to do analytically and we often resort to numerical integration methods. Here, we will approximate the integral by using the k -points which contribute the most to the sum. Recall the BZ figure from the previous section,



where the shaded region is a circle of radius k_c , an artificial cutoff that we introduce to perform the integral.

Let N_l be the number of sites in the l^{th} direction, so that $\mathcal{N} = N_1 N_2 \dots N_d$ is the total number of points. The volume of a single k -point in the reciprocal space is given by,

$$\frac{2\pi}{N_1 a} \cdot \frac{2\pi}{N_2 a} \cdot \dots \cdot \frac{2\pi}{N_d a} = \frac{(2\pi)^d}{\mathcal{N} a^d}$$

and this allows us to turn the sum into an integral in the $\mathcal{N} \rightarrow \infty$ limit,

$$\frac{(2\pi)^d}{\mathcal{N} a^d} \sum_{\vec{k}} \rightarrow \int_{-\pi/a}^{\pi/a} d^d k$$

and the order parameter takes the form,

$$\langle \cos \theta_j \rangle = \exp \left\{ -\frac{1}{2C} \frac{a^d}{(2\pi)^d} \int_{-\pi/a}^{\pi/a} d^d k \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \frac{1}{\omega_{\vec{k}}} \right\}$$

Recall that $\omega_{\vec{k}} \approx v_s |\vec{k}|$ for $|\vec{k}| \rightarrow 0$, and thus since $\omega_{\vec{k}}$ is small in that limit, we approximate the integral by changing the region to a sphere of radius k_c .

5.3 Long-range order

Finite Temperature

At finite temperatures, we can write,

$$n_B(\omega_{\vec{k}}) = \frac{1}{e^{\beta \omega_{\vec{k}}}} \approx \frac{1}{\beta \omega_{\vec{k}}} - \frac{1}{2} + \frac{\beta \omega_{\vec{k}}}{12} - \frac{(\beta \omega_{\vec{k}})^3}{720} + \dots$$

so that we have isolated the singular part of the function. The integral becomes,

$$\begin{aligned}
\int_{-\pi/a}^{\pi/a} d^d k \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \frac{1}{\omega_{\vec{k}}} &\approx \int_0^{k_c} d|\vec{k}| \frac{(2\pi)^{d/2}}{\Gamma(d/2)} |\vec{k}|^{d-1} \left(\frac{1}{\beta\omega_{\vec{k}}} + \dots \right) \frac{1}{\omega_{\vec{k}}} \\
&\sim \int_0^{k_c} d|\vec{k}| \frac{(2\pi)^{d/2}}{\Gamma(d/2)} |\vec{k}|^{d-3} \frac{T}{v_s^2} \\
&= \begin{cases} -\frac{1}{|\vec{k}|} \Big|_0^{k_c} \rightarrow \infty & d = 1 \\ -\ln |\vec{k}| \Big|_0^{k_c} \rightarrow \infty & d = 2 \\ \text{Regular} & d \geq 3 \end{cases}
\end{aligned}$$

In the cases of $d = 1$ and $d = 2$, the IR divergence at $\vec{k} = 0$ comes from the linearity of $\omega_{\vec{k}}$ which corresponds to the emergence of Goldstone bosons.

The surprising conclusion from this finite T analysis is that ferromagnetic **long-range order does not survive** under thermal or/and quantum fluctuations in dimensions $d = 1$ and $d = 2$!

Zero Temperature

At $T = 0$ the analysis is simpler. There is no thermal population of any excitations and thus $n_B(\omega_{\vec{k}}) \rightarrow 0$. The integral to compute is then,

$$\begin{aligned}
\int_{-\pi/a}^{\pi/a} d^d k \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \frac{1}{\omega_{\vec{k}}} &\approx \int_0^{k_c} d|\vec{k}| \frac{(2\pi)^{d/2}}{\Gamma(d/2)} |\vec{k}|^{d-1} \frac{1}{v_s |\vec{k}|} \\
&\sim \int_0^{k_c} d|\vec{k}| |\vec{k}|^{d-2} \\
&= \begin{cases} \ln |\vec{k}| \Big|_0^{k_c} \rightarrow \infty & d = 1 \\ \text{Regular} & d \geq 2 \end{cases}
\end{aligned}$$

The IR divergence exists only in $d = 1$ now. The conclusion now is that FM long-range order **does not** survive for $d = 1$ even at $T = 0$!

Altogether, we summarise the results of the Mermin-Wagner (Hohenberg, Coleman) theorem in the following:

There is no spontaneous breaking of a continuous symmetry when,

(i) $d = 1$ and $d = 2$ at $T > 0$.

(ii) $d = 1$ at $T = 0$.

6 Lecture 6: Kosterlitz-Thouless (KT) Phase

We continue now in studying the KT phase of the rotor model,

$$H = -J \sum_{\langle j,l \rangle} \cos(\theta_j - \theta_l) + \sum_j \frac{(\hat{L}_j^z)^2}{2C}$$

We found previously that,

- For $T > 0$: $\langle \cos(\hat{\theta}_j) \rangle = 0$ in $d = 1$ and $d = 2$
- For $T = 0$: $\langle \cos(\hat{\theta}_j) \rangle = 0$ in $d = 1$

which are manifestations of the statement that there is no long range order! We will investigate however the existence of any possible short range order. We thus define the correlation function, $C(\vec{R}_j, \vec{R}_l) = \langle \vec{I}_j, \vec{I}_l \rangle$ and compute it in the part integral formalism,

$$\begin{aligned} C(\vec{R}_j, \vec{R}_l) &= \langle \cos(\hat{\theta}_j - \hat{\theta}_l) \rangle \\ &= \frac{1}{Z} \int \mathcal{D}\theta(\tau) \cos(\theta_j - \theta_l) e^{-S[\theta]} \end{aligned}$$

where,

$$S[\theta] = \int_0^\beta d\tau \left[\sum_{j=1}^L \frac{C}{2} \dot{\theta}_j^2(\tau) - J \sum_{\langle j,l \rangle} \cos(\theta_j - \theta_l) \right]$$

and we chose $\tau = 0$, which is allowed by the translational symmetry of the model. We write, $\theta_j \approx \langle \theta_j \rangle + \delta\theta_j$, so that,

$$C(\vec{R}_j, \vec{R}_l) = \frac{1}{Z} \int \mathcal{D}\delta\theta \frac{1}{2} \left(e^{i(\theta_j - \theta_l)} + e^{-i(\theta_j - \theta_l)} \right) e^{-S[\delta\theta]}$$

with,

$$S[\delta\theta] = \sum_{\vec{k}, i\omega_n} \left[\frac{C}{2} (\omega_n^2 + \omega_k^2) \right] \theta^*(\vec{k}, i\omega_n) \theta(\vec{k}, i\omega_n)$$

In writing, $\delta\theta_j(\tau) = \frac{1}{\beta N} \sum_{\vec{k}, i\omega_n} \theta(\vec{k}, i\omega_n) e^{i\vec{k} \cdot \vec{R}_j - i\omega_n \tau}$, we have,

$$\begin{aligned} \langle e^{i(\hat{\theta}_j - \hat{\theta}_l)} \rangle &\approx \frac{1}{Z_0} \int \mathcal{D}\delta\theta e^{i(\delta\theta_j(\tau=0) - \delta\theta_l(\tau=0))} e^{-S_0[\delta\theta]} \\ &= \frac{1}{Z_0} \int \mathcal{D}\delta\theta e^{\frac{i}{\sqrt{\beta N}} \sum_{\vec{k}, i\omega_n} \theta(\vec{k}, i\omega_n) (e^{i\vec{k} \cdot \vec{R}_j} - e^{i\vec{k} \cdot \vec{R}_l})} e^{-S_0[\delta\theta]} \end{aligned}$$

At this point, we complete the sum and perform the Gaussian integral, as we did before. We can also perform the Matsubara sum and the result comes out as follows,

$$\begin{aligned}
 \langle e^{i(\theta_j - \theta_l)} \rangle &= \exp \left\{ -\frac{1}{C} \frac{1}{\beta N} \sum_{\vec{k}, i\omega_n} \frac{1}{\omega_n^2 + \omega^2} \left(1 - \cos(\vec{k} \cdot (\vec{R}_j - \vec{R}_l)) \right) \right\} \\
 &= \exp \left\{ -\frac{1}{C} \frac{1}{N} \sum_{\vec{k}} \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \cdot \frac{1}{\omega_{\vec{k}}} \left(1 - \cos(\vec{k} \cdot (\vec{R}_j - \vec{R}_l)) \right) \right\} \\
 &=_{N \rightarrow \infty} \exp \left\{ -\frac{1}{C} a^d \int_{-\pi/a}^{\pi/a} \frac{d^d \vec{k}}{(2\pi)^d} \left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \cdot \frac{1}{\omega_{\vec{k}}} \left(1 - \cos(\vec{k} \cdot (\vec{R}_j - \vec{R}_l)) \right) \right\}
 \end{aligned}$$

6.1 Asymptotic Behaviour at large distance

At large distances, $|\vec{R}_j - \vec{R}_l| \gg a$. We can compute the integrals asymptotically, both in the case of $T = 0$ and $T > 0$. We do this separately.

$T > 0$

$$\begin{aligned}
 I &= \int_{-\pi/a}^{\pi/a} \frac{d^d \vec{k}}{(2\pi)^d} \underbrace{\left[n_B(\omega_{\vec{k}}) + \frac{1}{2} \right] \cdot \frac{1}{\omega_{\vec{k}}}}_{\frac{1}{2}(\vec{k} \cdot (\vec{R}_j - \vec{R}_l))^2} \left(1 - \cos(\vec{k} \cdot (\vec{R}_j - \vec{R}_l)) \right) \\
 &\sim \int_0^{k_c \sim \pi/a} d|\vec{k}| |\vec{k}|^{d-1} \cdot \frac{T}{v^2 |\vec{k}|^2} \frac{1}{2} |\vec{k}|^2
 \end{aligned}$$

which is regular for all d . We can in fact approximate the parenthesis in the integral simply by a constant because in the large distance limit, cosine is rapidly oscillating leading to cancellations inside the integral. Thus,

$$I \sim \int_{|\vec{R}_j - \vec{R}_l|^{-1}}^{\pi/a} d|\vec{k}| |\vec{k}|^{d-3} \sim \begin{cases} \text{const.} + \text{const.} \cdot T |\vec{R}_j - \vec{R}_l| & d = 1 \\ \text{const.} + \text{const.} \cdot T \ln |\vec{R}_j - \vec{R}_l| & d = 2 \end{cases}$$

so that,

$$C(\vec{R}_j, \vec{R}_l) \sim \begin{cases} e^{\overbrace{\text{const.} \cdot T |\vec{R}_j - \vec{R}_l|}^{<0}} & d = 1 \\ e^{\text{const.} \cdot T \ln |\vec{R}_j - \vec{R}_l|} \sim |\vec{R}_j - \vec{R}_l|^{-\eta(T)} & d = 2 \end{cases}$$

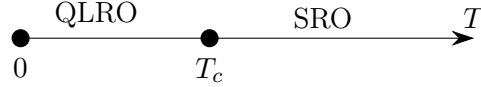
$T = 0$

$$\begin{aligned}
 I &\sim \int_{-\pi/a}^{\pi/a} d^d \vec{k} \frac{1}{2\omega_{\vec{k}}} \left(1 - \cos(\vec{k} \cdot (\vec{R}_j - \vec{R}_l)) \right) \\
 &\sim \int_{|\vec{R}_j - \vec{R}_l|^{-1}}^{\pi/a} d|\vec{k}| |\vec{k}|^{d-1} \frac{1}{|\vec{k}|} \sim \begin{cases} \text{const.} + \text{const.} \cdot \ln |\vec{R}_j - \vec{R}_l| & d = 1 \\ \text{const.} + \text{const.} \cdot |\vec{R}_j - \vec{R}_l|^{-1} & d = 2 \end{cases}
 \end{aligned}$$

Overall, the correlation functions read,

$$C(\vec{R}_j, \vec{R}_l) \sim \begin{cases} e^{-|\vec{R}_j - \vec{R}_l|/\xi(T)} & d = 1 \text{ and } T > 0 \\ |\vec{R}_j - \vec{R}_l|^{-\eta(T)} & d = 2 \text{ and } T > 0 \\ |\vec{R}_j - \vec{R}_l|^\delta & d = 1 \text{ and } T = 0 \end{cases}$$

The last two cases, having power law decay of the correlation function, signify the existence of a new kind of order dubbed Quasi-long range order (QSLO). We can sketch a phase diagram for those cases,



Note that both QLRO and short range order (SRO) have $\langle \theta \rangle = 0$, which means that the KT transition is not characterised by an order parameter field. One needs to go beyond Landau to fully study the transition.

6.2 Classical 2D XY model

We now investigate the KT phase of the 2D classical XY model, with Hamiltonian $H = -J \sum_{\langle ij \rangle} \cos(\theta_i - \theta_j)$. The partition function is given by,

$$Z = \int \prod_{i=1}^N d\theta_i e^{-\beta H}$$

We calculated before that,

$$\langle e^{i\theta_j} e^{-i\theta_l} \rangle \sim \begin{cases} |\vec{R}_j - \vec{R}_l|^{-\eta(T)} & \text{low } T \\ e^{-|\vec{R}_j - \vec{R}_l|/\xi(T)} & \text{high } T \end{cases}$$

We would like to find a continuum limit of the Hamiltonian of the 2D XY model. A way to do this is by first defining a spin variable that lives in the centre of the bond and then,

$$\begin{aligned} \cos(\theta_j - \theta_l) &= \cos[\theta(\vec{R}_j) - \theta(\vec{R}_l)] \\ &\approx 1 - \frac{1}{2} \left[\nabla \theta \left(\frac{\vec{R}_j + \vec{R}_l}{2} \right) \cdot (\vec{R}_j - \vec{R}_l) \right]^2 + \dots \\ &= \begin{cases} 1 - \frac{a^2}{2} (\partial_x \theta(\vec{R}))^2 \\ 1 - \frac{a^2}{2} (\partial_y \theta(\vec{R}))^2 \end{cases} \end{aligned}$$

depending on how we orientate the arbitrary $x - y$ -frame that we introduced. This allows us to take, $-J \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) \rightarrow \frac{J}{2} \int d^2 \vec{R} [\nabla \theta(\vec{R})]^2$ and therefore,

$$H \approx \frac{J}{2} \int d^2 \vec{R} [\nabla \theta(\vec{R})]^2$$

The action corresponding to the Hamiltonian is then,

$$\begin{aligned} S[\theta] &= \int_0^\beta d\tau \left[\frac{c}{2} \dot{\theta}(\tau)^2 - J \sum_{\langle j,l \rangle} \cos(\theta_j - \theta_l) \right] \\ &\rightarrow \int_0^\beta d\tau \left[\frac{c}{2} (\partial_\tau \theta(\tau))^2 - J (\partial_x \theta)^2 \right] \\ &\equiv \int_0^\beta (\nabla \theta(x, \tau))^2 \end{aligned}$$

Let us sketch how the XY model in 2 dimensions admits vortex configurations. First of all, a vortex configuration can be seen as a source field with positive divergence, which in a simple symmetric case is a vector field with the spins pointing in the outwards radial direction. The same configuration is equivalent to one which “winds” around the centre of the vortex in a clockwise fashion. One can see this by applying a transformation $\theta_j \rightarrow \theta_j + \frac{\pi}{2}$ for all j . In the continuum limit, we can consider the continuous function $\theta_{\text{vortex}}(\vec{R})$, from which we can define a **winding number**,

$$W = \oint_C \nabla \theta(\vec{R}) \cdot d\vec{l} = 2\pi n, n \in \mathbf{Z}$$

If two vortices have $n_1 = n_2$, they are topologically equivalent, in the sense that they can be smoothly connected to each other.

Example: Vortex with $n = 1$

We can model a vortex with $n = 1$ on the complex plane by writing $z = x + iy$. The vortex configuration is,

$$\theta(\vec{R}) = \text{Im} \ln z = \text{Im} [\ln \sqrt{x^2 + y^2} e^{i \tan^{-1}(y/x)}] = \tan^{-1}(y/x)$$

The gradient of the field is,

$$\nabla \theta(\vec{R}) = -\frac{y}{x^2 + y^2} \hat{x} + \frac{1}{x^2 + y^2} \hat{y}$$

A straightforward calculation of the line integral then gives, $n = 1$.

7 Lecture 7: KT Transition (I)

We continue in this lecture to study the KT transition of the 2D XY model. We start by considering fluctuations around vortex configurations, $\theta' = \theta_{\text{vortex}} + \delta\theta$. We first look at the saddle point solutions,

$$\begin{aligned}
 \delta H &= H(\theta + \delta\theta) - H(\theta) \\
 &= \frac{J}{2} \int d^2 \vec{R} [(\nabla\theta)^2 + 2(\nabla\theta \cdot \nabla\delta\theta) + (\nabla\delta\theta)^2 - (\nabla\theta)^2] \\
 &= J \int d^2 \vec{R} \nabla\theta \cdot \nabla\delta\theta \\
 &= J \int d^2 \vec{R} \underbrace{\partial_\mu \theta \partial_\mu \delta\theta}_{=\partial_\mu(\theta \partial_\mu \delta\theta) - (\partial_\mu^2 \theta) \delta\theta} \\
 &= -J \int d^2 \vec{R} \nabla^2 \theta(\vec{R}) \delta\theta(\vec{R}) = 0
 \end{aligned}$$

so that the field satisfies Laplace's equation in 2D,

$$\nabla^2 \theta(\vec{R}) = 0.$$

The vortex configuration given before, for $n = 1$, satisfies Laplace's equation. To check if vortices will proliferate, we must estimate their energy contribution. If we find that this contribution is too large, then we expect that the vortices will be exponentially suppressed.

Free energy of a single vortex

Let us calculate the free energy of a single vortex by using the configuration of $\theta_v = \text{Im} \ln z = \tan^{-1}(y/x)$. We first calculate the energy contribution which is the only one at $T = 0$.

$$E = \frac{J}{2} \int d^2 \vec{R} [(\nabla\theta_v)^2] = \frac{J}{2} \int d^2 \vec{R} \frac{1}{x^2 + y^2}$$

We must now introduce a lower cutoff (UV) a , provided by the smallest energy scale that the field theory can explain. This is naturally provided by the lattice spacing. Similarly, we introduce an upper (IR) cutoff, R_c . The integral then can be carried out to give,

$$E = \frac{J}{2} \int_a^{R_c} dR (2\pi R) \frac{1}{R^2} = \pi J \ln \left(\frac{R_c}{a} \right) \rightarrow \infty \text{ as } R_c \rightarrow \infty$$

The simple calculation seems to suggest that summing over all the vortex configurations will only add subleading terms to the calculation of the partition function (the Boltzmann weight $e^{-\beta H(\theta)}$ is subleading to the saddle point solution $e^{-\beta \theta_{\text{const.}}}$). But we should remember that this is just a calculation for a single vortex sitting at $r = 0$. The calculation is actually a hint that we should look at the entropic contribution of many vortices, which could in turn lower the free energy of the system.

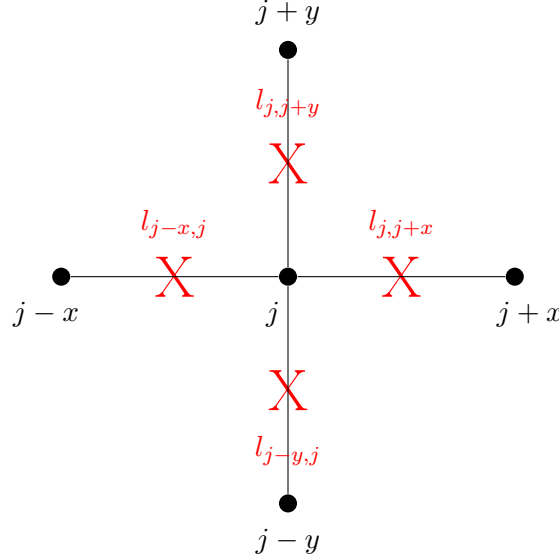
To add the entropic contribution, we use just a simple argument to count the number of such contributions. On a lattice with lattice spacing a , a vortex with radius R_c can be placed in the system by $\sim \left(\frac{R_c}{a}\right)^2$ ways. This means that the free energy is,

$$F_{\text{vortex}} = E_{\text{vortex}} - TS = (\pi J - 2T) \ln \left(\frac{R_c}{a} \right)$$

This provides a natural $T_c = \frac{\pi J}{2}$, such that for $T > T_c$, a single vortex is favoured due to its large entropy, and for $T < T_c$, a single vortex is not favoured because of its large energy. Note that this is simply an estimate and more elaborate methods give $T_c \approx 0.873J$.

7.1 Vortex analysis from the Lattice model

We go back to the lattice model and take a different approach to analyse the vortex configurations. To this end, we need to define new variables that live on a link that connects to lattice sites, so that if i is connected with j , with i on the left of j , the link variable is $l_{i,j}$.



The partition function is,

$$\begin{aligned}
 Z &= \int \prod_{i=1}^N d\theta_i e^{\beta J \sum_{\langle ij \rangle} \cos(\theta_i - \theta_j)} \\
 &= \int \prod_{i=1}^N d\theta_i \prod_{\langle ij \rangle} \underbrace{e^{\beta J \cos(\theta_i - \theta_j)}}_{\sum_{l_{ij}=-\infty}^{\infty} A_{l_{ij}} e^{il_{ij}(\theta_i - \theta_j)}} \\
 &\propto \sum_{\{l_{ij}\}=-\infty}^{\infty} \prod_{\langle ij \rangle} A_{l_{ij}} \left(\prod_j \delta_{\sum l_j=0} \right)
 \end{aligned}$$

where on the last step we have used the lattice version of Gauss' law,

$$\int_0^{2\pi} d\theta_j e^{i\theta_j (l_{j,j+x} + l_{j,j+y} - l_{j-x,j} - l_{j-y,j})} = 2\pi \delta_{l_{j,j+x} + l_{j,j+y} - l_{j-x,j} - l_{j-y,j}=0} \equiv 2\pi \delta_{\sum l_j=0}$$

Thus the partition function is equivalent to the product of the weights $A_{l_{ij}}$ of the link variables.

7.2 Villain's Approximation of the partition function

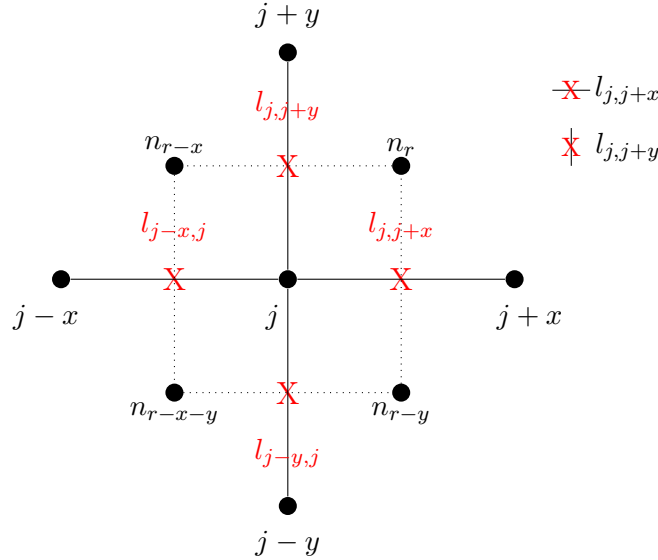
At low temperatures, it turns out that a good approximation to use is given by Villain, which states that,

$$A_{l_{ij}} \approx e^{-\frac{1}{2N} l_{ij}^2}$$

We will now rewrite this by defining new equivalent variables (in the sense that they satisfy Gauss' law),

$$\begin{aligned} l_{j,j+x} &= n_r - n_{r-y} \\ l_{j,j+y} &= n_{r-x} - n_r \end{aligned}$$

which can be understood via the diagram,



The new degrees of freedom, $n_r \in \mathbf{Z}$ and live on the dual lattice, at the centre of the plaquettes. With this, we can rewrite the partition function as,

$$\begin{aligned} Z &= \sum_{\{l_{ij}\}=-\infty}^{\infty} \left(\prod_{\langle ij \rangle} e^{-\frac{1}{2\beta J} l_{ij}^2} \right) \left(\prod_j \delta_{\sum l_j=0} \right) \\ &= \sum_{\{n_r\}=-\infty}^{\infty} \prod_{\langle r,r' \rangle} e^{-\frac{1}{2\beta J} (n_r - n_{r'})^2} \underbrace{(\dots)}_{\text{automatically fulfilled constraint!}} \\ &= \int_{-\infty}^{\infty} \left(\prod_r d\phi_r \right) \sum_{\{n_r\}=-\infty}^{\infty} \prod_r \delta(\phi_r - n_r) \prod_{\langle r,r' \rangle} e^{-\frac{1}{2\beta J} (\phi_r - \phi_{r'})^2} \end{aligned}$$

where on the last step we used the Poisson resummation formula. This is useful because we can now integrate the Gaussian integral of the ϕ_r ,

$$\begin{aligned} Z &= \int_{-\infty}^{\infty} \left(\prod_r d\phi_r \right) \sum_{\{m_r\}} \exp \left\{ -\frac{1}{2\beta J} \sum_{r,r'} \phi_r (G^{-1})_{r,r'} \phi_{r'} + 2\pi i \sum_r m_r \phi_r \right\} \\ &= \underbrace{\sqrt{\frac{(2\pi\beta J)^N}{\det G^{-1}}}}_{Z_{SW}} \underbrace{\sum_{\{m_r\}=-\infty}^{\infty} \exp \left\{ -2\pi^2 \beta J \sum_{r,r'} m_r G_{r,r'} m_{r'} \right\}}_{Z_{\text{vortex}}} \end{aligned}$$

and the Green's function is given by,

$$(G^{-1})_{r,r'} = 4\delta_{r,r'} - \sum_{\substack{\hat{e}=\pm a\hat{x} \\ \pm a\hat{y}}} \delta_{r,r+\hat{e}}$$

All in all, we see that the partition function nicely separates between a spin-wave contribution and a vortex contribution. The Green's function is an $N \times N$ matrix and to obtain a simpler form we can use the Fourier space decomposition in which it is diagonal. Once the Fourier space matrix is obtained, it is easy to invert since it is diagonal, and we can then we can Fourier transform back to real space. To see why the square root corresponds to the spin-wave part, we can carry out the integral for the fluctuations around an ordered phase with $\langle \theta \rangle = 0$ by writing, $\theta_j = \langle \theta \rangle + \delta\theta_j$, which fully decouples the interacting part.

8 Lecture 8: KT Transition (II)

We continue in this lecture with more considerations of the KT transition in the Villain formalism. We start by analysing the short and long range properties of the Green's function.

8.1 Asymptotic Behaviour

We will now find the asymptotic behaviour of the Green's function, by writing it in Fourier space,

$$G_{\vec{r}, \vec{r}'} = \frac{1}{N} \sum_{\vec{k}} \underbrace{\frac{1}{4 - 2 \cos(k_x a) - 2 \cos(k_y a)}}_{G_{\vec{k}, \vec{k}'}} e^{i\vec{k} \cdot (\vec{r}' - \vec{r})}$$

$$\rightarrow_{N \rightarrow \infty} a^2 \int \frac{d^2 k}{(2\pi)^2} \frac{1}{4 - 2 \cos(k_x a) - 2 \cos(k_y a)} e^{i\vec{k} \cdot (\vec{r}' - \vec{r})}.$$

At large distances, $|\vec{r} - \vec{r}'| \gg a$, the Green's function is determined by how $G_{\vec{k}, \vec{k}'}$ decays. We can expand around $(k_x, k_y) = (0, 0)$, where the spin wave has gapless excitations, and get $\sim \frac{1}{|\vec{k}|a^2}$ inside the integrand. This is the Fourier transform of the Coulomb potential in 2D. Therefore, the Green's function in real space is given by,

$$G_{\vec{r}, \vec{r}'} \sim -\ln \frac{|\vec{r} - \vec{r}'|}{a}$$

At short distances, we can take $\vec{r} = \vec{r}'$, so that the integrand needs regularization because of log-divergent terms if we consider very small k . Introducing a linear system size R_c , we get,

$$\int d^2 k \frac{1}{|\vec{k}|^2} = \int_{\pi/R_c}^{\pi/a} d|\vec{k}| \frac{1}{|\vec{k}|} = \ln |\vec{k}| \sim \ln \frac{R_c}{a}$$

a result which is divergent if the system size is infinite. The result is then,

$$G_{\vec{r}, \vec{r}'} = -\frac{1}{2\pi} \ln \frac{|\vec{r} - \vec{r}'|}{a} - \frac{1}{4} + G(0)$$

with the last term containing the divergent part, proportional to $\ln(R_c/a)$.

Considering now the low T limit of the partition function corresponding to the vortices, we split it into two contributions, $\vec{r} = \vec{r}'$ and $\vec{r} \neq \vec{r}'$.

$$Z_{\text{vortex}} = \sum_{\{m_r\}=-\infty}^{\infty} \exp \left\{ -2\pi^2 \beta J \sum_{r, r'} m_r G_{r, r'} m_{r'} \right\}$$

$$= \sum_{\{m_r\}=-\infty}^{\infty} \exp \left\{ -2\pi^2 \beta J G(0) \sum_r (m_r)^2 \right\} \exp \left\{ -2\pi^2 \beta J \sum_{r \neq r'} m_r G_{r, r'} m_{r'} \right\}$$

We now have a condition where the sum over the indices r is 0, so that it contributes substantially to the sum. This leads to the constraint,

$$\sum_r m_r = 0$$

to maximize the contribution to the partition function. This is called the “charge-neutral condition”. This leads to the simplification,

$$Z_{\text{vortex}} = \sum_{\substack{\{m_r\}=-\infty \\ \sum_r m_r=0}}^{\infty} \exp \left\{ \pi \beta J \sum_{r \neq r'} m_r \frac{\ln |\vec{r} - \vec{r}'|}{a} m_{r'} \right\}$$

valid in the low T limit. This shows that long-range order in fact causes confinement!

We can interpret this as a Coulomb gas. Vortices can be viewed as the electric charges in 2 dimensions, $m_r \leftrightarrow e$. In the low- T limit, we can only have vortex-antivortex pairs, which we can identify as dipoles. In the high- T limit, charge neutrality is no longer enforced, and we get proliferation of vortices, in the same way as a Coulomb gas has free charges. In fact, this analogy applies to a much larger class of models, not just the XY or modified Villain. In those cases, only the long-range form of the potential will change.

8.2 Alternative Derivation

Alternative to the derivation above, we can consider the starting Hamiltonian of the XY model in terms of the phase field θ , $H = \frac{J}{2} \int d^2 R [\nabla \theta(\vec{R})]^2$, and take a multivortex configuration,

$$\theta_{\text{vortex}}(\vec{R}) = \sum_{i=1}^n \text{Im} \ln(z - z_{\vec{r}_i}) m_{\vec{r}_i}$$

corresponding to n vortices with winding numbers $m_{\vec{r}_i}$. We can then calculate the energy of the configuration,

$$\begin{aligned} E_{\text{vortex}} &= \frac{J}{2} \int d^2 \vec{R} [\nabla \theta_{\text{vortex}}(\vec{R})]^2 \\ &= \frac{J}{2} \int d^2 \vec{R} \sum_{i,j} m_{\vec{r}_i} m_{\vec{r}_j} [\nabla \text{Im} \ln(z - z_{\vec{r}_i}) \cdot \nabla \text{Im} \ln(z - z_{\vec{r}_j})] \\ &= \frac{J}{2} \sum_i m_{\vec{r}_i}^2 \underbrace{\int d^2 \vec{R} [\nabla \text{Im} \ln(z - z_{\vec{r}_i})]^2}_{=2\pi \ln\left(\frac{R_c}{a}\right)} + \frac{J}{2} \sum_{i \neq j} m_{\vec{r}_i} m_{\vec{r}_j} \underbrace{\int d^2 \vec{R} [\nabla \text{Im} \ln(z - z_{\vec{r}_i}) \cdot \nabla \text{Im} \ln(z - z_{\vec{r}_j})]}_{=-4\pi \ln \frac{|\vec{r} - \vec{r}'|}{a}} \end{aligned}$$

with the first term corresponding to the energy of a single vortex, and the second describes the interaction and is an integral that can be calculated exactly.

As we saw before, the partition function is a product of the spin-wave and the vortex contributions, so that the free energy is a sum of the two. The point now is that the n -vortex configurations are the saddle point solutions that minimize the free energy. We can expand around any such saddle point configuration,

$$\theta(\vec{R}) = \theta_{\text{vortex}}(\vec{R}) + \delta\theta(\vec{R})$$

with $\delta\theta(\vec{R})$ being a trivial field configuration, without a winding number. The degrees of freedom then separate as follows,

$$H = \frac{J}{2} \int d^2 \vec{R} [(\nabla \theta_{\text{vortex}}(\vec{R}))^2 + (\nabla \delta\theta(\vec{R}))^2] + \int d^2 \vec{R} [2(\nabla \theta_{\text{vortex}}(\vec{R})) \cdot (\nabla \delta\theta(\vec{R}))]$$

The crossing term will calculate to 0 exactly because the vortex field has non-trivial winding numbers at certain positions while the fluctuation field is trivial so that $\nabla\delta\theta = 0$. The second term in the first integral is the spin-wave contribution. Recall from the previous calculation that,

$$Z_{\text{vortex}} = \sum_{\{m_r\}=-\infty}^{\infty} \exp\left\{-\frac{\pi^2}{2}\beta J \sum_r (m_r)^2\right\} \exp\left\{\pi\beta J \sum_{r \neq r'} m_r \frac{\ln|\vec{r}-\vec{r}'|}{a} m_{r'}\right\}$$

and we let $\ln y = -\pi^2/2\beta J$, with y the fugacity of the vortices. The fugacity has the property of being small at small T . We now see that the most important contributions are those with $m_{\vec{r}} = 0, \pm 1$, however we would still like to sum over the ones which do not contribute that much, with $|m_{\vec{r}}| > 1$. We will now argue that this is formally possible. Our goal is to rewrite the partition function in this form,

$$Z_{\text{vortex}} = \sum_{\{m_r\}=0,\pm 1} \exp\left\{\ln y' \sum_r (m_r)^2\right\} \exp\left\{\pi\beta J \sum_{r \neq r'} m_r \frac{\ln|\vec{r}-\vec{r}'|}{a} m_{r'}\right\}$$

Going back to the start, before integrating out the ϕ -SW fields,

$$\begin{aligned} Z_{\text{vortex}} &= \int \prod_r d\phi_{\vec{r}} \sum_{\{m_{\vec{r}}\}} e^{-\frac{1}{2\beta J} \sum_{\langle r, r' \rangle} (\phi_{\vec{r}} - \phi_{\vec{r}'})^2 + 2\pi i \sum_{\vec{r}} m_{\vec{r}} \phi_{\vec{r}}} \\ &= \int \prod_r d\phi_{\vec{r}} \sum_{m_{\vec{r}}=0,\pm 1} e^{-\frac{1}{2\beta J} \sum_{\langle r, r' \rangle} (\phi_{\vec{r}} - \phi_{\vec{r}'})^2 + 2\pi i \sum_{\vec{r}} m_{\vec{r}} \phi_{\vec{r}} + \ln y \sum_{\vec{r}} m_{\vec{r}}^2} \end{aligned}$$

where on the second step we have used the fact obtained above that the most important degrees of freedom are the ones for $m_r = 0, \pm 1$. The sum over m_r can now be done exactly,

$$\begin{aligned} \sum_{m_{\vec{r}}=0,\pm 1} e^{2\pi i \sum_{\vec{r}} m_{\vec{r}} \phi_{\vec{r}} + \ln y \sum_{\vec{r}} m_{\vec{r}}^2} &= \sum_{m_{\vec{r}}=0,\pm 1} \prod_{\vec{r}} e^{2\pi i m_{\vec{r}} \phi_{\vec{r}} + \ln y m_{\vec{r}}^2} \\ &= \prod_{\vec{r}} \left[1 + e^{\ln y} e^{2\pi i \phi_{\vec{r}}} e^{\ln y} e^{-2\pi i \phi_{\vec{r}}}\right] \\ &= \prod_{\vec{r}} [1 + 2y \cos(2\pi \phi_{\vec{r}})] \\ &\approx e^{\sum_{\vec{r}} 2y \cos(2\pi \phi_{\vec{r}})} \end{aligned}$$

Therefore,

$$Z_{\text{vortex}} = \int \prod_r d\phi_{\vec{r}} e^{-\frac{1}{2\beta J} \sum_{\langle r, r' \rangle} (\phi_{\vec{r}} - \phi_{\vec{r}'})^2 + \sum_{\vec{r}} 2y \cos(2\pi \phi_{\vec{r}})}$$

which is the field-theory that describes spin-waves, vortices and interactions in the XY-model. In the continuum limit, $a \rightarrow 0$, $\sum_{\langle r, r' \rangle} (\phi_{\vec{r}} - \phi_{\vec{r}'})^2 \approx a^2 \sum_{\vec{r}} (\partial_x \phi)^2 + (\partial_y \phi)^2 \approx a^2 \int d^2 \vec{r} (\nabla \phi)^2$. We note here that ϕ is not a compactified field and followed by performing a Poisson summation. The interaction term is, $\sum_{\vec{r}} \cos(2\pi \phi_{\vec{r}}) \approx \frac{1}{a^2} \int d^2 \vec{r} \cos(2\pi \phi_{\vec{r}})$. As $a \rightarrow 0$,

$$Z_{\text{vortex}} = \int \mathcal{D}\phi(\vec{r}) e^{-\int d^2 \vec{r} \left\{ \frac{1}{2\beta J} (\nabla \phi)^2 - 2y \cos(2\pi \phi) \right\}}$$

We also write, $\phi' = \frac{1}{\beta J} \phi$ and the partition function takes the form, $Z_{vortex} = \int \mathcal{D}\phi' e^{-S[\phi']}$, with the action being given by,

$$S[\phi] = \int d^2\vec{r} \left[\frac{1}{2} (\nabla \phi)^2 - \frac{2y}{a^2} \cos(2\pi\sqrt{\beta J} \phi) \right]$$

We identify this action as the Sine-Gordon field theory in 1 + 1D. For $y = 0$, this is a bosonic and conformal field theory in 2D.

The action has minima at $2\pi\sqrt{\beta J}\phi' = 2\pi\lambda$, for $\lambda \in \mathbb{Z}$. We can interpret this in the following way. At the minima of the cosine potential, the excitation is particle-like and localised at the minima (for $T > T_c$). When the excitation is wave-like ($T < T_c$), they are non-localised. The wavefunction extends to large regions and in essence, the cosine term can be ignored, we are dealing with a free-theory. At $T = T_c$, the field theory describes the KT-transition. The shallowness of the potential, which controls also how localised the excitations are, depends only on $\sqrt{\beta J}$. Thus, using the known result of Sine-Gordon theory,

$$2\pi\sqrt{\beta_c J} = \sqrt{8\pi} \iff \beta_c = \frac{2}{\pi J}$$

which is the same estimation got by using a single-vortex analysis! But this analysis ignores microscopic details, field theory only tells us universal characteristics.

8.3 RG Analysis of the KT-transition

We conclude the KT-transition analysis by some RG arguments. Recall that for a partition function of the form,

$$Z = \int \mathcal{D}\phi \exp(-[S_0 + S_1])$$

we can take S_0 as the action describing some critical theory and S_1 as a perturbation. The question we want to answer is whether or not the perturbation is relevant or irrelevant, and answer this by a ‘Poor’s man scaling’ approach. We call d the spatial dimension, and $S_1 = g \int d^d r A(\vec{r})$. We are after the correlation function,

$$\langle A(\vec{r}) A(\vec{r}') \rangle_0 = \frac{1}{Z_0} \int \mathcal{D}\phi A(\vec{r}) A(\vec{r}') e^{-S_0} \approx \frac{1}{|\vec{r} - \vec{r}'|^{2h}}$$

with h the scaling dimension.

1. For $h > d$, the perturbation is RG irrelevant.
2. For $h < d$, the perturbation is RG relevant.
3. For $h = d$, the perturbation is RG marginal.

In the Sine-Gordon theory in $d = 2$, $A(\vec{r}) = \cos(2\pi\sqrt{\beta J}\phi(\vec{r}))$, so that,

$$\langle A(\vec{r}) A(\vec{r}') \rangle \sim \langle e^{2\pi i \sqrt{\beta J} \phi(\vec{r})} e^{-2\pi i \sqrt{\beta J} \phi(\vec{r}')} \rangle$$

In $d = 2$ we can show that, $\langle e^{i\alpha\phi(\vec{r})} e^{-i\alpha\phi(\vec{r}')} \rangle = \frac{1}{|\vec{r} - \vec{r}'|} \alpha^{2/2\pi}$, so that $h = \alpha^2/4\pi$. In the Sine-Gordon theory, $h = \pi\beta J$, and thus, the KT transition at J_c when the perturbation is marginal. This gives again, $h_c = \pi\beta_c J = 2$, so that $T_c = \frac{\pi}{2} J$, coinciding with the single vortex result.

The free energy is then, $f \sim e^{-\text{const.} \sqrt{T - T_c}}$, hinting an infinite order transition. For a KT-type transition, there is no discontinuity or divergence in the observables.

For the correlation function in the θ -field,

$$\langle e^{i\theta_j} e^{-i\theta_j} \rangle \sim \frac{1}{|\vec{R}_i - \vec{R}_j|}^{1/4} \left[\ln \frac{|\vec{R}_i - \vec{R}_j|}{a} \right]^{1/8}$$

with the log-correction coming from the marginality of the cosine term. This shows that it is numerically very hard to distinguish between exponential and power law decay of correlation functions.

9 Lecture 9: Bosonic Systems

9.1 Functional Integral for Bosonic Systems

We now change gears and turn our attention to bosonic systems, with Hamiltonians of the form,

$$\hat{H} = \sum_{j,l}^N t_{jl} \hat{a}_j^\dagger \hat{a}_l + \frac{1}{2} \sum_{jlmn} V_{jlmn} \hat{a}_j^\dagger \hat{a}_l^\dagger \hat{a}_m \hat{a}_n$$

We want to develop a path-integral formulation of the partition function, which we will try to Trotterize as before,

$$Z = \text{Tr}[e^{-\beta \hat{H}}] = \text{Tr}[e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}}]$$

The operators are bosonic and satisfy $[\hat{a}, \hat{a}^\dagger] = 1$. With the vacuum state, $|0\rangle$, we can build excited states via $|n\rangle = \frac{1}{\sqrt{n!}} (\hat{a}^\dagger)^n |0\rangle$. The bosonic coherent states are given by,

$$|z\rangle = e^{za^\dagger} |0\rangle = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n!}} z^n |n\rangle$$

with $z \in \mathbb{C}$. These are defined via $a|z\rangle = z|z\rangle$, easy to verify with the above definition. Additionally, it is also true that $\langle z|$ is the left eigenvector of $a^\dagger$ with eigenvalue z^* , $\langle z| a^\dagger = \langle z| z^*$. This implies the overlap between two coherent states is,

$$\langle z|z'\rangle = e^{z^*z'}$$

The BCH formula, $e^{\hat{X}} e^{\hat{Y}} = e^{\hat{Y}} e^{\hat{X}} e^{[\hat{X}, \hat{Y}]}$, can be used to obtain the expectation value of a Hamiltonian function (assumed to be normal ordered) $\hat{H}(\hat{a}^\dagger, \hat{a})$,

$$\langle z| \hat{H}(\hat{a}^\dagger, \hat{a}) |z'\rangle = \langle z|z'\rangle H(z^*, z') = e^{z^*z'} H(z^*, z')$$

Using a polar coordinate representation of a complex number, $z = x + iy$, we may prove the overcompleteness relation for the coherent states of bosons as follows,

$$\int \frac{dz dz^*}{2\pi i} e^{-z^*z} |z\rangle \langle z| = \mathbb{I}$$

Similarly, the trace of an operator $\hat{A}$, is calculated by,

$$\begin{aligned} \text{Tr}[\hat{A}] &= \sum_{n=0}^{\infty} \langle n| \hat{A} |n\rangle = \sum_{n=0}^{\infty} \int d^2z e^{-z^*z} \langle n|z\rangle \langle z| \hat{A} |n\rangle \\ &= \int d^2z e^{-z^*z} \langle z|z\rangle \hat{A} \sum_{n=0}^{\infty} |n\rangle \langle n|z\rangle \\ &= \int d^2z e^{-z^*z} \langle z| \hat{A} |z\rangle \end{aligned}$$

Equipped with the basic facts for the coherent states stated above, we are in position to formulate the bosonic path integral. We will assume that $\hat{H}(\hat{a}^\dagger, \hat{a})$ to be normal ordered. The partition function is,

$$\begin{aligned} Z &= \text{Tr}[e^{-\beta \hat{H}}] = \int d^2z_0 \langle z_0| e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}} |z_0\rangle e^{-z_0^* z_0} \\ &= \int d^2z_0 \langle z_0| e^{-\Delta\tau \hat{H}} \dots \int d^2z_1 e^{-z_1^* z_1} |z_1\rangle \langle z_1| e^{-\Delta\tau \hat{H}} |z_0\rangle e^{-z_0^* z_0} \end{aligned}$$

We therefore need to compute $\langle z_{k+1} | e^{-\Delta\tau \hat{H}} | z_k \rangle$. This is computed as in all the previous cases we have encountered during trotterization,

$$\begin{aligned} \langle z_{k+1} | e^{-\Delta\tau \hat{H}} | z_k \rangle &= \langle z_{k+1} | (\mathbb{I} - \Delta\tau \hat{H}) | z_k \rangle \\ &= \langle z_{k+1} | z_k \rangle (1 - \Delta\tau H(z_{k+1}^*, z_k)) \\ &= e^{z_{k+1}^* z_k} e^{-\Delta\tau H(z_{k+1}^*, z_k)} \end{aligned}$$

The partition function is then,

$$Z = \int \prod_{k=0}^{N-1} d^2 z_k e^{-\sum_{k=0}^{N-1} z_k^* z_k} e^{\sum_{k=0}^{N-1} z_{k+1}^* z_k - \Delta\tau \sum_{k=0}^{N-1} H(z_{k+1}^*, z_k)}$$

where the first exponential is the identity resolution, and the second term is the matrix element computed above. In the limit $\Delta\tau \rightarrow 0$, $N \rightarrow \infty$ with $N\Delta\tau = \beta$ fixed, and PBC ($z_N = z_0$), the continuum limit of the mixed term is,

$$\begin{aligned} \sum_{k=0}^{N-1} z_{k+1}^* (z_{k+1} - z_k) &\rightarrow \int_0^\beta d\tau z^* \partial_\tau z + \mathcal{O}(\partial_\tau z^2) \\ \sum_{k=0}^{N-1} H(z_{k+1}^*, z_k) &\rightarrow \int_0^\beta d\tau (H(z^*, z) + \mathcal{O}(\Delta\tau)) \end{aligned}$$

Altogether, the bosonic partition function is given by,

$$\begin{aligned} Z &= \int_{\substack{z(0)=z(\beta) \\ z^*(0)=z^*(\beta)}} \mathcal{D}z^* \mathcal{D}z e^{-S[z^*, z]} \\ S &= \int_0^\beta d\tau [z^*(\tau) \partial_\tau z(\tau) + H(z^*, z)] \end{aligned}$$

The result is easily generalisable to many-bosons. We take operators which are site-indexed, $\hat{a}_j^\dagger, \hat{a}_j$, for $j = 1, \dots, N$. The indexed list $\{z_1, \dots, z_N\}$ is used to define the many-body coherent state via,

$$|\{z\}\rangle = e^{\sum_{j=1}^N z_j \hat{a}_j^\dagger} |0\rangle = \prod_{j=1}^N e^{z_j \hat{a}_j^\dagger} |0\rangle$$

and satisfies $\hat{a}_j |\{z\}\rangle = z_j |\{z\}\rangle$. A general interacting Hamiltonian, $\hat{H} = \frac{1}{2} \sum_{jl} t_{jl} \hat{a}_j^\dagger \hat{a}_l + H_I$, then has a partition function,

$$\begin{aligned} Z &= \int_{\substack{z_j(0)=z_j(\beta) \\ z_j^*(0)=z_j^*(\beta)}} \mathcal{D}z_j^* \mathcal{D}z_j e^{-S_0[z^*, z] + S_I} \\ S_0 &= \int_0^\beta d\tau \left[\sum_{j=1}^N z_j^*(\tau) \partial_\tau z_j(\tau) + H_0(z_j^*, z_j) \right] \\ S_I &= \int_0^\beta d\tau H_I(z^*, z) \end{aligned}$$

A possible way forward is to perform a perturbation expansion in S_I ,

$$Z = \int \mathcal{D}z^* \mathcal{D}z \left\{ 1 - S_I + \frac{1}{2} S_I^2 + \dots \right\} e^{-S_0[z^*, z]}$$

which is suitable for diagrammatic calculations.

10 Lecture 10: BEC and Superfluidity (I)

To study Bose condensation and superfluidity, we use the continuum version of the interacting Bose Hamiltonian,

$$H = \int d^d \vec{r} \left[\psi^\dagger(\vec{r}) \left(-\frac{\nabla^2}{2m} - \mu \right) \psi(\vec{r}) \right] + \int d^d \vec{r} d^d \vec{r}' \psi^\dagger(\vec{r}) \psi^\dagger(\vec{r}') V(\vec{r} - \vec{r}') \psi(\vec{r}') \psi(\vec{r})$$

where the orbitals $\psi(\vec{r})$ are related to the coherent states via $|\{z\}\rangle \rightarrow e^{\int d^d \vec{r} z(\vec{r}) \psi^\dagger(\vec{r})} |0\rangle$. We will consider short range repulsive potentials, and in particular a contact/pseudopotential,

$$V(\vec{r} - \vec{r}') = g \delta^d(\vec{r} - \vec{r}')$$

with $g \geq 0$. We want to consider weakly interacting bosons here, for simplicity. For strong interactions, the field theoretical treatment we will employ here will fail. But what is a “weak interaction”? For fermions there is a natural notion of “weakly” provided by the energy scale of interaction strengths $V_{\text{int}} \leq E_F$. Nevertheless, Bogoliubov’s theory will provide a notion of weak interactions.

The partition function is given by,

$$Z = \text{Tr}[e^{-\beta \hat{H}}] = \int \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-S[\bar{\psi}, \psi]}$$

$$S[\bar{\psi}, \psi] = \int_0^\beta d\tau \left[\int d^d \vec{r} \left\{ \bar{\psi} \partial_\tau \psi + \bar{\psi} \left(\frac{\nabla^2}{2m} - \mu \right) \psi \right\} + \int d^d \vec{r} \frac{g}{2} \underbrace{\bar{\psi} \psi \psi \psi}_{(\bar{\psi} \psi)^2} \right]$$

where the fact that the Hamiltonian was already normal ordered was used. The fields ψ are now complex valued and not operator valued.

10.1 Non-Interacting Limit: BEC

The simplest case is $g = 0$, which will serve as the starting point to study BEC. We will still need to understand the stability of the phase under interactions and its relation to superfluidity. The action is now,

$$S_0[\bar{\psi}, \psi] = \int_0^\beta d\tau \int d^d \vec{r} \left\{ \bar{\psi} \partial_\tau \psi + \bar{\psi} \left(\frac{\nabla^2}{2m} - \mu \right) \psi \right\}$$

conveniently quadratic, and therefore we can use the Fourier transformed fields to diagonalise the action. To this end we consider bosons in a box with PBC, so that their Fourier decomposition is,

$$\psi(\vec{r}, \tau) = \frac{1}{\sqrt{\beta V}} \sum_{\vec{k}, i\omega_n} a(\vec{k}, i\omega_n) e^{i\vec{k} \cdot \vec{r} - i\omega_n \tau}$$

and the action becomes,

$$S_0[\bar{\psi}, \psi] = \sum_{\vec{k}, i\omega_n} \left(-i\omega_n + \frac{|\vec{k}|^2}{2m} - \mu \right) \bar{a}(\vec{k}, i\omega_n) a(\vec{k}, i\omega_n)$$

For completeness we note that the Hamiltonian in Fourier space is $\hat{H} = \sum_{\vec{k}} \left(\frac{|\vec{k}|^2}{2m} - \mu \right) a_{\vec{k}}^\dagger a_{\vec{k}}$.

The particle number operator is $\hat{N} = \int d^d \vec{r} \psi^\dagger(\vec{r}) \psi(\vec{r})$, and we want to find the thermal average $\langle N \rangle_T = \int d^d \vec{r} \langle \psi^\dagger(\vec{r}) \psi(\vec{r}) \rangle_T$. The thermal average of the particle number is used to determine the value of the chemical potential μ . To this end, we define the thermal Green's function,

$$\begin{aligned} G(\vec{r}, \tau; \vec{r}', \tau') &\equiv -\langle T_\tau \hat{\psi}(\vec{r}, \tau) \hat{\psi}^\dagger(\vec{r}', \tau') \rangle_T \\ &= \begin{cases} -\langle \hat{\psi}(\vec{r}, \tau) \hat{\psi}^\dagger(\vec{r}', \tau') \rangle_T & \text{if } \tau > \tau' \\ -\langle \hat{\psi}^\dagger(\vec{r}', \tau') \hat{\psi}(\vec{r}, \tau) \rangle_T & \text{if } \tau' > \tau \end{cases} \end{aligned}$$

where T_τ is the time-ordering operator. We have taken the operators in the Heisenberg picture, explicitly, $\hat{A}(\tau) = e^{\hat{H}\tau} \hat{A}(0) e^{-\hat{H}\tau}$.

Let us now derive the Green's function for the non-interacting bosons and show that it is in fact a path integral. We take, without loss of generality, $\tau > \tau'$. We have,

$$\begin{aligned} -\langle \hat{\psi}(\vec{r}, \tau) \hat{\psi}^\dagger(\vec{r}', \tau') \rangle_T &= -\frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} \hat{\psi}(\vec{r}, \tau) \hat{\psi}^\dagger(\vec{r}', \tau') \right] \\ &= -\frac{1}{Z} \text{Tr} \left[e^{-\beta \hat{H}} e^{\hat{H}\tau} \hat{\psi}(\vec{r}) e^{-\hat{H}\tau} e^{\hat{H}\tau'} \hat{\psi}^\dagger(\vec{r}') e^{-\hat{H}\tau'} \right] \\ &= -\frac{1}{Z} \text{Tr} \left[e^{-(\beta-\tau)\hat{H}} \hat{\psi}(\vec{r}) e^{-(\tau-\tau')\hat{H}} \hat{\psi}^\dagger(\vec{r}') e^{-\hat{H}\tau'} \right] \end{aligned}$$

At this point we need to insert a resolution of identity using the bosonic coherent states,

$$\int d\bar{\psi}_k d\psi_k e^{-\bar{\psi}_k \psi_k} |\{\psi_k\}\rangle \langle \{\psi_k\}|$$

inserted at a particular time labeled by k . Inserting these into the trace above, the time evolution operator will act on them giving, and then the creation operator will hit them, producing a complex number,

$$|\{\psi_k\}\rangle \langle \{\psi_k\}| \hat{\psi}^\dagger(\vec{r}) = |\{\psi_k\}\rangle \langle \{\psi_k\}| \bar{\psi}_k(\vec{r})$$

This procedure continues, time evolving the state, until the annihilation operator hits the set of coherent states, now at k' , via,

$$\hat{\psi}(\vec{r}) |\{\psi_{k'}\}\rangle \langle \{\psi_{k'}\}| = \psi_{k'}(\vec{r}) |\{\psi_{k'}\}\rangle \langle \{\psi_{k'}\}|$$

The time ordering guarantees that we need not worry about the ordering of the fields at the different times k and k' . Therefore, we only need to find two suitable timesteps to perform the trace. Altogether, the thermal Green's function in the path-integral formalism becomes,

$$G(\vec{r}, \tau; \vec{r}', \tau') = -\frac{1}{Z} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \psi(\vec{r}, \tau) \bar{\psi}(\vec{r}', \tau') e^{-S_0[\bar{\psi}, \psi]}$$

10.2 Thermal Green's function for $g = 0$

We calculate now the thermal Green's function by expressing the fields in terms of the Fourier decompositions,

$$\begin{aligned} G(\vec{r}, \tau; \vec{r}', \tau') &= -\frac{1}{Z} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \psi(\vec{r}, \tau) \bar{\psi}(\vec{r}', \tau') e^{-S_0[\bar{\psi}, \psi]} \\ &= \frac{(-1)}{\beta V} \sum_{\vec{k}, i\omega_n} \sum_{\vec{k}', i\omega_{n'}} \int \mathcal{D}\bar{a} \mathcal{D}a a(\vec{k}, i\omega_n) \bar{a}(\vec{k}', i\omega_{n'}) e^{i\vec{k} \cdot \vec{r} - i\omega_n \tau} e^{-i\vec{k}' \cdot \vec{r}' + i\omega_{n'} \tau'} e^{-S_0[\bar{a}, a]} \end{aligned}$$

To perform the Gaussian integrals, we now decompose the complex variable a and $\bar{a}$ in terms of two real ones,

$$a = a' + ia'', \quad \bar{a} = a' - ia''$$

The action becomes,

$$S_0[\bar{a}, a] = \sum_{\vec{k}, i\omega_n} \left(-i\omega_n + \frac{|\vec{k}|^2}{2m} - \mu \right) \left\{ a_{\vec{k}, i\omega_n}'^2 + a_{\vec{k}, i\omega_n}''^2 \right\}$$

If the two momenta $\vec{k} \neq \vec{k}'$ then the integral over the a variables will be zero, simply because the integrand is odd (compare with $\int_{\mathbb{R}^2} dx x e^{-\alpha x^2} = 0$). Thus, the Green's function becomes,

$$\begin{aligned} G(\vec{r}, \tau; \vec{r}', \tau') &= G(\vec{r} - \vec{r}', \tau - \tau') \\ &= \frac{(-1)}{\beta V} \sum_{\substack{\vec{k}, i\omega_n \\ \vec{k}', i\omega_{n'}}} e^{i\vec{k} \cdot \vec{r} - i\omega_n \tau} e^{-i\vec{k}' \cdot \vec{r}' + i\omega_{n'} \tau'} \cdot \int \mathcal{D}a' \mathcal{D}a'' \left(a_{\vec{k}, i\omega_n}'^2 + a_{\vec{k}, i\omega_n}''^2 \right) \delta_{\vec{k}, \vec{k}'} \delta_{\omega_n, \omega_{n'}} e^{-S_0[\bar{a}, a]} \\ &= \frac{1}{\beta V} \sum_{\vec{k}, i\omega_n} \frac{1}{\underbrace{i\omega_n - \frac{|\vec{k}|^2}{2m} + \mu}_{G(\vec{k}, i\omega_n)}} e^{i\vec{k} \cdot (\vec{r} - \vec{r}') - i\omega_n (\tau - \tau')} \end{aligned}$$

To calculate the particle number, we may choose any time τ and perform the integral,

$$\begin{aligned} N &= \int d^d \vec{r} \langle \hat{\psi}^\dagger(\vec{r}) \psi(\vec{r}) \rangle_T \\ &= \int d^d \vec{r} \langle T_\tau \psi(\vec{r}, \tau) \hat{\psi}^\dagger(\vec{r}, \tau + 0^+) \rangle_T \end{aligned}$$

where we have added an infinitesimally small and positive time to make $\hat{\psi}^\dagger$ a little bit later in time. This will ensure convergence of the sum over the frequencies. Thus,

$$\begin{aligned} N &= - \int d^d \vec{r} G(\vec{r}, \tau; \vec{r}, \tau + 0^+) \\ &= - \underbrace{\int d^d \vec{r}}_V \frac{1}{\beta V} \sum_{\vec{k}, i\omega_n} \frac{1}{i\omega_n - \frac{|\vec{k}|^2}{2m} + \mu} e^{-i\omega_n (\tau - \tau + 0^+)} \\ &= - \frac{1}{\beta} \sum_{\vec{k}, i\omega_n} \frac{1}{i\omega_n - \frac{|\vec{k}|^2}{2m} + \mu} e^{-i\omega_n 0^+} \end{aligned}$$

Here ω_n are bosonic Matsubara frequencies because of the PBC. The sum over the frequencies can be performed by using the function $n_B(z) = \frac{1}{e^z - 1}$ and the residue theorem on the complex plane. This results from,

$$\begin{aligned} 0 &= \oint_C \frac{dz}{2\pi i} n_B(z) \frac{1}{z - \frac{|\vec{k}|^2}{2m} + \mu} e^{-z 0^+} \\ &= \frac{1}{\beta} \sum_{i\omega_n} \frac{1}{i\omega_n - \frac{|\vec{k}|^2}{2m} + \mu} e^{-i\omega_n 0^+} + \underbrace{\frac{1}{e^{\beta \left(\frac{|\vec{k}|^2}{2m} - \mu \right)}}}_{n_B\left(\frac{|\vec{k}|^2}{2m} - \mu\right)} \end{aligned}$$

Therefore,

$$N = \sum_{\vec{k}} n_B \left(\frac{|\vec{k}|^2}{2m} - \mu \right)$$

The last equation can be used to determine the BEC critical temperature.

10.3 Critical Temperature of BEC

We consider now bosons in $d = 3$. First we rewrite the equation for the mean particle number in terms of the density of states, $\rho(\epsilon)$,

$$\rho(\epsilon) = \frac{1}{V} \sum_{\vec{k}} \delta \left(\epsilon - \frac{|\vec{k}|^2}{2m} \right) = \int \frac{d^3 \vec{k}}{(2\pi)^3} \delta \left(\epsilon - \frac{|\vec{k}|^2}{2m} \right) \propto \sqrt{\epsilon}$$

Thus,

$$\begin{aligned} N &= \sum_{\vec{k}} n_B \left(\frac{|\vec{k}|^2}{2m} - \mu \right) = \int_{-\infty}^{\infty} \sum_{\vec{k}} \delta \left(\epsilon - \frac{|\vec{k}|^2}{2m} \right) \frac{1}{e^{\beta(\epsilon-\mu)} - 1} \\ &= \int_0^{\infty} d\epsilon \frac{V \rho(\epsilon)}{e^{\beta(\epsilon-\mu)} - 1} \\ &= \underbrace{N_0(T)}_{\vec{k}=0} + \sum_{\vec{k} \neq 0} n_B \left(\frac{|\vec{k}|^2}{2m} - \mu \right) \end{aligned}$$

where we split the contribution from the condensate state which will occur at $\vec{k} = 0$. This had to be done because its occupancy is macroscopic. Solving for $N_0(T)$ then,

$$N_0(T) = N \left[1 - \left(\frac{T}{T_{BEC}} \right)^{3/2} \right]$$

for $T_{BEC} \approx 2.31 \left(\frac{N}{V} \right)^{2/3} \frac{\hbar^2}{m}$, and the ratio N/V is the density of bosons in free space. This shows that, for $T \leq T_{BEC}$, $\mu = 0$, and in the $T \rightarrow 0$ limit, the ground state is the condensate,

$$|GS\rangle \propto \left(a_{\vec{k}=0}^\dagger \right)^N |0\rangle$$

10.4 Comment on $d = 1$ and $d = 2$

In $d = 1$ and $d = 2$ the DOS is given by,

$$\begin{aligned} \rho_{d=1}(\epsilon) &\propto \frac{1}{\sqrt{\epsilon}} \\ \rho_{d=2}(\epsilon) &\propto \text{const.} \end{aligned}$$

According to the equation that determines μ , $N/V = \int d\epsilon \frac{\rho(\epsilon)}{e^{\beta(\epsilon-\mu)} - 1}$, we see that the integral diverges in both cases when $\mu = 0$. This means in practice that μ cannot reach 0 for any $T > 0$, implying that there can be no BEC in $d = 1$ or $d = 2$ for the *ideal Bose gas*.

11 Lecture 11: BEC and Superfluidity (II)

We continue our considerations of the BEC and superfluidity from the previous lecture. Recall that the diagonalised action is,

$$S_0[\bar{a}, a] = \sum_{\vec{k}, i\omega_n} \left(-i\omega_n + \frac{|\vec{k}|^2}{2m} - \mu \right) \bar{a}(\vec{k}, i\omega_n) a(\vec{k}, i\omega_n)$$

with $\omega_n = \frac{2\pi}{\beta}n$, for $n \in \mathbb{Z}$. We can isolate the contribution that will form the condensate, i.e. the $\vec{k} = 0$ and $i\omega_n = 0$ part, which will form the condensate, i.e. is such that,

$$a_0(\vec{k} = 0, i\omega_n) \sim \sqrt{N}, \bar{a}_0(\vec{k} = 0, i\omega_n) \sim \sqrt{N}$$

and is referred to as the “condensate fraction”. In the BEC phase, for $T < T_{BEC}$, $\mu = 0$ and therefore,

$$N = N_0 + \sum_{\vec{k} \neq 0} n_B \left(\frac{|\vec{k}|^2}{2m} \right)$$

is the total number of bosons.

11.1 Mean Field Treatment

We proceed by a mean field treatment by adding fluctuations on top of the condensate solution (true below T_{BEC}),

$$\begin{aligned} \psi(\vec{r}, \tau) &= \frac{1}{\sqrt{\beta V}} \sum_{\vec{k}, i\omega_n} a(\vec{k}, i\omega_n) e^{i\vec{k} \cdot \vec{r} - i\omega_n \tau} \\ &= \frac{1}{\sqrt{\beta V}} a_0 + \psi_1(\vec{r}, \tau) \\ &\equiv \psi_0 + \psi_1 \\ \bar{\psi}(\vec{r}, \tau) &= \bar{\psi}_0 + \bar{\psi}_1 \end{aligned}$$

where ψ_1 are the fluctuation fields above the condensate and are such that $\psi_1(\vec{r}, 0) = \psi_1(\vec{r}, \tau)$. In terms of the fluctuations, we can rewrite the action as follows,

$$\begin{aligned} S_0[\bar{\psi}, \psi] &= \int_0^\beta d\tau \int d^d \vec{r} \left\{ [\bar{\psi}_0 + \bar{\psi}_1] \left(\partial_\tau - \frac{\nabla^2}{2m} + \mu \right) [\psi_0 + \psi_1] \right\} \\ &= -\mu \beta |\psi_0|^2 - \mu \int_0^\beta d\tau \int d^d \vec{r} (\bar{\psi}_0 + \bar{\psi}_1 \psi_0) \\ &\quad + \int_0^\beta d\tau \int d^d \vec{r} \bar{\psi}_1 \left(\partial_\tau - \frac{\nabla^2}{2m} + \mu \right) \psi_1 \\ S_1[\bar{\psi}, \psi] &= \frac{g}{2} \int_0^\beta d\tau \int d^d \vec{r} (\psi_0 + \psi_1)^2 (\bar{\psi}_0 + \bar{\psi}_1)^2 \\ &= \frac{g}{2} \beta V |\psi_0|^4 + \frac{g}{2} \int_0^\beta d\tau \int d^d \vec{r} (2\psi_0^2 \psi_0 \psi_1 + 2\bar{\psi}_0 \bar{\psi}_1 \psi_0^2) \\ &\quad + \frac{g}{2} \int_0^\beta d\tau \int d^d \vec{r} (\bar{\psi}_0^2 \psi_1^2 + \bar{\psi}_1^2 \psi_0^2 + 4|\psi_0|^2 \bar{\psi}_1 \psi_1) \end{aligned}$$

and therefore to leading order we have,

$$\begin{aligned} S &= -\mu\beta V|\psi_0|^2 + \frac{g}{2}\beta V|\psi_0|^4 + \dots \\ &= \beta V \left(-\mu|\psi_0|^2 + \frac{g}{2}|\psi_0|^4 \right) + \dots \end{aligned}$$

where we remind ourselves that $g > 0$ and $\mu > 0$.

In the mean field theory, we minimise the action,

$$\frac{\partial S}{\partial |\psi_0|^2} = 0 \implies |\psi_0| = \sqrt{\frac{\mu}{g}}$$

ψ_0 is a complex field which has a minima on a whole circle on $\mathbb{C}^2$. This is the $U(1)$ symmetry of the field which is spontaneously broken. In the non-interacting case, there is no symmetry breaking.

With the saddle-point solution given above, the action for the fluctuations takes the form,

$$S[\bar{\psi}_1, \psi_1] = \int_0^\beta d\tau \int d^d\vec{r} \left\{ \bar{\psi}_1 \left(\partial_\tau - \frac{\nabla^2}{2m} \right) \psi_1 + \frac{1}{2}g\psi_0^2 (\bar{\psi}_1 + \psi_1)^2 \right\}$$

There is a very natural Hamiltonian formulation given by,

$$\hat{H}_{eff} = \sum_{\vec{k} \neq 0} \left(\frac{|\vec{k}|^2}{2m} + g\psi_0^2 \right) a_{\vec{k}}^\dagger a_{\vec{k}} + \frac{g}{2} \sum_{\vec{k} \neq 0} \psi_0^2 \left(a_{\vec{k}}^\dagger a_{-\vec{k}}^\dagger + a_{-\vec{k}} a_{\vec{k}} \right)$$

which is diagonalised via a Bogoliubov transformation.

We will diagonalise the action above, by writing $\psi_1 = A + iP$, $\bar{\psi}_1 = A - iP$, with suitable boundary conditions. We use,

$$\begin{aligned} \int d\tau A \partial_\tau A &= \int d\tau \partial_\tau (A^2) - \underbrace{[(\partial_\tau A)A]}_{\rightarrow 0} \\ \int d^3\vec{r} (P \nabla^2 A - A \nabla^2 P) &= 0 = \int \nabla \cdot [A \nabla P - P \nabla A] \end{aligned}$$

which give,

$$\begin{aligned} S[A, P] &= \int_0^\beta d\tau \int d^3\vec{r} \left[(A - iP) \left(\partial_\tau - \frac{\nabla^2}{2m} \right) (A + iP) + \frac{g}{2}\psi_0^2 (2A)^2 \right] \\ &= \int_0^\beta d\tau \int d^3\vec{r} \left[(iA \partial_\tau P - iP \partial_\tau A) + A \left(-\frac{\nabla^2}{2m} + 2g\psi_0^2 \right) A + P \left(-\frac{\nabla^2}{2m} \right) P \right] \end{aligned}$$

which shows that A and P are now conjugate variables. We use Fourier and Matsubara decompositions for both A and P , which bring the action into the form,

$$S[A, P] = \sum_{\vec{k}, i\omega_n} \begin{pmatrix} A(-\vec{k}, -i\omega_n) & P(-\vec{k}, -i\omega_n) \end{pmatrix} \underbrace{\begin{pmatrix} \frac{|\vec{k}|^2}{2m} + 2g\psi_0^2 & \omega_n \\ -\omega_n & \frac{|\vec{k}|^2}{2m} \end{pmatrix}}_{G^{-1}(\vec{k}, i\omega_n)} \begin{pmatrix} A(\vec{k}, i\omega_n) \\ P(\vec{k}, i\omega_n) \end{pmatrix}$$

In general, the Green's function corresponding to the Hamiltonian may be found as follows,

$$\begin{aligned}
G(\vec{k}, i\omega_n) &\equiv -\langle \begin{pmatrix} A(\vec{k}, i\omega_n) \\ P(\vec{k}, i\omega_n) \end{pmatrix} \begin{pmatrix} A(-\vec{k}, -i\omega_n) & P(-\vec{k}, -i\omega_n) \end{pmatrix} \rangle \\
&= -\begin{pmatrix} \langle A(\vec{k}, i\omega_n) A(-\vec{k}, -i\omega_n) \rangle & \langle A(\vec{k}, i\omega_n) P(-\vec{k}, -i\omega_n) \rangle \\ \langle P(\vec{k}, i\omega_n) A(-\vec{k}, -i\omega_n) \rangle & \langle P(\vec{k}, i\omega_n) P(-\vec{k}, -i\omega_n) \rangle \end{pmatrix} \\
&= -\frac{1}{2} \begin{pmatrix} \frac{|\vec{k}|^2}{2m} + 2g\psi_0^2 & \omega_n \\ -\omega_n & \frac{|\vec{k}|^2}{2m} \end{pmatrix}^{-1} \\
&= -\frac{1}{\omega_n^2 + \underbrace{\left[\left(\frac{|\vec{k}|^2}{2m} + 2g\psi_0^2 \right) \frac{|\vec{k}|^2}{2m} \right]}_{\omega_k^2}} \begin{pmatrix} \frac{|\vec{k}|^2}{2m} & -\omega_n \\ -\omega_n & \frac{|\vec{k}|^2}{2m} + 2g\psi_0^2 \end{pmatrix}
\end{aligned}$$

The dispersion relation is now,

$$\omega_{\vec{k}} = \sqrt{\left(\frac{|\vec{k}|^2}{2m} + 2g\psi_0^2 \right) \frac{|\vec{k}|^2}{2m}} = \begin{cases} \sqrt{\frac{g\psi_0^2}{m}} |\vec{k}| & \frac{|\vec{k}|^2}{2m} \ll 2g\psi_0^2 \\ \frac{|\vec{k}|^2}{2m} & \frac{|\vec{k}|^2}{2m} \gg 2g\psi_0^2 \end{cases}$$

In the first case we have SSB of the U(1) symmetry of the field. The excitations are gapless phonons, with only one branch because the U(1) symmetry has only one generator. This is reminiscent of superfluidity which also has a phonon mode. In the high energy limit of the second case, the excitations are free bosons.

The specific heat is $c_v \sim T^3$, similar to the Debye specific heat for which we also have $\omega_{\vec{k}} \sim |\vec{k}|$. In the ideal Bose gas, where $\omega_{\vec{k}} \sim |\vec{k}|^2$, we find instead $c_v \sim T^{3/2}$. This shows that small interactions which spontaneously break the U(1) symmetry drastically change c_v . We note though that we ignored phase fluctuations. This would bring extra gapped excitations which are anyway suppressed in the low-energy regime. We instead assumed a fixed phase.

We should however justify the approximation by checking that the fluctuations are indeed small. To this end we will calculate condensate and non-condensate fraction.

$$\begin{aligned}
N &= \int d^d \vec{r} \langle \hat{\psi}^\dagger(\vec{r}) \hat{\psi}(\vec{r}) \rangle_T \\
&= \int d^d \vec{r} \langle \bar{\psi}(\vec{r}, \tau) \psi(\vec{r}, \tau + 0^-) \rangle_T \\
&= \underbrace{V|\psi_0|^2}_{N_0} + \underbrace{\int d^d \vec{r} \langle \bar{\psi}_1(\vec{r}, \tau) \psi_1(\vec{r}, \tau + 0^-) \rangle_T}_{N_1 = \frac{1}{\beta} \sum_{\vec{k}, i\omega_n} \langle \bar{\psi}_1(\vec{k}, i\omega_n) \psi_1(\vec{k}, i\omega_n) \rangle e^{-i\omega_n 0^-}}
\end{aligned}$$

where $\psi_1(\vec{k}, i\omega_n) = A_{\vec{k}, i\omega_n} + iP_{\vec{k}, i\omega_n}$ and $\bar{\psi}_1(\vec{k}, i\omega_n) = A_{\vec{k}, i\omega_n} - iP_{\vec{k}, i\omega_n}$. To calculate the thermal averages, we use the Green's function from before. We find,

$$N_1 = \sum_{\vec{k}} \left\{ \frac{1}{\omega_{\vec{k}}} \left(n_B(\omega_{\vec{k}}) + \frac{1}{2} \right) \left(\frac{|\vec{k}|^2}{2m} + g|\psi_0|^2 - \omega_{\vec{k}} \right) + n_B(\omega_{\vec{k}}) \right\}$$

In the low T limit, $n_B(\omega_{\vec{k}}) \sim \frac{T}{\omega_{\vec{k}}}$ and $\omega_{\vec{k}} \sim v|\vec{k}|$ but let us first consider the $T = 0$ limit. In that case,

$$\begin{aligned} N_1 &= V \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{\omega_{\vec{k}}} \left(\frac{|\vec{k}|^2}{2m} + g|\psi_0|^2 - \omega_{\vec{k}} \right) \\ &= \frac{1}{24\pi^2} V k_0^3 \end{aligned}$$

with $k_0 = 2\sqrt{mg}\psi_0$. Thus, $N_1 \ll N_0 = V\psi_0^2$ when $\psi_0 \sim \sqrt{\frac{N}{V}} \ll (mg)^{-3/2}$, where N/V is the density of bosons in real space. Therefore, the Bogoliubov approximation is valid in the low density limit and when the interactions are weak. At least one condition is needed to justify the use of the Gaussian perturbation expansion.

11.2 $d = 1$ and $d = 2$

When $d = 1$, $\langle \psi \rangle = \psi_0 = 0$. This will lead to power-law correlation functions for bosons, and a Luttinger liquid phase.

When $d = 2$, we can write $\psi(\vec{r}, \tau) = \rho(\vec{r}, \tau)e^{i\theta(\vec{r}, \tau)}$, with θ and $\theta + 2\pi$ identified. This allows us to study fluctuations around the radial field, $\rho(\vec{r}, \tau) = \rho_0 + \delta\rho(\vec{r}, \tau)$ and write the action in terms of $\delta\rho$. Doing that one finds a superfluid phase.

12 Lecture 12: Fermionic Systems

We now move to the study of fermionic systems. To develop the corresponding functional integral we need to introduce Grassmann variables.

12.1 Grassmann Algebra and Calculus

We start by defining the Grassmann algebra and state some basic properties without derivations. Consider a general Hamiltonian,

$$H = \sum_{j,l} t_{jl} f_j^\dagger f_l + \sum_{jlmn} V_{jlmn} f_j^\dagger f_l^\dagger f_m f_n$$

where $\{f_i\}$ are fermion variables which satisfy the anticommutation relations,

$$\begin{aligned} \{\hat{f}_j, \hat{f}_l^\dagger\} &= \delta_{jl} \\ \{\hat{f}_j^\dagger, \hat{f}_l^\dagger\} &= \{\hat{f}_j, \hat{f}_l\} = 0 \end{aligned}$$

The Fock space is comprised by the set of states,

$$|n_1, n_2, \dots, n_N\rangle = \left(\hat{f}_1^\dagger\right)^{n_1} \dots \left(\hat{f}_N^\dagger\right)^{n_N} |0\rangle, \quad n_j = 0, 1 \quad \forall j$$

To define the Grassmann algebra we consider just a single fermionic mode, $\hat{f}^\dagger, \hat{f}$, and the Grassmann variables $\bar{\eta}, \eta$. Notice that these are not complex conjugate of each other. They satisfy the following properties,

$$\begin{aligned} \bar{\eta}^2 = \eta^2 &= 0 & \eta \hat{f} &= -\hat{f} \eta \\ \bar{\eta} \eta &= -\eta \bar{\eta} & \eta \hat{f}^\dagger &= -\hat{f}^\dagger \eta \end{aligned}$$

and calculus defined by,

$$\begin{aligned} \frac{\partial}{\partial \eta} 1 &= 0, \quad \frac{\partial}{\partial \eta} \eta = 1, \quad \frac{\partial}{\partial \eta} \bar{\eta} = 0 \\ \int d\eta \eta &\equiv \frac{\partial}{\partial \eta} \eta = 1 \\ \int d\eta d\bar{\eta} \eta \bar{\eta} &= \frac{\partial}{\partial \eta} \frac{\partial}{\partial \bar{\eta}} \eta \bar{\eta} = \frac{\partial}{\partial \eta} \frac{\partial}{\partial \bar{\eta}} \bar{\eta} \eta = \frac{\partial}{\partial \eta} \eta = -1 \end{aligned}$$

We also define the Grassmann coherent states,

$$\begin{aligned} |\eta\rangle &\equiv e^{\hat{f}^\dagger \eta} |0\rangle \rightarrow \hat{f} |\eta\rangle = \eta |0\rangle \\ \langle \bar{\eta}| &\equiv \langle 0| e^{\bar{\eta} \hat{f}} \rightarrow \langle \bar{\eta}| \hat{f}^\dagger = \langle 0| \bar{\eta} \end{aligned}$$

so that we may correspondingly think of them as left (right) eigenvectors of $\hat{f}^\dagger$ ($\hat{f}$). The overlap between two coherent states is calculated simply as,

$$\begin{aligned} \langle \bar{\eta} | \eta \rangle &= \langle 0 | e^{\bar{\eta} \hat{f}} e^{\hat{f}^\dagger \eta} | 0 \rangle \\ &= \langle 0 | (1 + \bar{\eta} \hat{f}) (1 + \hat{f}^\dagger \eta) | 0 \rangle \\ &= 1 + \langle 0 | \bar{\eta} \hat{f} \hat{f}^\dagger \eta | 0 \rangle \\ &= 1 + \bar{\eta} \eta = e^{\bar{\eta} \eta} \end{aligned}$$

The matrix element of a normal ordered operator $\hat{N} = \hat{f}^\dagger \hat{f}$ is,

$$\langle \bar{\eta} | \hat{N} | \eta \rangle = \bar{\eta} \eta e^{\bar{\eta} \eta}$$

and for $\hat{A} = \alpha \hat{f}^\dagger + \beta \hat{f}$,

$$\langle \bar{\eta} | \hat{A} | \eta \rangle = (\alpha \bar{\eta} + \beta \eta) e^{\bar{\eta} \eta}$$

The resolution of identity is given by,

$$\int d\bar{\eta} d\eta e^{-\bar{\eta} \eta} | \eta \rangle \langle \bar{\eta} | = \mathbb{I}$$

and the trace,

$$\text{Tr} \{ \hat{A} \} = \int d\bar{\eta} d\eta e^{-\bar{\eta} \eta} \langle -\bar{\eta} | \hat{A} | \eta \rangle$$

Note that $\hat{A}$ here involves an even number of fermions (in other words, it conserves fermion parity), since in a different case $\text{Tr} \{ \hat{A} \} = 0$.

The generalization of the coherent states to a many-fermion system is explicitly,

$$\begin{aligned} |\{ \eta \} \rangle &\equiv |\eta_1, \eta_2, \dots, \eta_N \rangle = e^{\sum_{j=1}^N \hat{f}_j^\dagger \eta_j} |0 \rangle \\ \langle \{ \bar{\eta} \} | &\equiv \langle \bar{\eta}_1, \bar{\eta}_2, \dots, \bar{\eta}_N | = \langle 0 | e^{\sum_{j=1}^N \bar{\eta}_j \hat{f}_j} \end{aligned}$$

so that,

$$\begin{aligned} \hat{f}_j |\{ \eta \} \rangle &= \eta_j |\{ \eta \} \rangle \\ \langle \{ \eta \} | \{ \bar{\eta} \} \rangle &= e^{\sum_{j=1}^N \bar{\eta}_j \eta_j} = e^{\bar{\eta}^T \eta} \end{aligned}$$

where $\eta = (\eta_1, \dots, \eta_N)^T$ and $\bar{\eta} = (\bar{\eta}_1, \dots, \bar{\eta}_N)^T$. The Hamiltonian matrix elements for a free Hamiltonian $H_0 = \sum_{jl} t_{jl} \hat{f}_j^\dagger \hat{f}_l$ are given by,

$$\langle \{ \eta \} | H_0 | \{ \bar{\eta} \} \rangle = \sum_{jl} t_{jl} \bar{\eta}_j \eta_l e^{\bar{\eta}^T \eta}$$

The integral measure is defined as $\int d\bar{\eta} d\eta \equiv \int d\bar{\eta}_1 d\eta_1 \dots d\bar{\eta}_N d\eta_N$ and the completeness relation is then,

$$\int d\bar{\eta} d\eta e^{-\bar{\eta}^T \eta} |\{ \eta \} \rangle \langle \{ \bar{\eta} \} | = \mathbb{I}_{2^N \times 2^N}$$

Finally, the trace is given by,

$$\text{Tr}(\hat{A}) = \int d\bar{\eta} d\eta \langle \{ -\bar{\eta} \} | \hat{A} | \{ \eta \} \rangle e^{-\bar{\eta}^T \eta}$$

12.2 Gaussian Integrals over Grassmann Fields

Recall that for a complex N -component field $z \equiv (z_1, z_2, \dots, z_N)^T$ and $\hat{A}$ a hermitian matrix, we have an exact expression for the Gaussian integral,

$$\int dz^* dz e^{-z^\dagger \hat{A} z} = \frac{1}{\det(\hat{A})}$$

which follows from being able to diagonalize $\hat{A}$ and by Hermiticity, the invariance of the Jacobian. For a Grassmann field, $\bar{\eta}, \eta$, we can do the same integral without having any restriction on $\hat{A}$,

$$\int d\bar{\eta} d\eta e^{-\bar{\eta}^T \hat{A} \eta} = \det(\hat{A})$$

The proof proceeds as follows,

$$\begin{aligned} \int d\bar{\eta} d\eta e^{-\bar{\eta}^T \hat{A} \eta} &= \int d\bar{\eta} d\eta e^{-\sum_{i,j=1}^N \bar{\eta}_i A_{ij} \eta_j} \\ &= \int d\bar{\eta} d\eta \prod_{i,j=1}^N (1 + \eta_j A_{ji}^T \bar{\eta}_i) \\ &= \int d\bar{\eta} d\eta [(\eta_1 A_{11} \bar{\eta}_1) \dots (\eta_N A_{NN} \bar{\eta}_N) + \dots] \\ &= \int d\bar{\eta} d\eta \sum_{\sigma \in S_N} \eta_1 A_{1,\sigma(1)}^T \bar{\eta}_{\sigma(1)} \cdot \eta_2 A_{2,\sigma(2)}^T \bar{\eta}_{\sigma(2)} \cdot \dots \cdot \eta_N A_{N,\sigma(N)}^T \bar{\eta}_{\sigma(N)} \\ &= \int \underbrace{d\bar{\eta} d\eta}_{=\partial_{\bar{\eta}_N} \partial_{\eta_N} \dots \partial_{\bar{\eta}_1} \partial_{\eta_1}} \sum_{\sigma \in S_N} \text{sign}(\sigma) \eta_1 A_{1,\sigma(1)}^T \bar{\eta}_1 \cdot \eta_2 A_{2,\sigma(2)}^T \bar{\eta}_{\sigma(2)} \cdot \dots \cdot \eta_N A_{N,\sigma(N)}^T \bar{\eta}_N \\ &= \sum_{\sigma \in S_N} \text{sign}(\sigma) A_{1,\sigma(1)}^T \cdot \eta_2 A_{2,\sigma(2)}^T \cdot \dots \cdot \eta_N A_{N,\sigma(N)}^T \\ &= \det(\hat{A}^T) = \det(\hat{A}) \end{aligned}$$

The Jacobian of Grassmann integrals is,

$$\int d\eta_{2N} d\eta_{2N-1} \dots d\eta_1 = \int d\theta_{2N} d\theta_{2N-1} \dots d\theta_1 \det(J)^{-1}$$

where $\eta = J\theta$.

Finally, we can consider a generic fermionic Hamiltonian with interactions,

$$\hat{H} = \sum_{jl} t_{jl} \hat{f}_j^\dagger \hat{f}_l + \sum_{jlmn} V_{jlmn} \hat{f}_j^\dagger \hat{f}_l^\dagger \hat{f}_m \hat{f}_n$$

and calculate the partition function, $Z = \text{Tr } e^{-\beta \hat{H}}$ as follows,

$$\begin{aligned}
Z &= \text{Tr } e^{-\beta \hat{H}} \\
&= \int d\bar{\eta}_0 d\eta_0 e^{-\bar{\eta}_0^T \eta_0} \langle \{-\bar{\eta}_0\} | e^{-\Delta\tau \hat{H}} \dots e^{-\Delta\tau \hat{H}} | \{\eta_0\} \rangle \\
&= \int \prod_{k=0}^{N-1} d\bar{\eta}_k d\eta_k e^{-\sum_{k=0}^{N-1} \bar{\eta}_k^T \eta_k} \langle \{-\bar{\eta}_0\} | e^{-\Delta\tau \hat{H}} | \{\eta_{N-1}\} \rangle \times \langle \{-\bar{\eta}_{N-1}\} | \dots | \{\eta_1\} \rangle \langle \{-\bar{\eta}_1\} | e^{-\Delta\tau \hat{H}} | \{\eta_0\} \rangle \\
&= \int \prod_{k=0}^{N-1} d\bar{\eta}_k d\eta_k e^{-\sum_{k=0}^{N-1} \bar{\eta}_{k+1}^T (\eta_{k+1} - \eta_k)} e^{-\Delta\tau \sum_{k=0}^{N-1} H(\bar{\eta}_{k+1}, \eta_k)} \\
&\rightarrow \int_{\substack{\eta(0)=-\eta(\beta) \\ \bar{\eta}(0)=-\bar{\eta}(\beta)}} \mathcal{D}\bar{\eta} \mathcal{D}\eta \exp \left(- \int_0^\beta d\tau [\bar{\eta}^T \partial_\tau \eta + H(\bar{\eta}, \eta)] \right)
\end{aligned}$$

where we have used antiperiodic boundary conditions (APBC) along the imaginary axis. In the second step we defined $\bar{\eta}_N = -\bar{\eta}_0$ and expanded the exponential as before,

$$\begin{aligned}
\langle \{-\bar{\eta}_1\} | e^{-\Delta\tau \hat{H}} | \{\eta_0\} \rangle &= \langle \{-\bar{\eta}_1\} | 1 - \Delta\tau \hat{H} | \{\eta_0\} \rangle \\
&= e^{-\Delta\tau H(\bar{\eta}_1, \eta_0)} \langle \{-\bar{\eta}_1\} | \{\eta_0\} \rangle \\
&= e^{-\Delta\tau H(\bar{\eta}_1, \eta_0)} \bar{\eta}_1^T \eta_0
\end{aligned}$$

13 Lecture 13: BCS Theory

We are now in position to study the BCS theory in the path integral representation. The Hamiltonian we consider is,

$$\hat{H} = \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} - \frac{g}{V} \sum_{\vec{k}, \vec{k}'} c_{\vec{k}, \uparrow}^\dagger c_{-\vec{k}, \downarrow}^\dagger c_{-\vec{k}', \downarrow} c_{\vec{k}', \uparrow}$$

where $g > 0$ for an attractive potential and $\epsilon_{\vec{k}} = \frac{|\vec{k}|^2}{2m} - \mu$. The partition function can be computed by introducing Grassman variables,

$$Z = \text{Tr} e^{-\beta \hat{H}} = \int_{\substack{\eta(0)=-\eta(\beta) \\ \bar{\eta}(0)=-\bar{\eta}(\beta)}} \mathcal{D}\bar{\eta} \mathcal{D}\eta \exp \left(- \int_0^\beta d\tau \left[\sum_{\vec{k}, \sigma} \bar{\eta}_{\vec{k}, \sigma} \partial_\tau \eta_{\vec{k}, \sigma} + H(\bar{\eta}, \eta) \right] \right)$$

where,

$$H(\bar{\eta}, \eta) = \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \bar{\eta}_{\vec{k}, \sigma} \eta_{\vec{k}, \sigma} - \frac{g}{V} \sum_{\vec{k}, \vec{k}'} \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} \eta_{-\vec{k}', \downarrow} \eta_{\vec{k}', \uparrow}$$

The Grassmann values don't have expectation values and therefore no saddle point approach is directly possible (note however that a pair of Fermi fields does acquire a non-zero expectation value, which is the point of BCS!). Also, directly doing perturbation theory for H in powers of g (like in a conventional Fermi liquid) won't work here because of convergence issues. Instead, we need to perform a Hubbard-Stratonovich transformation by introducing an auxiliary field. The goal is to get,

$$\begin{aligned} & \exp \left(- \int_0^\beta d\tau \sum_{\vec{k}, \vec{k}'} \left(-\frac{g}{V} \right) \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} \eta_{-\vec{k}', \downarrow} \eta_{\vec{k}', \uparrow} \right) \\ & \propto \underbrace{\int \mathcal{D}\Delta^* \mathcal{D}\Delta e^{-\int_0^\beta d\tau \frac{V}{g} \left(\Delta^* + \frac{g}{V} \sum_{\vec{k}} \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} \right) \left(\Delta + \frac{g}{V} \sum_{\vec{k}} \eta_{-\vec{k}, \downarrow} \eta_{\vec{k}, \uparrow} \right)}}_{=\text{const.}} \times e^{\int_0^\beta d\tau \frac{g}{V} \sum_{\vec{k}, \vec{k}'} \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} \eta_{-\vec{k}', \downarrow} \eta_{\vec{k}', \uparrow}} \\ & = \int \mathcal{D}\Delta^* \mathcal{D}\Delta \exp \left(- \int_0^\beta d\tau \frac{V}{g} \left[|\Delta|^2 + \sum_{\vec{k}} (\Delta \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} + \Delta^* \eta_{-\vec{k}, \downarrow} \eta_{\vec{k}, \uparrow}) \right] \right) \end{aligned}$$

The partition function will now be written in terms of both the Δ and η fields,

$$\begin{aligned} Z &= \int \mathcal{D}\bar{\eta} \mathcal{D}\eta \mathcal{D}\Delta^* \mathcal{D}\Delta e^{-S[\bar{\eta}, \eta, \Delta^*, \Delta]}, \\ S[\bar{\eta}, \eta, \Delta^*, \Delta] &= \int_0^\beta d\tau \left[\sum_{\vec{k}, \sigma} \bar{\eta}_{\vec{k}, \sigma} (\partial_\tau + \epsilon_{\vec{k}}) \eta_{\vec{k}, \sigma} + \frac{V}{g} |\Delta|^2 + \sum_{\vec{k}} \left(\Delta \bar{\eta}_{\vec{k}, \uparrow} \bar{\eta}_{-\vec{k}, \downarrow} + \Delta^* \eta_{-\vec{k}, \downarrow} \eta_{\vec{k}, \uparrow} \right) \right] \end{aligned}$$

What we have thus achieved is that the Grassmann integration in the η fields is Gaussian, and in doing so we have to pay the price of an additional functional integral over Δ^*, Δ . Note that in deriving that we used,

$$\bar{\eta}_{\vec{k}, \downarrow} (\partial_\tau + \epsilon_{\vec{k}}) \eta_{\vec{k}, \downarrow} = -(\partial_\tau \bar{\eta}_{\vec{k}, \downarrow}) \eta_{\vec{k}, \downarrow} = +\eta_{\vec{k}, \downarrow} \partial_\tau \bar{\eta}_{\vec{k}, \downarrow}$$

The partition function takes the nice form,

$$\begin{aligned}
Z &= \int \mathcal{D}\bar{\eta} \mathcal{D}\eta \mathcal{D}\Delta^* \mathcal{D}\Delta \prod_{\vec{k}} \det \underbrace{\begin{pmatrix} \partial_\tau + \epsilon_{\vec{k}} & \Delta \\ \Delta^* & \partial_\tau - \epsilon_{\vec{k}} \end{pmatrix}}_{\equiv M(\tau', \sigma'; \tau, \sigma) \in \infty \text{ dim space}} \exp \left(- \int_0^\beta \frac{V}{g} |\Delta|^2 \right) \\
&= \int \mathcal{D}\bar{\eta} \mathcal{D}\eta \mathcal{D}\Delta^* \mathcal{D}\Delta \exp \left(- \int_0^\beta d\tau \frac{V}{g} |\Delta|^2 + \sum_{\vec{k}} \ln \det \begin{pmatrix} \partial_\tau + \epsilon_{\vec{k}} & \Delta \\ \Delta^* & \partial_\tau - \epsilon_{\vec{k}} \end{pmatrix} \right)
\end{aligned}$$

To solve this, we use Matsubara modes,

$$\eta_{\vec{k}, \sigma} = \frac{1}{\sqrt{\beta}} \sum_{\vec{k}} \eta_{\vec{k}, \sigma}(i\omega_n) e^{-i\omega_n \tau}, \quad \omega_n = \frac{2\pi}{\beta} \left(n + \frac{1}{2} \right)$$

under which we have the identification $\partial_\tau \leftrightarrow -i\omega_n$. In this form, the inverse Green's function in Matsubara space is $M(i\omega'_n, \sigma'; i\omega_n, \sigma)$, and the partition function is simply,

$$Z = \int \mathcal{D}\bar{\eta} \mathcal{D}\eta \mathcal{D}\Delta^* \mathcal{D}\Delta \exp(-S[\Delta^*, \Delta])$$

This form is particularly useful for QMC as long as it is guaranteed that $S[\Delta^*, \Delta] > 0$. Note that,

$$\ln \prod_{\vec{k}} \det M = \sum_{\vec{k}} \ln \det M$$

and we can perform a gradient expansion for $\ln M$.

13.1 BCS Mean Field Theory

In the mean field theory, we will only seek the saddle point solution ($\delta S_{eff}(\Delta^*, \Delta) = 0$), which is precisely given by the condensation of the fields,

$$\Delta^*(\tau) \rightarrow \Delta^* \quad \Delta(\tau) \rightarrow \Delta$$

so that they are now static. Their value needs to be computed self-consistently.

This approach is equivalent to the original BCS approach. In the BCS theory, one starts from the BCS Hamiltonian with a static Δ ,

$$\begin{aligned}
\hat{H}_{MF} &= \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} c_{\vec{k}, \sigma}^\dagger c_{\vec{k}, \sigma} + \sum_{\vec{k}} \Delta c_{\vec{k}, \uparrow}^\dagger c_{-\vec{k}, \downarrow}^\dagger + \Delta^* c_{-\vec{k}', \downarrow} c_{\vec{k}', \uparrow} + \frac{V}{g} |\Delta|^2 \\
&= \sum_{\vec{k}} \begin{pmatrix} c_{\vec{k}, \uparrow}^\dagger & c_{-\vec{k}, \downarrow} \end{pmatrix} \begin{pmatrix} \epsilon_{\vec{k}} & \Delta \\ \Delta^* & -\epsilon_{\vec{k}} \end{pmatrix} \begin{pmatrix} c_{\vec{k}, \uparrow} \\ c_{-\vec{k}, \downarrow}^\dagger \end{pmatrix} + \frac{V}{g} |\Delta|^2 + \sum_{\vec{k}} \epsilon_{\vec{k}}
\end{aligned}$$

where we assumed a band inversion symmetry, $\epsilon_{-\vec{k}} = \epsilon_{\vec{k}}$. We can diagonalise the Hamiltonian by introducing a matrix $M_{\vec{k}}$ so that,

$$M_{\vec{k}}^\dagger \begin{pmatrix} \epsilon_{\vec{k}} & \Delta \\ \Delta^* & -\epsilon_{\vec{k}} \end{pmatrix} M_{\vec{k}} = \begin{pmatrix} E_{\vec{k}} & 0 \\ 0 & -E_{\vec{k}} \end{pmatrix}, \quad E_{\vec{k}} = \sqrt{\epsilon_{\vec{k}}^2 + |\Delta|^2}$$

This form is justified by the particle-hole symmetry of the problem. The new basis is related to the original creation and annihilation operators via,

$$M_{\vec{k}}^{\dagger} \begin{pmatrix} c_{\vec{k}\uparrow} \\ c_{-\vec{k}\downarrow}^{\dagger} \end{pmatrix} = \begin{pmatrix} \alpha_{\vec{k}} \\ \beta_{\vec{k}} \end{pmatrix}, \quad \begin{pmatrix} c_{\vec{k}\uparrow}^{\dagger} & c_{-\vec{k}\downarrow} \end{pmatrix} M_{\vec{k}} = \begin{pmatrix} \alpha_{\vec{k}}^{\dagger} & \beta_{\vec{k}}^{\dagger} \end{pmatrix}$$

The matrix M is determined in a gauge that makes $u_{\vec{k}}$ real in the following,

$$M_{\vec{k}} = \begin{pmatrix} u_{\vec{k}} & -v_{\vec{k}}^* \\ v_{\vec{k}} & u_{\vec{k}}^* \end{pmatrix}, \quad u_{\vec{k}} = \sqrt{\frac{1}{2} \left(1 + \frac{\epsilon_{\vec{k}}}{E_{\vec{k}}} \right)}, \quad v_{\vec{k}} = \frac{|\Delta|}{\Delta} \sqrt{\frac{1}{2} \left(1 - \frac{\epsilon_{\vec{k}}}{E_{\vec{k}}} \right)}$$

This provides us with the Bogoliubov transformation which explicitly reads,

$$\begin{aligned} \alpha_{\vec{k}}^{\dagger} &= u_{\vec{k}} c_{\vec{k}\uparrow}^{\dagger} + v_{\vec{k}} c_{-\vec{k}\downarrow} \\ \beta_{\vec{k}}^{\dagger} &= -v_{\vec{k}}^* c_{\vec{k}\uparrow}^{\dagger} + u_{\vec{k}}^* c_{-\vec{k}\downarrow} \end{aligned}$$

with $\{\alpha_{\vec{k}}, \alpha_{\vec{k}'}^{\dagger}\} = \{\beta_{\vec{k}}, \beta_{\vec{k}'}^{\dagger}\} = \delta_{\vec{k}, \vec{k}'}$ and $\{\alpha_{\vec{k}}, \beta_{\vec{k}}^{\dagger}\} = 0$.

Altogether, the mean field Hamiltonian is,

$$H_{MF} = \sum_{\vec{k}} E_{\vec{k}} \left(\alpha_{\vec{k}}^{\dagger} \alpha_{\vec{k}} - \beta_{\vec{k}}^{\dagger} \beta_{\vec{k}} \right) + \frac{V}{g} |\Delta|^2 + \text{const.}$$

At MF level, the ground state is the vacuum of $\alpha_{\vec{k}}^{\dagger}$ and full occupation of $\beta_{\vec{k}}^{\dagger}$ so that,

$$\begin{aligned} \alpha_{\vec{k}} |\text{MF}\rangle &= 0 \quad \forall \quad \vec{k} \\ \beta_{\vec{k}}^{\dagger} |\text{MF}\rangle &= 0 \quad \forall \quad \vec{k} \end{aligned}$$

Thus, the new operators $\alpha_{\vec{k}}^{\dagger}$ and $\beta_{\vec{k}}$ create Bogoliubov quasiparticles on top of the $|\text{MF}\rangle$. The *Cooper pair wavefunction* is given by,

$$\begin{aligned} |\text{MF}\rangle &= \prod_{\vec{k}} \left(u_{\vec{k}}^* - v_{\vec{k}}^* c_{\vec{k}\uparrow}^{\dagger} c_{-\vec{k}\downarrow}^{\dagger} \right) |0\rangle \\ &\propto \prod_{\vec{k}} \left(1 - \frac{v_{\vec{k}}^*}{u_{\vec{k}}^*} c_{\vec{k}\uparrow}^{\dagger} c_{-\vec{k}\downarrow}^{\dagger} \right) |0\rangle \\ &= \prod_{\vec{k}} \exp \left\{ -\frac{v_{\vec{k}}^*}{u_{\vec{k}}^*} c_{\vec{k}\uparrow}^{\dagger} c_{-\vec{k}\downarrow}^{\dagger} \right\} |0\rangle \end{aligned}$$

The BCS wavefunction describes s -wave superconductivity.

13.2 Gap Equation

We see that if $|\Delta| \neq 0$, a superconducting gap opens, and experimentally it can be measured by tunneling experiments. To determine it within the mean field theory, we need to first compute the

Free energy as follows.

$$\begin{aligned}
Z_{MF} &= \text{Tr} e^{-\beta H_{MF}} = e^{-\beta \frac{V}{g} |\Delta|^2} \text{Tr} \left(\prod_{\vec{k}} e^{-\beta E_{\vec{k}} (\alpha_{\vec{k}}^\dagger \alpha_{\vec{k}} - \beta_{\vec{k}}^\dagger \beta_{\vec{k}})} \right) \\
&= e^{-\beta \frac{V}{g} |\Delta|^2} \prod_{\vec{k}} \text{tr}_{n_{\alpha_{\vec{k}}}=0,1}^{n_{\beta_{\vec{k}}}=0,1} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-\beta E_{\vec{k}}} & 0 & 0 \\ 0 & 0 & e^{-\beta E_{\vec{k}}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= e^{-\beta \frac{V}{g} |\Delta|^2} \prod_{\vec{k}} \left(4 \cosh^2 \frac{\beta E_{\vec{k}}}{2} \right)
\end{aligned}$$

The Free energy, $F = -\frac{1}{\beta} \ln Z_{MF}$ is then,

$$\begin{aligned}
F_{MF} &= \frac{V}{g} |\Delta|^2 - \frac{2}{\beta} \sum_{\vec{k}} \ln \cosh \left(\frac{\beta E_{\vec{k}}}{2} \right) \\
&= \frac{V}{g} |\Delta|^2 - \frac{2V}{\beta} \int \frac{d^3 \vec{k}}{(2\pi)^3} \ln \cosh \left(\frac{\beta E_{\vec{k}}}{2} \right)
\end{aligned}$$

We have specified to $d = 3$ spatial dimensions in the continuum limit. In conventional superconductivity, the attractive interaction is mediated due to acoustic phonons, and therefore there is a natural cutoff which provides regularization $|\epsilon_{\vec{k}}| < \omega_D$. This means that it is only in this range that the interaction is attractive. One typically has $\epsilon_F \sim 10^4 \sim 10^5 K$ so that $\omega_D \sim 10^2 K$. Thus, we consider a thin shell of width $\sim \omega_D$ around the Fermi surface.

To obtain the gap equation that self-consistently determines $|\Delta|$, we note that the saddle point solution corresponds to the minimization of the free energy wrt $|\Delta|^2$. We get,

$$\frac{1}{g} = \int_{|\epsilon_{vk}| < \omega_D} \frac{d^3 \vec{k}}{(2\pi)^3} \frac{\tanh \left(\frac{\beta E_{\vec{k}}}{2} \right)}{2E_{\vec{k}}}$$

At $T = 0K$, we rewrite the gap equation in terms of the DOS which we approximate as $D(E) \approx D(0)\theta(-|\epsilon_{\vec{k}}| + \omega_D)$,

$$\frac{1}{g} = \int_{-\infty}^{\infty} d\epsilon \int_{|\epsilon_{vk}| < \omega_D} \frac{d^3 \vec{k}}{(2\pi)^3} \delta(\epsilon - \epsilon_{\vec{k}}) \frac{1}{2E_{\vec{k}}} \approx D(0) \int_{-\omega_D}^{\omega_D} d\epsilon \frac{1}{2\sqrt{\epsilon^2 + |\Delta|^2}}$$

The latter equation is solved by,

$$|\Delta| = 2\omega_D \exp \left\{ -\frac{1}{gD(0)} \right\}$$

For BCS superconductors, $|\Delta| \sim 10^{-4} - 10^{-3} eV$ or $1 - 10K$.

The superconducting transition temperature T_c is given by solving $|\Delta| = 0$ at $T = T_c$,

$$\frac{1}{g} = \int_{-\omega_D}^{\omega_D} d\epsilon \frac{\tanh \left(\frac{|\epsilon_{\vec{k}}|}{2T_c} \right)}{2|\epsilon|} \implies T_c \approx 1.13\omega_D \exp \left\{ -\frac{1}{gD(0)} \right\}$$

T_c is essentially determined by the thermal energy which is required to break a Cooper pair. Considering the electron-phonon interaction, McMillan was able to show that there is an upper limit of $T_c \sim 40K$ provided for the BCS theory. Note also that the ratio of the gap to T_c is universal, which was experimentally known before,

$$\frac{2|\Delta|}{T_c} \approx 3.54$$

A test to whether or not the superconductor is of BCS type or of different unconventional pairing is provided by the *isotope effect*. The logic is as follows. The Debye frequency is scaling with the mass of the atom as $\omega_D \sim 1/\sqrt{m}$. Thus, $T_c\sqrt{m} \sim \text{const.}$. Confirming a relation like that gives direct evidence that the interaction is mediated by acoustic phonons.

14 Lecture 14: Majorana zero mode & Ising anyon

14.1 Hamiltonian Diagonalisation

In this lecture we will study the BdG Hamiltonian which describes inhomogeneous Fermi superfluids,

$$\hat{H}_{BdG} = \sum_{j,l}^N t_{jl} a_j^\dagger a_l + \frac{1}{2} \sum_{j,l}^N \left(\Delta_{jl} a_j^\dagger a_l^\dagger + \Delta_{jl}^* a_l a_j \right)$$

with $t_{jl} = t_{jl}^*$ and $\Delta_{jl} = -\Delta_{lj}$. We denote these parameters in matrix form by $t = t^\dagger$ and $\Delta = -\Delta^T$. Notice that the Hamiltonian doesn't conserve particle number but it does conserve the fermion parity (i.e. particle number mod 2). Specifically, we let the parity operator,

$$\hat{P} = (-1)^{\sum_j a_j^\dagger a_j}$$

with eigenvalues ± 1 , so that $[\hat{P}, \hat{H}] = 0$. The Hamiltonian can be rewritten in terms of the vector $\mathbf{a}^\dagger \equiv (a_1^\dagger \ a_2^\dagger \ \dots \ a_N^\dagger)$, as,

$$\hat{H} = \frac{1}{2} \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} t & \Delta \\ \Delta^\dagger & -t^T \end{pmatrix} \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix} + \frac{1}{2} \text{Tr}(t)$$

and the Hamiltonian $\mathcal{H} = \begin{pmatrix} t & \Delta \\ \Delta^\dagger & -t^T \end{pmatrix}$ is the single particle BdG. $\mathcal{H}$ is Hermitian and has a particle-hole symmetry,

$$\Sigma \mathcal{H} \Sigma = -\mathcal{H}, \quad \Sigma = \begin{pmatrix} 0 & \mathbb{I}_{N \times N} \\ \mathbb{I}_{N \times N} & 0 \end{pmatrix} \mathcal{K}$$

where $\mathcal{K}$ denotes complex conjugation and thus satisfies $\mathcal{K}^2 = \mathbb{I}$. The PH symmetry is easy to prove,

$$\begin{aligned} \text{LHS} &= \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \begin{pmatrix} t & \Delta \\ \Delta^\dagger & -t^T \end{pmatrix} \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \\ &= \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \begin{pmatrix} \Delta & t \\ -t^T & -\Delta^* \end{pmatrix} \mathcal{K} \\ &= \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \begin{pmatrix} \Delta^* & t^* \\ -t^\dagger & -\Delta \end{pmatrix} \mathcal{K}^2 \\ &= - \begin{pmatrix} t & \Delta \\ \Delta^\dagger & -t^T \end{pmatrix} \end{aligned}$$

where we used t -hermiticity, $t^* = t^T$. The idea now is that for $\mathcal{H}$ which is particle hole symmetric, then if $\mathcal{H} \begin{pmatrix} u \\ v \end{pmatrix} = E \begin{pmatrix} u \\ v \end{pmatrix}$ where $u \equiv (u_1, \dots, u_N)$ and $v \equiv (v_1, \dots, v_N)$, we can generate another

eigenvector as $\begin{pmatrix} v^* \\ u^* \end{pmatrix}$ with energy $-E$. This follows from,

$$\begin{aligned} \mathcal{H} \begin{pmatrix} v^* \\ u^* \end{pmatrix} &= -\Sigma \mathcal{H} \Sigma \begin{pmatrix} v^* \\ u^* \end{pmatrix} \\ &= -\begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \mathcal{H} \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \begin{pmatrix} v^* \\ u^* \end{pmatrix} \\ &= -\begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \mathcal{K} \mathcal{H} \begin{pmatrix} u \\ v \end{pmatrix} \\ &= -E \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \begin{pmatrix} u^* \\ v^* \end{pmatrix} \\ &= -E \begin{pmatrix} v^* \\ u^* \end{pmatrix} \end{aligned}$$

We can diagonalise the Hamiltonian as follows. We first consider a matrix,

$$M = \left(\begin{array}{c|c} U & V^* \\ \hline V & U^* \end{array} \right), \quad MM^\dagger = \mathbb{I}_{2N \times 2N}, \quad M^\dagger \mathcal{H} M = \left(\begin{array}{ccc|ccc} E_1 & \dots & 0 & & & \\ \vdots & \ddots & \vdots & & & \\ 0 & \dots & E_N & & & \\ \hline & & & -E_1 & \dots & 0 \\ & & & \vdots & \ddots & \vdots \\ & & & 0 & \dots & -E_N \end{array} \right)$$

Note that the above ordering specifies the ordering of the eigenvectors (column vectors) of M . This diagonalization changes the basis into the “Bogoliubov quasiparticle operators”,

$$\begin{pmatrix} \mathbf{d}^\dagger & \mathbf{d} \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} M, \quad \begin{pmatrix} \mathbf{d} \\ \mathbf{d}^\dagger \end{pmatrix} = M^\dagger \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix}$$

which in component form becomes,

$$\begin{aligned} d_m^\dagger &= \sum_j \left(a_j^\dagger U_{jm} + a_j V_{jm} \right) \\ d_m &= \sum_j \left(a_j^\dagger V_{jm} + a_j U_{jm}^* \right) \end{aligned}$$

The Bogoliubov quasiparticles satisfy the anticommutation relations, $\{d_m^\dagger, d_n\} = \delta_{mn}$ and $\{d_m, d_n\} = \{d_m^\dagger, d_n^\dagger\} = 0$. In terms of these, the Hamiltonian is then,

$$H = \sum_{m=1}^N E_m d_m^\dagger d_m = \frac{1}{2} \sum_{m=1}^N E_m + \frac{1}{2} \text{Tr}\{t\}$$

The ground state is the Bogoliubov vacuum which is defined by $|0\rangle_d$ with $d_m |0\rangle_d = 0$ for all m .

A small remark: If U^{-1} exists, then $|0\rangle_d \propto \exp\left(\frac{1}{2} \sum_{jl} a_j^\dagger Z_{jl} a_l^\dagger\right) |0\rangle_a$ with $Z = (VU^{-1})^*$. We can show this by calculating $d_m |0\rangle_d = 0$ explicitly. What the ground state shows is that all fermions are paired in Cooper pairs in the vacuum state $|0\rangle_d$. If U^{-1} does not exist, then there exist left-over unpaired fermions.

14.2 Zero Modes

Let us write the Hamiltonian for simplicity as $H = \sum_{m=1}^N E_m d_m^\dagger d_m + \text{const.}$ and consider the special case where $E_1 = 0$ and $E_{m \geq 2} > 0$. Then there are two degenerate ground states which are protected by a gap,

$$|0\rangle, \quad d_1^\dagger |0\rangle_d$$

We have $[H, d_1^\dagger] = [H, d_1] = 0$ for the $m = 1$ mode. Let us use these to build Majorana fermions via,

$$c_1 = d_1^\dagger + d_1 \quad c_2 = i(d_1^\dagger - d_1)$$

which also satisfy $[H, c_1] = [H, c_2] = 0$. Here, c_1 and c_2 are unpaired Majorana fermions. In the special case that they correspond to Ising anyons, they are also known as zero-modes. The latter have spatially localised wavefunctions of the form $d_1 = \sum_j (a_j^\dagger V_{j1}^* + a_j U_{j1}^*)$, where V_{j1}, U_{j1} are local.

The Majorana fermions (or majoranas) satisfy the following algebra,

$$\begin{aligned} c_1^2 &= c_2^2 = 1 \\ \{c_1, c_2\} &= 0 \\ (ic_1 c_2)^2 &= 1 \implies ic_1 c_2 = \pm 1 \end{aligned}$$

Additionally, from $ic_1 c_2 = 2d_1^\dagger d_2 - 1$ we find,

$$\begin{aligned} ic_1 c_2 |0\rangle_d &= -|0\rangle_d \\ ic_1 c_2 (d_1^\dagger |0\rangle_d) &= d_1^\dagger |0\rangle_d \end{aligned}$$

so that the $ic_1 c_2$ operator may be interpreted as the fermion parity operator in the d -basis.

14.3 Kitaev's Majorana chain (cond-mat/0010440)

We consider here a special case for a BdG Hamiltonian, namely the Kitaev (Majorana) chain, which describes a chain of spinless fermions with OBC. Explicitly, we write,

$$H = -t \sum_{j=1}^{N-1} (a_j^\dagger a_{j+1} + \text{h.c.}) + -t \sum_{j=1}^{N-1} (a_j^\dagger a_{j+1}^\dagger + \text{h.c.}) - \mu \sum_{j=1}^N (a_j^\dagger a_j - 1)$$

and without loss of generality we consider $t > 0$ and $\mu > 0$. There are two phases corresponding to the Kitaev chain Hamiltonian: a) a trivial phase and b) a topological phase which supports Majorana zero modes.

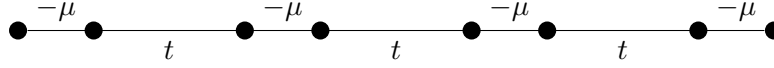
We solve this by rewriting the Hamiltonian in terms of majoranas at each site j ,

$$\begin{aligned} a_j &= \frac{1}{2} (c_{2j-1} + ic_{2j}) \\ a_j^\dagger &= \frac{1}{2} (c_{2j-1} - ic_{2j}) \end{aligned}$$

and obtain,

$$H = t \sum_{j=1}^{N-1} i(c_{2j} c_{2j+1}) - \mu \sum_{j=1}^{N-1} i(c_{2j-1} c_{2j})$$

In the majorana language, the original tight-binding model admits the following interpretation,



Let us first consider two special limits.

- $\mu = 0$: In this case, c_1 and c_{2N} are zero modes which we may pair as follows,

$$d_1 = \frac{1}{2} (c_1 + ic_{2N}) \quad d_1^\dagger = \frac{1}{2} (c_1 - ic_{2N})$$

The two majoranas at the right and left boundaries are decoupled from the chain and have zero eigenvalues. The rest of the Hamiltonian is,

$$H = t \sum_{m=2}^N (2d_m^\dagger d_m - 1)$$

It's clear from this that the ground state is two fold degenerate: $|0\rangle_d$ and $d_1^\dagger |0\rangle_d$. In this case, the majoranas are topologically protected which means that their degeneracy is very hard to split. The two ground states are explicitly given by,

$$\begin{aligned} |0\rangle_d &= \left(1 + a_1^\dagger a_2^\dagger + \dots + a_1^\dagger a_2^\dagger a_3^\dagger a_4^\dagger + \dots \right) |0\rangle_a = e^{\sum_{i<j} a_i^\dagger a_j} |0\rangle_a \\ d_1^\dagger |0\rangle_d &= \left(a_1^\dagger + a_2^\dagger + \dots + a_1^\dagger a_2^\dagger a_3^\dagger + \dots \right) |0\rangle_a \end{aligned}$$

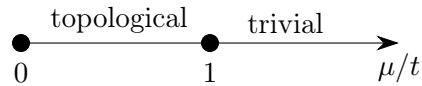
so that the vacuum includes all the possible fermion pairings, and the two ground states have different parities.

- $t = 0$: The Hamiltonian is simply,

$$H = -\mu \sum_{j=1}^N (ic_{2j-1} c_{2j})$$

Here we have a unique GS, $|0\rangle_d$ and no unpaired modes.

The phase diagram is simply given,



14.4 Braiding of spatially localized Majorana zero modes

Spatially localized Majorana zero modes appear for example as vortices in the $p_x + ip_y$ superconductors.

15 Lecture 15: Majorana field theory & Jackiw-Rebbi mechanism

In this lecture we will develop a Majorana field theory in $(1+1)$ D and review the Jackiw-Rabbi mechanism. Recall that in the Majorana chain we saw that there are two phases: a) for $|t| > |\mu|$ a topological phase, which supports Majorana zero modes at the boundary (assuming Open Boundary Conditions), b) for $|t| < |\mu|$ a trivial phase, with no majorana zero-modes, and at $|t| = |\mu|$ the theory is critical.

15.1 Critical Point

The theory we want to develop is for the critical point,

$$H_0 = \int_0^L \left[-\frac{iv}{2} (\chi^R \partial_x \chi^R - \chi^L \partial_x \chi^L) \right]$$

where the Majorana fields in the continuum satisfy the anticommutation relation,

$$\{\chi^{R(L)}, \chi^{R(L)}\} = \delta(x - x')$$

and the boundary conditions can be one of the two,

$$\chi^{R(L)}(x + L) = \begin{cases} \chi^{R(L)}(x) & \text{PBC} \\ -\chi^{R(L)}(x) & \text{APBC} \end{cases}$$

imposed on the allowed momenta. Decomposing into Fourier modes, $\chi^{R(L)} = \frac{1}{\sqrt{L}} \sum_k \chi_k^{R(L)} e^{ikx}$, we find that the two kinds of boundary conditions give rise to different sorts of allowed momenta,

$$\text{PBC} \implies k = 0, \pm \frac{2\pi}{L}, \dots$$

$$\text{APBC} \implies k = \pm \frac{\pi}{L}, \pm \frac{3\pi}{L}, \dots$$

and the commutation between the different Fourier components becomes,

$$\{\chi_k^{R(L)}, \chi_{k'}^{R(L)}\} = \delta_{k', -k}$$

We may now interpret $\chi_{k>0}^{R(L)}$ and $\chi_{k<0}^{R(L)}$ as a fermionic pair of creation and annihilation operators respectively. For PBC, there is a leftover $k = 0$ mode. To get around that, we can combine χ_0^R with χ_0^L into a single Dirac mode,

$$d_0 = \frac{1}{2} (\chi_0^R + i\chi_0^L), \quad d_0^\dagger = \frac{1}{2} (\chi_0^R - i\chi_0^L)$$

The quantised version of the field theory is then as follows,

$$\begin{aligned} H_0 &= \frac{v}{2} \sum_k k (\chi_{-k}^R \chi_k^R - \chi_{-k}^L \chi_k^L) \\ &= \frac{v}{2} \sum_{k>0} k \left(\underbrace{\chi_{-k}^R}_{(\chi_k^R)^\dagger} \chi_k^R - \chi_{-k}^L \underbrace{\chi_k^L}_{(\chi_{-k}^L)^\dagger} \right) + \frac{v}{2} \sum_{k>0} (-k) \left(\chi_k^R \underbrace{\chi_{-k}^R}_{(\chi_k^R)^\dagger} - \underbrace{\chi_k^L}_{(\chi_{-k}^L)^\dagger} \chi_{-k}^L \right) \\ &= v \sum_{k>0} k \left[(\chi_k^R)^\dagger \chi_k^R + (\chi_{-k}^L)^\dagger \chi_{-k}^L \right] - v \underbrace{\sum_{k>0} k}_{\equiv E_0} \end{aligned}$$

We thus find a linear dispersion relation, $\epsilon_k = v|k|$, which indicates gaplessness. The $k > 0$ modes are the χ^R which we dub right-movers, and the $k < 0$ modes are the χ^L left-movers. Since we are working on a compactified circle, by virtue of the chosen boundary conditions, only discrete points are allowed and the theory is split into two sectors,

- **PBC:** $k = \frac{2\pi}{L}n$ corresponds to the R -sector (Ramond), with a two-fold degenerate ground state given by, $|0\rangle_{PBC}$ and $d_0^\dagger |0\rangle_{PBC}$. The ground state energy is,

$$E_{PBC} = -v \sum_{k>0} k = \frac{-2\pi v}{L} \sum_{n=0}^{\infty} n = \frac{\pi v}{6L}$$

- **APBC:** $k = \frac{2\pi}{L}(n + \frac{1}{2})$ corresponds to the NS -sector (Neveu-Schwarz), with a unique ground state $|0\rangle_{APBC}$. The ground state energy is,

$$E_{APBC} = -v \sum_{k>0} k = \frac{-2\pi v}{L} \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right) = -\frac{\pi v}{12L}$$

The calculation of the ground-state energies involved a Riemann-zeta function regularization scheme which is justified as follows. We can go back to the lattice model with $t = \mu$ and diagonalize the system at hand. We will find that for a certain L ,

$$E(L) = \mathcal{E}_\infty L - \frac{\pi V}{6L} c$$

where c is the *central charge* ($c^{PBC} = -1$ and $c^{APBC} = -\frac{1}{2}$). This is vacuum energy which any CFT defined on a circle has.

15.2 Massive Majorana Field Theory

Now we will gap out some of the degrees of freedom by adding a massive Majorana term. The Hamiltonian is,

$$H = H_0 + \int dx \, im \chi^R(x) \chi^L(x)$$

for $m \neq 0$. Carrying the same Fourier decomposition as before, we find that the mass term becomes,

$$\begin{aligned} H_m^{APBC} &= im \sum_{k>0} \left[\chi_k^R \chi_{-k}^L + (\chi_k^R)^\dagger (\chi_{-k}^L)^\dagger \right] \\ H_m^{PBC} &= im \sum_{k>0} \left[\chi_k^R \chi_{-k}^L + (\chi_k^R)^\dagger (\chi_{-k}^L)^\dagger \right] + im \chi_k^R \chi_k^L \end{aligned}$$

We thus see that the mass term breaks the ground-state degeneracy in the PBC case. Altogether, the Hamiltonian gives,

$$H = \sum_{k>0} \begin{pmatrix} (\chi_k^R)^\dagger & \chi_{-k}^L \end{pmatrix} \begin{pmatrix} vk & im \\ -im & -vk \end{pmatrix} \begin{pmatrix} \chi_k^R \\ (\chi_{-k}^L)^\dagger \end{pmatrix}$$

and the dispersion relation becomes,

$$E_k = \pm \sqrt{(vk)^2 + m^2}$$

Depending on the sign of m , there are two distinct phases which are both gapped. For the Kitaev majorana chain, one of these describes the trivial and the other the topological phase (it is a matter of convention which is which). Also, to justify the continuum limit approximation, the field theory is applicable then only close to the critical point, or more concretely, we require the correlation length to be much larger than the lattice spacing.

15.3 Jackiw-Rabbi Mechanism

Let us now allow the mass term to be instead of a constant, position dependent, $m(x)$,

$$H = H_0 + \int dx \, im(x) \chi^R(x) \chi^L(x)$$

Suppose that at $x = 0$, $m(x)$ changes sign so that we are describing a domain wall. The claim we want to prove is that there is a Majorana zero mode at the edge of this mass domain wall.

To do that, we first look for an eigenmode $\gamma^\dagger$, $[H, \gamma^\dagger] = \epsilon \gamma^\dagger$,

$$H |\psi\rangle = E |\psi\rangle \implies H \gamma^\dagger |\psi\rangle = (\epsilon + E) \gamma^\dagger |\psi\rangle$$

A strong majorana mode has $\epsilon = 0$ and therefore both $\gamma^\dagger |\psi\rangle$ and $|\psi\rangle$ have the same energy but different parity. We start with an ansatz for $\gamma^\dagger$,

$$\gamma^\dagger = \int dx \, [u(x) \chi^R(x) + v(x) \chi^L(x)]$$

and seek the functions $u(x), v(x)$ that satisfy the commutation relation above. We calculate by brute force,

$$\begin{aligned} [H, \gamma^\dagger] &= \dots \\ &= \int dx \, [-iv(\partial_x u) \chi^R(x) + iv(\partial_x v) \chi^L(x) - im(x)u(x) \chi^L(x) + im(x)v(x) \chi^R(x)] \\ &\stackrel{!}{=} \epsilon \gamma^\dagger \end{aligned}$$

We thus get a system of equations to solve,

$$\begin{aligned} -iv\partial_x u + im(x)v(x) &= \epsilon u(x) \\ iv\partial_x u - im(x)u(x) &= \epsilon v(x) \end{aligned}$$

Writing, $\phi(x) = (u(x), v(x))^T$, the system of equation takes the matrix form,

$$-i\sigma^z \partial_x \phi(x) - m(x) \sigma^y \phi(x) = \epsilon \phi(x)$$

By using the anticommutation relation $\{\sigma^z, \sigma^x\} = \{\sigma^z, \sigma^y\} = 0$, we conclude that if $\phi(x)$ is a solution with energy ϵ , then $\sigma^x \phi(x)$ is also a solution with energy $-\epsilon$. This makes sense as the particle-hole symmetry of the BdG system.

For $\epsilon = 0$ we can take linear combinations of the solutions,

$$\begin{aligned} \sigma^x (\phi(x) \pm \sigma^x \phi(x)) &= \pm (\phi(x) \pm \sigma^x \phi(x)) \\ &\equiv \begin{pmatrix} 1 \\ -1 \end{pmatrix} \phi_0(x) \end{aligned}$$

Plugging in the equation above, the equation reduces to the scalar equation,

$$-iv\partial_x\phi_0 + im(x)\phi_0 = 0$$

The ODE is separable and solved via,

$$\phi_0(x) = \phi_0(0)e^{\pm \int_0^x \frac{m(x')}{v} dx'}$$

which indeed describes a bound state solution which we associate with the majorana zero mode. For this to be the case, we need strictly exponentially decaying function inside the bulk, and we thus need to choose the signs accordingly. Taking $\phi(x) = \begin{pmatrix} 1 & 1 \end{pmatrix}^T \phi_0(x)$, we find,

$$\gamma^\dagger = \phi_0(0) \int dx [\chi^R(x) + \chi^L(x)] e^{-\int_0^x dx' m(x')/v}$$

which satisfies $\gamma^\dagger = \gamma$.

This allows for the following construction: take a wire with consecutive regions of topological and trivial phases. This means that there is a sign change of $m(x)$ at every boundary, and by the Jackiw-Rabbi mechanism, a majorana zero mode!

16 Lecture 16: Schwinger boson & spin coherent state

In this lecture we will develop the path integral formalism for a spin system. Let us quickly recall the spin algebra. We consider a spin $\vec{S} = (S^x, S^y, S^z)$ with commutation relations $[S^\alpha, S^\beta] = i\epsilon^{\alpha\beta\gamma} S^\gamma$. For each dimension d , the $SU(2)$ algebra has exactly one unitary irreducible representation, the spin- S representation with dimension is $d = 2s + 1$. The commuting operators $\{\vec{S}^2, S^z\}$ provide the irreducible multiplets $|s, m\rangle$ such that,

$$\begin{aligned} S^2 |s, m\rangle &= s(s+1) |s, m\rangle \\ S^z |s, m\rangle &= m |s, m\rangle \end{aligned}$$

The finite dimensionality of the representations means that we have a highest weight state, namely $|s, s\rangle$. By writing $S^\pm = S^x \pm iS^y$, we can move between multiplets as follows,

$$S^+ |s, m\rangle = \sqrt{s(s+1) - m(m+1)} |s, m+1\rangle \quad S^- |s, m\rangle = \sqrt{s(s+1) - m(m-1)} |s, m-1\rangle$$

16.1 Schwinger Boson Representation

The spin algebra resembles the character of the usual bosons in the sense that spins at the same site do not commute with each other (unlike two spins at two different sites). We introduce bosons a, b . Then, for any spin S , we can represent the spin algebra exactly as follows.

$$\mathbf{S} = \frac{1}{2}$$

The states are given by,

$$|\uparrow\rangle = a^\dagger |0\rangle \quad |\downarrow\rangle = b^\dagger |0\rangle$$

and the operators,

$$S^+ = a^\dagger b \quad S^- = b^\dagger a \quad S^z = \frac{1}{2}(a^\dagger a - b^\dagger b)$$

These bosons are constraint such that $a^\dagger a + b^\dagger b = 1$, which means that the Hilbert space is restricted to those states which correspond to single occupancy. One can immediately see that such a representation, when acted on the states as defined above, obtains the correct matrix elements (eg $S^+ |\downarrow\rangle = |\uparrow\rangle$).

$$\mathbf{S} = 1$$

The states are given by,

$$|S_z = +1\rangle = \frac{1}{\sqrt{2}}(a^\dagger)^2 |0\rangle_{bv} \quad |S_z = 0\rangle = a^\dagger b^\dagger |0\rangle_{bv} \quad |S_z = -1\rangle = \frac{1}{\sqrt{2}}(b^\dagger)^2 |0\rangle_{bv}$$

where to avoid confusion we wrote $|0\rangle_{bv}$ to denote the bosonic vacuum as opposed to $|S_z = 0\rangle$ to denote the spin state with $S_z = 0$. The constraint now that removes the unphysical states in the bosonic subspace is $a^\dagger a + b^\dagger b = 2$. and the operators,

$$S^+ = a^\dagger b \quad S^- = b^\dagger a \quad S^z = \frac{1}{2}(a^\dagger a - b^\dagger b)$$

General S

In general, the representation is elegantly written as follows,

$$S^\alpha = \frac{1}{2} \begin{pmatrix} a^\dagger & b^\dagger \end{pmatrix} \sigma^\alpha \begin{pmatrix} a \\ b \end{pmatrix}$$

with constraint,

$$n_a + n_b = a^\dagger a + b^\dagger b = 2s$$

The states in the bosonic subspace (which satisfy the above constraint) are,

$$|n_a, n_b\rangle = \frac{1}{n_a! n_b!} (a^\dagger)^{n_a} (b^\dagger)^{n_b} |0\rangle$$

The quantum numbers for the corresponding spin state are given by,

$$S^z |n_a, n_b\rangle = \frac{1}{2} (a^\dagger a - b^\dagger b) |n_a, n_b\rangle = \frac{1}{2} (\hat{n}_a - \hat{n}_b) |n_a, n_b\rangle$$

We thus identify that $m = \frac{1}{2}(n_a - n_b)$. The Casimir element (eigenvalue of S^2) is found as follows, making use of representations of operators above,

$$\begin{aligned} S^2 |n_a, n_b\rangle &= \left[\frac{1}{2} (S^+ S^- + S^- S^+) + (S^z)^2 \right] |n_a, n_b\rangle \\ &= \left[\frac{1}{2} a^\dagger b b^\dagger + \frac{1}{2} b^\dagger a a^\dagger b + \frac{1}{4} (a^\dagger a - b^\dagger b)^2 \right] |n_a, n_b\rangle \\ &= \frac{1}{2} (n_a + n_b) \left[\frac{1}{2} (n_a + n_b) + 1 \right] |n_a, n_b\rangle \\ &= S(S+1) |n_a, n_b\rangle \end{aligned}$$

since $2s = n_a + n_b$. Therefore, we have the identification $|n_a, n_b\rangle = |s, m\rangle$ with

$$n_a = s + m \quad n_b = s - m$$

16.2 Spin Coherent States

To establish the path integral of the spin system, we need to introduce the coherent state corresponding to a spin. For classical spins, the spin is an $O(3)$ vector with three components,

$$\vec{\Omega} = (\Omega^x, \Omega^y, \Omega^z)$$

with the constraint $|\vec{\Omega}|^2 = 1$, so that geometrically, this is a vector on the surface of a unit sphere. It can then be parametrised by two angles θ, ϕ ,

$$\begin{aligned} \Omega^x &= \sin \theta \cos \phi \\ \Omega^y &= \sin \theta \sin \phi \\ \Omega^z &= \cos \theta \end{aligned}$$

The idea now is that the same picture would apply to a quantum spin, as can be demonstrated using the Schwinger boson representation. We introduce two component bosons $a^\dagger, b^\dagger$. The coherent state is then given by,

$$|z_1, z_2\rangle = e^{z_1 a^\dagger + z_2 b^\dagger} |0\rangle$$

which leaves us however with redundant degrees of freedom for the spin Hilbert space. In order to restrict the Hilbert space, we introduce the projection operator $\hat{P}_{n_a+n_b=2s}$ which acts as to keep only those states which exist in the $2s$ subspace.

$$\hat{P}_{n_a+n_b=2s} |z_1, z_2\rangle = \hat{P}_{n_a+n_b=2s} \left[1 + (z_1 a^\dagger + z_2 b^\dagger) + \frac{1}{2}(z_1 a^\dagger + z_2 b^\dagger)^2 + \dots \right] |0\rangle$$

The idea now is that for each S , the only state that survives the action of the projector is the $(2s)$ th one (eg, for $S = 1$, it is the square term). We thus define the coherent state as,

$$|\vec{\Omega}\rangle = \frac{1}{\sqrt{(2s)!}} (z_1 a^\dagger + z_2 b^\dagger)^{(2s)} |0\rangle$$

and fix normalisation as follows. We first rewrite the coherent state,

$$\begin{aligned} |\vec{\Omega}\rangle &= \frac{1}{\sqrt{(2s)!}} (z_1 a^\dagger + z_2 b^\dagger)^{(2s)} |0\rangle \\ &= \frac{1}{\sqrt{(2s)!}} \sum_{n_a=0}^{2s} \frac{(2s)!}{n_a!(2s-n_a)!} z_1^{n_a} z_2^{2s-n_a} \underbrace{(a^\dagger)^{n_a} (b^\dagger)^{2s-n_a} |0\rangle}_{=\sqrt{n_a!(2s-n_a)!} |n_a, 2s-n_a\rangle} \\ &= \sum_{n_a=0}^{2s} \sqrt{\frac{(2s)!}{n_a!(2s-n_a)!}} z_1^{n_a} z_2^{2s-n_a} |n_a, 2s-n_a\rangle \\ &= \sum_{m=-s}^s \sqrt{\frac{(2s)!}{n_a!(2s-n_a)!}} z_1^{s+m} z_2^{s-m} |s, m\rangle \end{aligned}$$

where we used the previously obtained results of $s = 1/2(n_a + n_b)$ and $m = 1/2(n_a - n_b)$. We then check that,

$$\langle \vec{\Omega} | \vec{\Omega} \rangle = \sum_{m=-s}^s \frac{(2s)!}{n_a!(2s-n_a)!} (|z_1|^2)^{s+m} (|z_2|^2)^{s-m} = (|z_1|^2 + |z_2|^2)^{(2s)} \stackrel{!}{=} 1$$

The above normalisation condition has the simple solution of $|z_1|^2 + |z_2|^2 = 1$, which we treat as a constraint on the complex plane of three independent variables. Such a constraint corresponds to the $\mathbb{CP}^1$ manifold, which has the known parametrisation,

$$\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = e^{i\chi} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

where χ is an overall phase. Substituting this in the definition of the coherent phase we see that the state has an overall phase of $e^{i2s\chi}$. We remove this factor by performing a gauge fixing of $|\vec{\Omega}\rangle$ by choosing $\chi = \phi/2$, making z_1 purely real. Thus, the parametrisation that we take becomes,

$$\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}$$

which is essentially a diffeomorphism from $\mathbb{CP}^1$ to the sphere S^2 . The remaining parameters, (θ, ϕ) parametrize the unit sphere, as was the case for the classical spin.

Resolution of identity

$$\mathbb{I} = \frac{(2s+1)}{4\pi} \int d\vec{\Omega} \left| \vec{\Omega} \right\rangle \left\langle \vec{\Omega} \right| = \sum_{m=-s}^{m=s} |s, m\rangle \langle s, m|$$

with $d\vec{\Omega} = \sin\theta d\theta d\phi$, $\theta \in [0, \pi]$, $\phi \in [0, 2\pi]$.

Matrix Elements

The matrix elements of the spin operators in the coherent state can also be calculated. Indeed, upon calculation, one finds the nice expression,

$$\langle \vec{\Omega} | S^\alpha | \vec{\Omega} \rangle = S \Omega^\alpha$$

In addition,

$$\begin{aligned} \langle \vec{\Omega} | \vec{S} \cdot \vec{S} | \vec{\Omega} \rangle &= \sum_{\alpha} \langle \vec{\Omega} | S^\alpha | \vec{\Omega} \rangle \Omega^\alpha \\ &= \sum_{\alpha} S (\Omega^\alpha)^2 = S |\vec{\Omega}|^2 = S \end{aligned}$$

which shows that $|\vec{\Omega}\rangle$ is fully polarised along the $|\vec{\Omega}\rangle$ direction of the Bloch sphere.

We focus here on $S = 1/2$ so that $|\vec{\Omega}\rangle = (z_1 a^\dagger + z_2 b^\dagger) |0\rangle$, in which case the matrix elements are explicitly,

$$\begin{aligned} \langle \vec{\Omega} | S^+ | \vec{\Omega} \rangle &= \langle 0 | (z_1^* a + z_2^* b) a^\dagger b (z_1 a^\dagger + z_2 b^\dagger) | 0 \rangle \\ &= z_1^* z_2 = \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{i\phi} = \frac{1}{2} \sin \theta e^{i\phi} \\ \langle \vec{\Omega} | S^+ | \vec{\Omega} \rangle &= \frac{1}{2} \sin \theta e^{-i\phi} \end{aligned}$$

so that,

$$\langle \vec{\Omega} | S^x | \vec{\Omega} \rangle = \frac{1}{2} \sin \theta \cos \theta = S \Omega^x$$

The fact that not all states are orthogonal to each other, $\langle \vec{\Omega}' | \vec{\Omega} \rangle \neq 0$ is seen easily by the above mapping: they are vectors on the surface of the Bloch sphere. However, this geometric representation is still a very good representation to work with.

16.3 Path Integral for a single spin

We first establish the path integral for a single spin and then generalise to the case of multiple spins. We start from a general Hamiltonian H and calculate the partition function by trotterizing using the spin-coherent states,

$$Z = \text{Tr} \left\{ e^{-\beta H} \right\} = \frac{2s+1}{4\pi} \int d\Omega_0 \text{Tr} \left\{ \left| \vec{\Omega}_0 \right\rangle \left\langle \vec{\Omega}_0 \right| e^{-\beta H} \right\}$$

To calculate the trace we use the decomposition,

$$\begin{aligned}\mathrm{Tr}\left\{\left|\vec{\Omega}_0\right\rangle\left\langle\vec{\Omega}_0\right|e^{-\beta H}\right\} &= \sum_{m=-s}^s \left\langle m\left|\vec{\Omega}_0\right\rangle\left\langle\vec{\Omega}_0\right|e^{-\beta H}\left|m\right\rangle = \sum_{m=-s}^s \left\langle\vec{\Omega}_0\right|e^{-\beta H}\left|m\right\rangle\left\langle m\left|\vec{\Omega}_0\right\rangle\right. \\ &= \left\langle\vec{\Omega}_0\right|e^{-\beta H}\left|\vec{\Omega}_0\right\rangle\end{aligned}$$

so that the partition function becomes,

$$\begin{aligned}Z &= \mathrm{Tr}\left\{e^{-\beta H}\right\} = \frac{2s+1}{4\pi} \int d\Omega_0 \left\langle\vec{\Omega}_0\right|e^{-\beta H}\left|\vec{\Omega}_0\right\rangle \\ &= \frac{2s+1}{4\pi} \int d\Omega_0 \left\langle\vec{\Omega}_0\right|e^{-\Delta\tau H} \dots e^{-\Delta\tau H}\left|\vec{\Omega}_0\right\rangle\end{aligned}$$

and we insert a set of $N-1$ resolutions of identity in between each factor of $e^{-\Delta\tau H}$. We only need to calculate the matrix element of each smaller piece, as we did many times before.

Alternatively, we can first write the Hamiltonian in terms of Schwinger bosons, $H(a^\dagger, a, b^\dagger, b)$, and then immediately use the already developed bosonic path integral formalism. The catch is that we must restrict to the physical subspace given by the constraint $a^\dagger a + b^\dagger b = 2s$, so that on every path is allowed.

$$Z = \int \mathcal{D}[a^*, a] \mathcal{D}[b^*, b] \delta(a^* a + b^* b - 2s) e^{-S[a^*, a, b^*, b]}$$

where the delta-function screens out the unphysical paths. The bosonic action is,

$$S[a^*, a, b^*, b] = \int_0^\beta [a^* \partial_\tau a + b^* \partial_\tau b + H(a^*, a, b^*, b)]$$

with the Hamiltonian being normal ordered in the bosons.

We now want to parametrise the path integral using the S^2 spherical variables. This is done by choosing,

$$\begin{aligned}a &= \sqrt{2s} z_1 = \sqrt{2s} \cos \frac{\theta}{2} \\ b &= \sqrt{2s} z_2 = \sqrt{2s} \sin \frac{\theta}{2} e^{i\phi}\end{aligned}$$

with the $\mathbb{CP}^1$ constraint automatically being satisfied, $|z_1|^2 + |z_2|^2 = 1$. The boundary conditions for the bosonic fields in the path integral are,

$$\begin{aligned}a(\tau=0) &= a(\tau=\beta), & b(\tau=0) &= b(\tau=\beta), \\ a^*(\tau=0) &= a^*(\tau=\beta), & b^*(\tau=0) &= b^*(\tau=\beta)\end{aligned}$$

so that the corresponding boundary conditions of the θ and ϕ variables are,

$$\begin{aligned}\theta(\tau=0) &= \theta(\tau=\beta) \mod \pi \\ \phi(\tau=0) &= \phi(\tau=\beta) \mod 2\pi\end{aligned}$$

We must further impose the condition that $\vec{\Omega}(\tau=0) = \vec{\Omega}(\tau=\beta)$. This means that upon imaginary time evolution by the small $e^{-\beta\Delta\tau}$, $\vec{\Omega}_0$ returns to its original position.

Berry phase

The Berry phase can be calculated as follows,

$$\begin{aligned}
 \int_0^\beta (a^* \partial_\tau a + b^* \partial_\tau b) &= 2s \int_0^\beta d\tau \left[\cos \frac{\theta}{2} \partial_\tau \cos \frac{\theta}{2} + \sin \frac{\theta}{2} e^{-i\phi} \partial_\tau (\sin \frac{\theta}{2} e^{i\phi}) \right] \\
 &= 2s \int_0^\beta d\tau \left[-\frac{1}{2} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \partial_\tau \theta + \sin \frac{\theta}{2} e^{-i\phi} \left(\frac{1}{2} \cos \frac{\theta}{2} e^{i\phi} \dot{\theta} + i \sin \frac{\theta}{2} e^{i\phi} \dot{\phi} \right) \right] \\
 &= 2s \int_0^\beta d\tau i \sin^2 \frac{\theta}{2} \dot{\phi} \\
 &= is \int_0^\beta d\tau (1 - \cos \theta) \dot{\phi}
 \end{aligned}$$

The first comment in place is that this term is purely imaginary, since ϕ and θ are real variables! We can give it a geometrical interpretation. We place the vector $\vec{\Omega}(\tau)$ and imagine its position on the Bloch sphere after time $\delta\tau$, $\vec{\Omega}(\tau + \delta\tau)$. Consider moving the vector $\vec{\Omega}(\tau = 0)$ from $\theta = 0$ to $\theta(\tau)$ and from $\phi(\tau)$ to $\phi(\tau + \Delta\tau)$. The integral,

$$\int_0^\theta \sin \theta' d\theta' \int_{\phi(\tau)}^{\phi(\tau+\delta\tau)} d\phi' = (1 - \cos \theta) \dot{\phi} \delta\tau$$

is the area under the path traced by $\vec{\Omega}$.

Suppose that the vector $\vec{\Omega}(\tau = 0)$ is the initial position of the vector, and after the path integral for a time $\Delta\tau = \beta$ it returns to its initial position. This means that calculating,

$$\int_0^\beta (1 - \cos \theta) \dot{\phi} d\tau$$

will give us the area of the upper (or lower) part of the solid angle of the sphere. The area of the sphere is 4π so that,

$$e^{is4\pi} = +1$$

for all $s = 0, \frac{1}{2}, 1, \dots$. This is because the area is only defined module 4π . This means that the Berry phase is always $+1$ for spins.

16.4 Classical Limit

The classical limit of the problem corresponds to the limit $s \rightarrow \infty$. This follows from the fact that the commutator between two components of the spin,

$$\left[\frac{1}{s} S^\alpha, \frac{1}{s} S^\beta \right] = i\hbar \epsilon_{\alpha\beta\gamma} \frac{S^\gamma}{s} \frac{1}{s} \rightarrow 0$$

so that the commutator vanishes $[S^\alpha, S^\beta] \rightarrow 0$. In the path integral formalism, and restoring the correct factors of $\hbar$,

$$Z = \int \mathcal{D}\vec{\Omega} e^{-S/\hbar}$$

However, the Berry phase is a purely quantum mechanical term, which would cause a lot of interference effects in the large s limit. Therefore, it cannot simply be ignored.

17 Lecture 17: Haldane's conjecture

Consider the Heisenberg Hamiltonian for a one-dimensional chain,

$$H = J \sum_{j=1}^N \vec{S}_j \cdot \vec{S}_{j+1}$$

for general spin S , an antiferromagnetic coupling $J > 0$, an even number of sites N (with PBC) and lattice spacing a . The physics of the chain crucially depends on the value of the spin S .

- When $S = \frac{1}{2}$ the Hamiltonian is exactly solvable. The solution was first found by Hans Bethe, via the Bethe ansatz method. The drawback of the method is that it cannot calculate correlation functions. Based on bosonisation methods (Luttinger liquid phenomenology) we can get the correlation functions. Altogether, the two methods can be combined to conclude that the chain is gapless.
- When $S = 1$, Affleck-Kennedy-Lieb and Tasaki have perturbed the Hamiltonian by including a biquadratic term,

$$H = J \sum_{j=1}^N (\vec{S}_j \cdot \vec{S}_{j+1}) - J' \sum_{j=1}^N (\vec{S}_j \cdot \vec{S}_{j+1})^2$$

The Hamiltonian is now $SO(3)$ symmetric (for $S = 1$, or integer in general) and $\mathbb{Z}_2$ time-reversal symmetric. By Bethe Ansatz, one can show that it is also gapless. This led people to believe that the system is gapless for all s , not just integer. A few more words about this case. The system is frustration-free, and we can write down a unique ground state. Let $(J, J') = (\cos \theta, \sin \theta)$,

- $\theta \in (\frac{3\pi}{4}, \frac{3\pi}{2})$ the chain is a ferromagnet
- $\theta \in (-\frac{\pi}{2}, -\frac{\pi}{4})$ the chain is in a trimerised phase
- $\theta \in (-\frac{\pi}{4}, \frac{\pi}{4})$ the chain is in the Haldane phase, possessing topological order.
- $\theta \in (\frac{\pi}{4}, \frac{3\pi}{4})$ the chain is in a dimerised phase
- Haldane conjectured that the excitation spectrum of the antiferromagnetic Heisenberg chain is gapless for half-integer spins and gapped for integer spins. This is now rigorously proven for $S = 1$, in which case there is topological order (not characterised by a local order parameter) on top of the gapped spectrum.

17.1 Coherent state path integral for many spins

We briefly generalise the path integral formulation developed in the previous section for the case of many spins. The coherent state is now given by,

$$|\vec{\Omega}\rangle = \prod_{j=1}^N |\Omega_j\rangle$$

The partition function is,

$$Z = \text{Tr} \left\{ e^{-\beta H} \right\} = \int \mathcal{D}\vec{\Omega} e^{-A[\vec{\Omega}]}$$

with,

$$A[\vec{\Omega}] = \sum_{j=1}^N i s \int_0^\beta d\tau (1 - \cos \theta_j) \dot{\phi}_j + \underbrace{\int_0^\beta \sum_{j=1}^N S^2 \vec{\Omega}_j \cdot \vec{\Omega}_{j+1}}_{H(\{\vec{\Omega}_j\})}$$

where we used the fact that $\langle \vec{\Omega} | \vec{S} | \vec{\Omega} \rangle = S \vec{\Omega}$. The Berry phase, being purely quantum, complicates the situation (as compared to the 2D XY model for example, where a transfer matrix approach could be used). We will thus need to obtain an effective description by approximations.

We first need to figure out which paths contribute most to the path integrals. We can imagine doing a mean-field theory, however, this is not very helpful here. A classical ground state would be the Neel $\uparrow\downarrow\uparrow\downarrow \dots$ state, however, in $d = 1$ Mermin-Wagner prohibits the development of a true long-range order. A naive saddle point solution of a static $\vec{\Omega}$ is thus not possible.

Haldane's idea was to split the $\vec{\Omega}$ vector into fast and slow modes, by writing,

$$\vec{\Omega}_j = (-1)^j \vec{n}_j + \frac{a}{S} \vec{L}_j$$

where the first part is the Neel order parameter piece, and the second correspond to the fluctuations around the order parameter, with a chosen normalisation. The most important contributions would then come from the cases where $\vec{\Omega}_{j+1}$ is close to $-\vec{\Omega}_j$, and therefore a dominant contribution is one where all vectors are closed to a configuration with antiferromagnetically ordered vectors.

A simple mean-field solution would now proceed by taking $\vec{n}_j$ as τ and space independent. But here we treat it as a dynamical field, $\vec{n}_j(\tau)$ and take fluctuations about it. The key point now is that if $\vec{n}_j$ is slowly varying, we can take the continuum limit with $\vec{n}_j \rightarrow \vec{n}(\vec{x})$.

The constraints to be imposed are,

$$\begin{aligned} |\vec{\Omega}_j| &= 1 && \text{unit vector} \\ |\vec{n}_j| &= 1 \end{aligned}$$

This means that $\vec{L}_j$ is a fluctuating field. This imposes a further constraint on $\vec{L}_j$,

$$\begin{aligned} |\vec{\Omega}_j|^2 &= \left| (-1)^j \vec{n}_j + \frac{a}{S} \vec{L}_j \right|^2 \\ &= 1 + 2 \frac{a}{S} (-1)^j \vec{n}_j \cdot \vec{L}_j + \frac{a^2}{S^2} |\vec{L}_j|^2 \\ &= 1 + \mathcal{O}(a^2/S^2) \end{aligned}$$

so that we must have $\vec{n}_j \cdot \vec{L}_j = 0$, so that fluctuations are orthogonal to the order parameter field. The prefactor in the last term means that equality holds strictly in the large S , $S \rightarrow \infty$ limit, and is therefore a semi-classical approximation.

This allows us now to perform the path integral by calculating an effective theory,

$$Z = \int \mathcal{D}\vec{n} \mathcal{D}\vec{L} \delta(\vec{n}_j \cdot \vec{L}_j) e^{-A[\vec{n}, \vec{L}]} \equiv \int \mathcal{D}\vec{n} e^{-A_{eff}[\vec{n}; \vec{L}]}$$

and the Hamiltonian decomposes in,

$$\begin{aligned} H(\vec{\Omega}) &= JS^2 \sum_{j=1}^N \vec{\Omega}_j \cdot \vec{\Omega}_{j+1} \\ &= -JS^2 \sum_{j=1}^N \vec{n}_j \cdot \vec{n}_{j+1} + JSa \sum_{j=1}^N (-1)^j \left(\vec{n}_j \cdot \vec{L}_{j+1} - \vec{n}_{j+1} \cdot \vec{L}_j \right) + Ja^2 \sum_{j=1}^N \vec{L}_j \cdot \vec{L}_{j+1} \end{aligned}$$

To go to the continuum limit, we write $x = ja$ so that

$$\vec{n}(x+a) \approx \vec{n}(x) + \frac{\partial \vec{n}(x)}{\partial x} a + \frac{1}{2} \frac{\partial^2 \vec{n}(x)}{\partial x^2} a^2$$

Order by order we have,

- 0th order: Since $\vec{n}_j \cdot \vec{n}_j = (-1)$, this is the classical ground state energy.
- 1st order: vanishes from the fact that $|\vec{n}_j|^2 = 1$, which implies $(\partial_x \vec{n}) \cdot \vec{n} = 0$.
- 2nd order: the most important term

The second term of the Hamiltonian simplifies as follows,

$$\begin{aligned} (-1)^j (\vec{n}_j \cdot \vec{L}_{j+1} - \vec{n}_{j+1} \cdot \vec{L}_j) &\rightarrow (-1)^j (\vec{n}_j \cdot \vec{L}_{j+1} + \vec{n}_j \cdot \vec{L}_{j-1}) \\ &= (-1)^j \vec{n}_j \cdot (\vec{L}_{j+1} + \vec{L}_{j-1}) \\ &\approx (-1)^j \vec{n}_j \cdot \left(2\vec{L}_j + 2 \cdot \frac{1}{2} \frac{\partial^2 \vec{L}_j}{\partial x^2} a^2 + \mathcal{O}(a^4) \right) \\ &= (-1)^j \vec{n}_j \cdot \frac{\partial^2 \vec{L}_j}{\partial x^2} a^2 \end{aligned}$$

The Hamiltonian's most dominant terms become,

$$H(\vec{\Omega}) = -JS^2 a^2 \sum_{j=1}^N \vec{n}_j \cdot \frac{1}{2} \frac{\partial^2 \vec{n}(x)}{\partial x^2} + Ja^2 \sum_{j=1}^N \vec{L}_j \cdot \vec{L}_{j+1}$$

The first can be further simplified by invoking the expression $\partial_x \left(\vec{n}_j \cdot \frac{\partial \vec{n}(x)}{\partial x} \right) - (\partial_x \vec{n}_j)(\partial_x \vec{n}_j) = -(\partial_x \vec{n}_j)^2$ and ignore the first term since we assume PBC. Finally, with $L = Na$, we convert the lattice sum to a continuous integral by $a \sum_{j=1}^N \rightarrow \int_0^L dx$, so that the Hamiltonian is,

$$H(\vec{n}, \vec{L}) = \frac{1}{2} JS^2 a^2 \int_0^L dx (\partial_x \vec{n})^2 + Ja \int_0^L dx |\vec{L}|^2$$

What remains to complete the above calculation is the Berry phase. For a single spin, the Berry phase is $\omega(\vec{\Omega}) = \int_0^\beta d\tau (1 - \cos \theta) \dot{\phi}$. We again rewrite the $\vec{\Omega}$ field in terms of the slowly varying staggered field and a fluctuation field, $\vec{\Omega}_j = (-1)^j \vec{n}_j + \frac{a}{S} \vec{L}_j$, and expand $\omega(\{\vec{\Omega}\})$ as a Taylor series in $\{\vec{n}_j\}$.

$$iS \sum_{j=1}^N \omega(\{\vec{\Omega}\}) = iS \sum_{j=1}^N \left[\omega(\{(-1)^j \vec{n}_j\}) + \int_0^\beta d\tau \frac{\delta \omega(\{\vec{\Omega}\})}{\delta \vec{\Omega}_j} \bigg|_{\vec{\Omega}_j = (-1)^j \vec{n}_j} \cdot \frac{a}{S} \vec{L}_j + \dots \right]$$

A lengthy calculation shows that in fact the first order term can be simplified to,

$$\frac{\delta \omega(\{\vec{\Omega}\})}{\delta \vec{\Omega}_j} \bigg|_{\vec{\Omega}_j = (-1)^j \vec{n}_j} = (\partial_\tau \vec{\Omega}_j) \times \vec{\Omega}_j$$

So that the Berry phase term is now,

$$iS \sum_{j=1}^N \omega(\{\vec{\Omega}\}) = iS \sum_{j=1}^N \left[(-1)^j \omega(\{\vec{n}_j\}) + \int_0^\beta d\tau (\partial_\tau \vec{\Omega}_j) \times \vec{\Omega}_j \cdot \frac{a}{S} \vec{L}_j + \dots \right]$$

and by going to the continuum space limit,

$$iS \omega(\{\vec{\Omega}\}) = iS \sum_{j=1}^N (-1)^j \omega(\{\vec{n}_j\}) + i \int_0^\beta d\tau \int_0^L dx (\partial_\tau \vec{n} \times \vec{n}) \cdot \vec{L}$$

At this point, we can integrate out the $\vec{L}$ modes by completing the square in $\vec{L}$ and performing a Gaussian integration. The partition function is,

$$Z = \int \mathcal{D}\vec{n} \mathcal{D}\vec{L} \delta(\vec{n}_j \cdot \vec{L}_j) e^{-A[\vec{n}, \vec{L}]} \equiv \int \mathcal{D}\vec{n} e^{-A_{eff}[\vec{n}, \vec{L}]}$$

where now,

$$\begin{aligned} A[\vec{n}, \vec{L}] &= iS \sum_j (-1)^j \omega(\{\vec{n}_j\}) + i \int d\tau \int dx (\partial_\tau \vec{n} \times \vec{n}) \cdot \vec{L} \\ &\quad + \int_0^\beta d\tau \int_0^L dx \frac{1}{2} JS^2 a^2 \left[(\partial_x \vec{n})^2 + Ja \int_0^L dx |\vec{L}|^2 \right] \end{aligned}$$

The easiest way to do is to complete the square so that,

$$\begin{aligned} A[\vec{n}, \vec{L}] &= iS \sum_j (-1)^j \omega(\{\vec{n}_j\}) + \int_0^\beta d\tau \int_0^L dx \frac{1}{2} JS^2 a^2 (\partial_x \vec{n})^2 \\ &\quad + \int_0^\beta d\tau \int_0^L dx \left\{ Ja \left[\vec{L} + \frac{i}{2Ja} (\partial_\tau \vec{n} \times \vec{n}) \right]^2 + \frac{1}{4Ja} (\partial_\tau \vec{n} \times \vec{n})^2 \right\} \end{aligned}$$

We can shift the variable by writing,

$$\vec{L}' = \vec{L} + \frac{i}{2Ja} (\partial_\tau \vec{n} \times \vec{n})$$

which satisfies automatically the constraint, $\vec{L}' \cdot \vec{n} = 0$. We can then perform the Gaussian integral over $\vec{L}'$. In addition, the new term that arises simplifies via $(\partial_\tau \vec{n} \times \vec{n})^2 = (\partial_\tau \vec{n})^2$. The effective action thus becomes,

$$A_{eff}[\vec{n}, \vec{L}] = iS \sum_j (-1)^j \omega(\{\vec{n}_j\}) + \int d\tau \int dx \left[\frac{1}{2} JS^2 a^2 (\partial_x \vec{n})^2 + \frac{1}{4Ja^2} (\partial_\tau \vec{n})^2 \right]$$

The coordinates can be rescaled such that the 'spin-wave velocity' is 1. Thus we let $x_0 = c\tau$ and $x_1 = x$, so that,

$$A_{eff}[\vec{n}] = iS \sum_j (-1)^j \omega(\{\vec{n}_j\}) + \frac{1}{g} \int_0^{\beta c} dx_0 \int dx_1^L (\partial_\mu \vec{n})^2$$

with $c = \sqrt{2}JSa$ and coupling strength $g = \frac{2\sqrt{2}}{s}$. We can also transform the first term in the continuum as follows,

$$\begin{aligned}
iS \sum_j (-1)^j \omega(\{\vec{n}_j\}) &= \frac{iS}{2} \sum_{j=1}^N [\omega(\{\vec{n}_{2j}\}) - \omega(\{\vec{n}_{2j-1}\})] \\
&= \frac{iS}{2} \sum_{j=1}^N \int_0^\beta d\tau \underbrace{\frac{\delta\omega(\{\vec{n}\})}{\delta\vec{n}}}_{\partial_\tau \vec{n} \times \vec{n}} \bigg|_{\vec{n}=\vec{n}_{2j}} \cdot \underbrace{(\vec{n}_{2j} - \vec{n}_{2j-1})}_{a\partial_x \vec{n}} \\
&= \frac{iS}{2} \int_0^\beta d\tau \int_0^L dx (\partial_\tau \vec{n} \times \vec{n}) \cdot (\partial_x \vec{n}) \\
&= -\frac{iS}{2} \int_0^\beta d\tau \int_0^L dx \vec{n} \cdot (\partial_\tau \vec{n} \times \partial_x \vec{n}) \\
&= -\frac{iS}{2} \int_0^{\beta c} dx_0 \int_0^L dx_1 \vec{n} \cdot (\partial_{x_0} \vec{n} \times \partial_{x_1} \vec{n}) \\
&= -2\pi i S \mathcal{Q}(\vec{n})
\end{aligned}$$

where,

$$\mathcal{Q}(\vec{n}) = \frac{1}{4\pi} \int_0^{\beta c} dx_0 \int_0^L dx_1 \vec{n} \cdot (\partial_{x_0} \vec{n} \times \partial_{x_1} \vec{n}) \in \mathbb{Z}$$

This shows that the Berry phase is now a 4π integer, and the overall term is simplified as follows,

$$e^{-\frac{i}{2}4\pi\mathcal{Q}(\vec{n})} = e^{-2\pi i\mathcal{Q}(\vec{n})} = \begin{cases} +1 & S \in \mathbb{Z} \\ (-1)^{\mathcal{Q}(\vec{n})} & S \in \frac{1}{2}\mathbb{Z} \end{cases}$$

Concluding Remarks

We can now go back to the Haldane conjecture that states that half-integer spin Heisenberg antiferromagnetic chains are gapless, and integer spin ones are gapped. The mapping between the original 1d antiferromagnetic chain to a classical $O(3)$ non-linear sigma model with a Berry phase term is able to explain this. Let us take the two cases:

- Spin S is an integer: In this case, the effective action is simply,

$$A_{eff}[\vec{n}] = \frac{1}{g} \int_0^{\beta c} dx_0 \int dx_0^L (\partial_\mu \vec{n})^2$$

which the $O(3)$ NLSM. This is known to have a mass gap with exponentially decaying correlation functions,

$$\langle \vec{n}(\vec{r}) \cdot \vec{n}(\vec{r}') \rangle \sim e^{-|\vec{r}-\vec{r}'|/\xi}$$

with $\xi \sim 1/\Delta$ and Δ the size of the gap.

- Spin S is a half-integer: In this case we have a classical $O(3)$ NLSM and a $\theta = \pi$ topological term. The idea here is that under RG flow (making S smaller), the topological term doesn't change while the coupling strength g increases (see Fradkin). It is therefore sufficient to understand the spin-1/2 chain and then generalise to higher half-integer spins. The former is integrable and

known to have gapless spinon excitations. This is however still a conjecture. The absence of analytical and numerical methods to access larger spins does not allow one to obtain reliable conclusions.

Note that Affleck gives an alternative explanation in PRL 56-408 (1986) by considering the effect of vortices upon adding an anisotropic mass term in the NLSM.

18 Lecture 18: AKLT Model

Having analysed the Heisenberg Hamiltonian for $S = 1/2$, a valid question to ask is what is the ground state and possible phases of the $S = 1$ Hamiltonian,

$$H = \sum_j \vec{S}_j \cdot \vec{S}_{j+1}$$

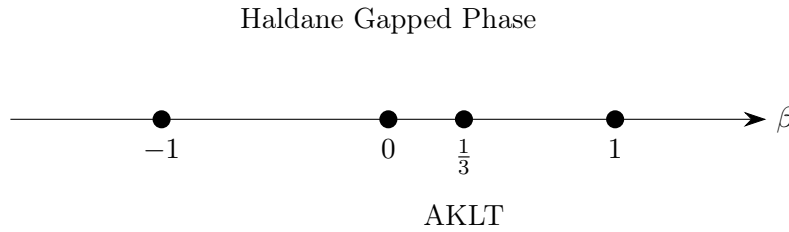
in one dimension. For classical spins, an antiferromagnetic interaction leads to a Neel state with ordering. For quantum spins, Mermin-Wagner prohibits ordering at $T = 0$ and long range order is reduced to power law correlation functions. The $S = 1/2$ is gapless, and it was thought that the same holds for $S = 1$. However, Haldane proved that it is gapped and moreover it supports symmetry protected topological order.

We consider here the most general $SU(2)$ invariant Hamiltonian for spin-1,

$$H = \sum_{j=1}^N \left[\vec{S}_j \cdot \vec{S}_{j+1} + \beta \left(\vec{S}_j \cdot \vec{S}_{j+1} \right)^2 \right]$$

in one spatial dimension with PBC (and we will later comment on OBC). What we will show below is that the addition of the second quadratic term with $\beta = 1/3$ makes the term identical to a projection to the total spin 2 sector for the two spin 1's at site j and $j + 1$.

We already know that for $\beta = 0$, H reduces to the Heisenberg model. The phase diagram of H for varying β is as follows,



At $\beta = -1$ the state is given by a product of dimers on adjacent sites,



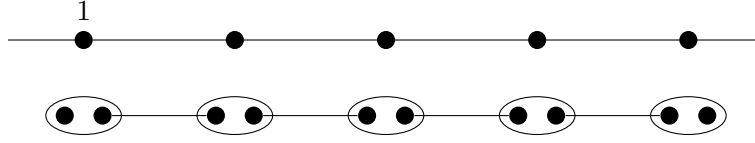
To solve the model, we can use the Schwinger boson representation for $S = 1$, so that the three different spin one basis states are,

$$\begin{aligned} |1\rangle &= \frac{1}{\sqrt{2}} \left(a^\dagger \right)^2 |\text{vac}\rangle \rightarrow |\uparrow\uparrow\rangle \\ |0\rangle &= a^\dagger b^\dagger |\text{vac}\rangle \rightarrow |\uparrow\downarrow\rangle \\ |-1\rangle &= \frac{1}{\sqrt{2}} \left(b^\dagger \right)^2 |\text{vac}\rangle \rightarrow |\downarrow\downarrow\rangle \end{aligned}$$

with the constraint, $a^\dagger a + b^\dagger b = 2$. A valence bond of two spin 1/2 at adjacent sites is then given by,

$$\begin{aligned} |VB\rangle_{j,j+1} &= \frac{1}{\sqrt{2}} \left(|\uparrow\rangle_j |\downarrow\rangle_{j+1} - |\downarrow\rangle_j |\uparrow\rangle_{j+1} \right) \\ &= \frac{1}{\sqrt{2}} \left(a_j^\dagger b_{j+1}^\dagger - b_j^\dagger a_{j+1}^\dagger \right) |\text{vac}\rangle \end{aligned}$$

The AKLT construction proceeds by considering each $S = 1$ at each site as a Schwinger boson, and therefore two spin $1/2$ s. Schematically this looks like,



with PBC. Explicitly, the AKLT state is comprised by a totally symmetrised state of two $s = 1/2$, and adjacent states are connected via a singlet. The state,

$$|\Psi\rangle = \prod_{j=1}^N \left(a_j^\dagger b_{j+1}^\dagger - b_j^\dagger a_{j+1}^\dagger \right) |\text{vac}\rangle$$

with the constraint $a_j^\dagger a_j + b_j^\dagger b_j = 2$ for all j includes no unphysical components.

18.1 Parent Hamiltonian

In the field of many-body physics, given a Hamiltonian H , the main task is to determine its ground state. Here we take the opposite approach: given a ground state of the form of the AKLT wavefunction, what is the Hamiltonian that produces it?

To answer the question we consider two spin 1 s. The total angular momentum of the two spins is given by the decomposition,

$$\mathbf{1} \otimes \mathbf{1} = \mathbf{0} \oplus \mathbf{1} \oplus \mathbf{2}$$

giving a total of 9 states. On the other hand, the total angular momentum of two spin $\frac{1}{2}$ is given by $\frac{1}{2} \otimes \frac{1}{2} = \mathbf{0} \oplus \mathbf{1}$. Finally, consider four spin $\frac{1}{2}$ s, and choose the total spin 0, $\mathbf{0}$, representation for the middle two,

$$\frac{1}{2} \otimes \underbrace{\frac{1}{2} \otimes \frac{1}{2}}_{\mathbf{0}} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{1}{2} \otimes \mathbf{0} \otimes \frac{1}{2} = \mathbf{0} \oplus \mathbf{1}$$

This means that by enforcing the virtual $S = 0$ spin singlet in the ground state wavefunction, we eliminate the $S = 2$ state.

This means that the ground state is such that, using the projector onto the $S = 2$ subspace be $\hat{P}_{S_{tot}=2}(j, j+1)$,

$$\hat{P}_{S_{tot}=2}(j, j+1) |\psi\rangle = 0$$

In general, the projector is a positive semi-definite operator, $\hat{P}_{S_{tot}=2}(j, j+1) \geq 0$. Writing,

$$H = \sum_j \hat{P}_{S_{tot}=2}(j, j+1) \implies H |\psi\rangle = 0$$

which shows that $|\psi\rangle$ is indeed a ground state, as the lowest energy state! This is essentially the proof of existence of such a Hamiltonian. Proving uniqueness requires some extra work.

We now find expressions for the projectors onto the subspaces, in terms of the interaction term $\vec{S}_j \cdot \vec{S}_{j+1}$. The starting point is,

$$\hat{P}_{S_{tot}=0} + \hat{P}_{S_{tot}=1} + \hat{P}_{S_{tot}=2} = \mathbf{I}$$

which holds always. Further we consider the interaction operator itself and decompose it into the orthogonal projector subspaces,

$$\begin{aligned}
\vec{S}_j \cdot \vec{S}_{j+1} &= \frac{1}{2} \left[(\vec{S}_j + \vec{S}_{j+1})^2 - \vec{S}_j^2 - \vec{S}_{j+1}^2 \right] \\
&= \frac{1}{2} \left[(\vec{S}_j + \vec{S}_{j+1})^2 - 1(1+1) - 1(1+1) \right] \\
&= \frac{1}{2} \left[(\vec{S}_j + \vec{S}_{j+1})^2 - 4 \right] \\
&= \sum_{S_{tot}=0,1,2} \frac{1}{2} [S_{tot}(S_{tot}+1) - 4] \hat{P}_{S_{tot}} \\
&= \frac{1}{2} [0(0+1) - 4] \hat{P}_{S_{tot}=0} + \frac{1}{2} [1(1+1) - 4] \hat{P}_{S_{tot}=1} + \frac{1}{2} [2(2+1) - 4] \hat{P}_{S_{tot}=2} \\
&= -2\hat{P}_{S_{tot}=0} - \hat{P}_{S_{tot}=1} + \hat{P}_{S_{tot}=2}
\end{aligned}$$

If we square the above expression, we get yet another equation,

$$\begin{aligned}
(\vec{S}_j \cdot \vec{S}_{j+1})^2 &= \left(-2\hat{P}_{S_{tot}=0} - \hat{P}_{S_{tot}=1} + \hat{P}_{S_{tot}=2} \right)^2 \\
&= 4\hat{P}_{S_{tot}=0} + \hat{P}_{S_{tot}=1} + \hat{P}_{S_{tot}=2}
\end{aligned}$$

Thus, we have three equations for the three projectors $\{\hat{P}_{S_{tot}=0}, \hat{P}_{S_{tot}=1}, \hat{P}_{S_{tot}=2}\}$, which we solve as follows,

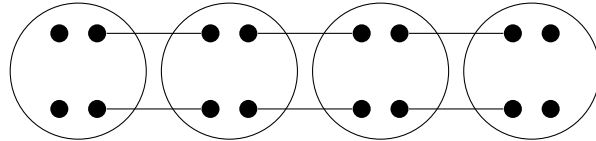
$$\begin{aligned}
\hat{P}_{S_{tot}=0} &= \frac{1}{3}(\vec{S}_j \cdot \vec{S}_{j+1})^2 - \frac{1}{3} \\
\hat{P}_{S_{tot}=1} &= -\frac{1}{2}(\vec{S}_j \cdot \vec{S}_{j+1}) - \frac{1}{2}(\vec{S}_j \cdot \vec{S}_{j+1})^2 + 1 \\
\hat{P}_{S_{tot}=2} &= \frac{1}{2}(\vec{S}_j \cdot \vec{S}_{j+1}) + \frac{1}{6}(\vec{S}_j \cdot \vec{S}_{j+1})^2 + \frac{1}{3}
\end{aligned}$$

In particular, the expression of $\hat{P}_{S_{tot}=2}$ completes the construction of the spin-1 AKLT Hamiltonian,

$$H_{\text{AKLT}} = \sum_{j=1}^N \left[\frac{1}{2}(\vec{S}_j \cdot \vec{S}_{j+1}) + \frac{1}{6}(\vec{S}_j \cdot \vec{S}_{j+1})^2 + \frac{1}{3} \right]$$

Generalization of the 1D AKLT Hamiltonian

It is possible to construct parent Hamiltonians for higher spin- S chains. For $S = 2$ for example,



The ground state, in Schwinger fermion representation, is given by,

$$|\psi\rangle = \prod_{j=1}^N \left(a_j^\dagger b_{j+1}^\dagger - b_j^\dagger a_{j+1}^\dagger \right) |\text{vac}\rangle$$

and the corresponding parent Hamiltonian is,

$$H = \sum_j \left[J_3 \hat{P}_{S_t=3}(j, j+1) + J_4 \hat{P}_{S_t=4}(j, j+1) \right]$$

with $J_3, J_4 > 0$.

It's also possible to construct a spin-3/2 AKLT ground state on the honeycomb lattice,

$$|\psi\rangle = \prod_{\langle ij \rangle} \left(a_i^\dagger b_j^\dagger - b_i^\dagger a_j^\dagger \right) |\text{vac}\rangle$$

and parent Hamiltonian $H = \sum_{\langle ij \rangle} \hat{P}_{S_t=3}(i, j)$.

18.2 Matrix Product Form of spin-1 AKLT State

Consider the ground state of the AKLT Hamiltonian,

$$\begin{aligned} |\psi\rangle &= \prod_{j=1}^N \left(a_j^\dagger b_{j+1}^\dagger - b_j^\dagger a_{j+1}^\dagger \right) |\text{vac}\rangle \\ &= \prod_{j=1}^N \left[\begin{pmatrix} a_j^\dagger & b_j^\dagger \end{pmatrix} \begin{pmatrix} b_{j+1}^\dagger \\ -a_{j+1}^\dagger \end{pmatrix} \right] |\text{vac}\rangle \\ &= \left(\dots \begin{pmatrix} a_{j-1}^\dagger & b_{j-1}^\dagger \end{pmatrix} \underbrace{\begin{pmatrix} b_j^\dagger \\ -a_j^\dagger \end{pmatrix} \begin{pmatrix} a_j^\dagger & b_j^\dagger \end{pmatrix} \begin{pmatrix} b_{j+1}^\dagger \\ -a_{j+1}^\dagger \end{pmatrix}}_{\begin{pmatrix} b_j^\dagger a_j^\dagger & (b_j^\dagger)^2 \\ -(a_j^\dagger)^2 & -a_j^\dagger b_j^\dagger \end{pmatrix}} \begin{pmatrix} a_{j+1}^\dagger & b_{j+1}^\dagger \end{pmatrix} \dots \right) |\text{vac}\rangle \\ &= \text{Tr} \left\{ \prod_{j=1}^N \begin{pmatrix} a_j^\dagger b_j^\dagger & (b_j^\dagger)^2 \\ -(a_j^\dagger)^2 & -a_j^\dagger b_j^\dagger \end{pmatrix} |\text{vac}\rangle_j \right\} \\ &= \text{Tr}[g_1 g_2 \dots g_N] \end{aligned}$$

where, upon the identification $|1\rangle = \frac{1}{\sqrt{2}}(a^\dagger)^2 |\text{vac}\rangle$, $|0\rangle = a^\dagger b^\dagger |\text{vac}\rangle$ and $|-1\rangle = \frac{1}{\sqrt{2}}(b^\dagger)^2 |\text{vac}\rangle$,

$$g_j = \begin{pmatrix} a_j^\dagger b_j^\dagger |\text{vac}\rangle_j & (b_j^\dagger)^2 |\text{vac}\rangle_j \\ -(a_j^\dagger)^2 |\text{vac}\rangle_j & -a_j^\dagger b_j^\dagger |\text{vac}\rangle_j \end{pmatrix} = \begin{pmatrix} |0\rangle_j & \sqrt{2}|-1\rangle_j \\ -\sqrt{2}|-1\rangle_j & -|0\rangle_j \end{pmatrix}$$

Consider the product of two consecutive g 's inside the trace,

$$\begin{aligned} g_j \cdot g_{j+1} &= \begin{pmatrix} |0\rangle_j & \sqrt{2}|-1\rangle_j \\ -\sqrt{2}|-1\rangle_j & -|0\rangle_j \end{pmatrix} \begin{pmatrix} |0\rangle_{j+1} & \sqrt{2}|-1\rangle_{j+1} \\ -\sqrt{2}|-1\rangle_{j+1} & -|0\rangle_{j+1} \end{pmatrix} \\ &= \begin{pmatrix} |0\rangle_j |0\rangle_{j+1} - 2|-1\rangle_j |1\rangle_{j+1} & \sqrt{2}|0\rangle_j |-1\rangle_{j+1} - \sqrt{2}|-1\rangle_j |0\rangle_{j+1} \\ -\sqrt{2}|1\rangle_j |0\rangle_{j+1} + \sqrt{2}|0\rangle_j |1\rangle_{j+1} & -2|1\rangle_j |-1\rangle_{j+1} + |0\rangle_j |0\rangle_{j+1} \end{pmatrix} \end{aligned}$$

which confirms that such a ground state has support only on the four states of the $\mathbf{0}$ and $\mathbf{1}$ subspaces of $\mathbf{1} \otimes \mathbf{1}$.

Let's focus on a chain with open boundary conditions. Notice that this means that the first and last spin-1/2 degrees of freedom will behave as edge states, leading to a four-fold degenerate ground state! The ground state will be,

$$|\psi\rangle = a_1^\dagger \prod_{j=1}^{N-1} \left(a_j^\dagger b_{j+1}^\dagger - b_j^\dagger a_{j+1}^\dagger \right) a_N^\dagger |\text{vac}\rangle$$

or equivalently with $b_1^\dagger, b_N^\dagger$ replacing the first and last operators. Let,

$$|\psi\rangle_{\alpha\beta} = (g_1 g_2 \dots g_N)_{\alpha\beta}, \quad \alpha, \beta = 1, 2$$

Observable Calculation using MPS

We can rewrite the matrix g_j in terms of a tensor $A^{[s_j]}$ as follows,

$$g_j = \begin{pmatrix} |0\rangle_j & \sqrt{2}|-1\rangle_j \\ -\sqrt{2}|-1\rangle_j & -|0\rangle_j \end{pmatrix} = \sum_{s_j=0,\pm 1} A^{[s_j]} |s_j\rangle$$

with,

$$\begin{aligned} A^{[s=+1]} &= \begin{pmatrix} 0 & 0 \\ -\sqrt{2} & 0 \end{pmatrix} = -\sqrt{2}\sigma^- \\ A^{[s=0]} &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma^z \\ A^{[s=-1]} &= \begin{pmatrix} 0 & \sqrt{2} \\ 0 & 0 \end{pmatrix} = \sqrt{2}\sigma^+ \end{aligned}$$

so that the ground state takes the form,

$$|\psi\rangle = \sum_{s_1, \dots, s_N=0,\pm 1} \text{Tr} \left(A^{[s_1]} A^{[s_2]} \dots A^{[s_N]} \right) |s_1 \dots s_N\rangle$$

The norm is then easily calculated,

$$\begin{aligned} \langle \psi | \psi \rangle &= \sum_{s_1, \dots, s_N=0,\pm 1} \text{Tr} \left(\bar{A}^{[s_1]} \dots \bar{A}^{[s_N]} \right) \left(A^{[s_1]} \dots A^{[s_N]} \right) \\ &= \sum_{s_1, \dots, s_N=0,\pm 1} \text{Tr} \left((\bar{A}^{[s_1]} \otimes A^{[s_1]}) \dots (\bar{A}^{[s_N]} \otimes A^{[s_N]}) \right) \\ &= \text{Tr}(T^N) \end{aligned}$$

where $T = \sum_{s_j=0,\pm 1} \bar{A}^{[s_j]} \otimes A^{[s_j]}$. There is natural way to graphically represent this operation using *tensor networks*.

One may also calculate (exercise) the correlation function and show that they decay exponentially,

$$\lim_{N \rightarrow \infty} \langle S_i^z S_j^z \rangle \sim e^{-|j-i|/\xi}$$

with $\xi = 1/\ln 3$, indicating a gapped phase. Furthermore, there is hidden string order, i.e. the existence of a string operator that attains a finite expectation value in the ground state...

The idea now is to rewrite the Heisenberg interaction in terms of projector operators. A projector $\hat{P}^{(s)}$ onto the spin subspace s is defined by its action: its eigenvalue is one if the many spin state it acts on has total spin s and 0 otherwise. For a collection of N spins, the projector is built by taking the product $(\sum_i^N \vec{S}_i) \cdot (\sum_i^N \vec{S}_i) - N(N+1)$. For example for two spin $1/2$,

$$\begin{aligned}\hat{P}^{(0)}(\vec{S}_1 + \vec{S}_2) &= \frac{1}{2} \left((\vec{S}_1 + \vec{S}_2) \cdot (\vec{S}_1 + \vec{S}_2) - 0 \right) = \frac{1}{4} - \vec{S}_1 \cdot \vec{S}_2 \\ \hat{P}^{(1)}(\vec{S}_1 + \vec{S}_2) &= \frac{1}{2} \left((\vec{S}_1 + \vec{S}_2) \cdot (\vec{S}_1 + \vec{S}_2) - 2 \right) = \frac{3}{4} + \vec{S}_1 \cdot \vec{S}_2 \\ \implies \mathbf{I} &= \hat{P}^{(0)} + \hat{P}^{(1)}\end{aligned}$$

19 Lecture 19: Parton Construction

In this lecture we will develop the Parton construction for quantum states of the spin-1/2 Heisenberg antiferromagnet,

$$H = J \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j$$

One motivation for doing this is that, the classical Neel antiferromagnetic state is **not** a ground state of the Hamiltonian. One way to see this is by considering even two spins,

$$H = J \vec{S}_1 \cdot \vec{S}_2$$

for which we know that the ground state is the singlet state, and not the $|\uparrow\downarrow\rangle$. In general it is very hard to find the exact ground state on a specific lattice and we often resort to numerical techniques (DMRG or exact diagonalization for example).

19.1 Abrikosov Representation

The starting point is to represent the spins using (Abrikosov) fermions,

$$|\uparrow\rangle = f_{\uparrow}^{\dagger} |0\rangle \quad |\downarrow\rangle = f_{\downarrow}^{\dagger} |0\rangle$$

which has the disadvantage that the 2-dimensional Hilbert space of the SU(2) spins, is now replaced by the four-dimensional Hilbert space of the fermions. To solve this problem, we enforce the constraint $\hat{Q} = \sum_{\alpha} f_{\alpha}^{\dagger} f_{\alpha} = 1$, so that all eigenstates of H will have to satisfy $\hat{Q} |\psi\rangle = |\psi\rangle$. In terms of fermions, the spin operators become,

$$S^+ = f_{\uparrow}^{\dagger} f_{\downarrow} \quad S^- = f_{\downarrow}^{\dagger} f_{\uparrow} \quad S^z = (f_{\uparrow}^{\dagger} f_{\uparrow} - f_{\downarrow}^{\dagger} f_{\downarrow})$$

and in general $\vec{S}_j = \frac{1}{2} \sum_{\alpha\beta} f_{j\alpha}^{\dagger} \vec{\sigma}_{\alpha\beta} f_{j\beta}$.

To express the Heisenberg interactions in terms of the Abrikosov fermions, we will make use of the SU(2) completeness relation,

$$\vec{\sigma}_{\alpha\beta} \cdot \vec{\sigma}_{\gamma\delta} = 2\delta_{\alpha\delta}\delta_{\gamma\beta} - \delta_{\alpha\beta}\delta_{\gamma\delta}$$

so that,

$$\begin{aligned} \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} (f_{i\alpha}^{\dagger} \vec{\sigma}_{\alpha\beta} f_{i\beta}) \cdot (f_{j\alpha}^{\dagger} \vec{\sigma}_{\gamma\delta} f_{j\delta}) \\ &= \frac{1}{4} (f_{i\alpha}^{\dagger} f_{i\beta}) (f_{j\gamma}^{\dagger} f_{j\delta}) [2\delta_{\alpha\delta}\delta_{\gamma\beta} - \delta_{\alpha\beta}\delta_{\gamma\delta}] \\ &= \frac{1}{4} [2(f_{i\alpha}^{\dagger} f_{i\beta})(f_{j\beta}^{\dagger} f_{j\alpha}) - (f_{i\alpha}^{\dagger} f_{i\alpha})(f_{j\gamma}^{\dagger} f_{j\gamma})] \\ &= \frac{1}{4} [2(f_{i\alpha}^{\dagger} f_{i\beta})(\delta_{\alpha\beta} - f_{j\alpha} f_{j\beta}^{\dagger}) - (f_{i\alpha}^{\dagger} f_{i\alpha})(f_{j\gamma}^{\dagger} f_{j\gamma})] \\ &= \frac{1}{4} [2(f_{i\alpha}^{\dagger} f_{i\alpha}) - 2f_{i\alpha}^{\dagger} f_{j\alpha} f_{j\alpha}^{\dagger} f_{i\beta} - (f_{i\alpha}^{\dagger} f_{i\alpha})(f_{j\gamma}^{\dagger} f_{j\gamma})] \\ &= -\frac{1}{2} (f_{i\alpha}^{\dagger} f_{j\alpha}) (f_{j\beta}^{\dagger} f_{i\beta}) + \underbrace{\frac{1}{2} n_i^f - \frac{1}{4} n_i^f n_j^f}_{+\frac{1}{4}} \\ &= -\frac{1}{2} (f_{i\alpha}^{\dagger} f_{j\alpha}) (f_{j\beta}^{\dagger} f_{i\beta}) + \frac{1}{4} \\ &= -\frac{1}{4} [(f_{i\alpha}^{\dagger} f_{j\alpha}) (f_{j\beta}^{\dagger} f_{i\beta}) + \text{h.c.}] + \frac{1}{4} \end{aligned}$$

so that the Hamiltonian is,

$$H = -\frac{J}{4} \sum_{\langle ij \rangle} \left[(f_{i\alpha}^\dagger f_{j\alpha}) (f_{j\beta}^\dagger f_{i\beta}) + \text{h.c.} \right] + \text{const.}$$

where the specific way of writing is chosen for convenience. The issue now is that we have a quartic term which makes the Hamiltonian analytically not solvable. The way we make progress is therefore a suitable mean field theory.

We do this again in the path integral formalism by using a set of Grassman variables $\{f_j, \bar{f}_j\}$ and remember to enforce the constraint of single f -fermion occupancy,

$$Z = \int \mathcal{D}[\bar{f}, f] \delta \left(\sum_{\alpha} \bar{f}_{j\alpha} f_{j\alpha} - 1 \right) e^{-S[f, \bar{f}]}$$

where,

$$S[f, \bar{f}] = \int_0^\beta d\tau \left(\sum_{j\alpha} \bar{f}_{j\alpha} \partial_\tau f_{j\alpha} - \frac{J}{2} \sum_{\langle ij \rangle} \left[(f_{i\alpha}^\dagger f_{j\alpha}) (f_{j\beta}^\dagger f_{i\beta}) \right] \right)$$

To handle the constraint we use the δ -function representation as a functional integral over a new field λ , $\delta(x) = 1/(2\pi) \int_{-\infty}^{\infty} d\lambda e^{i\lambda x}$, so that the functional integral takes the form,

$$Z = \int \mathcal{D}[\bar{f}, f] \mathcal{D}\lambda e^{-S[f, \bar{f}] + i \int_0^\beta \sum_j \lambda_j (\sum_{\alpha} \bar{f}_{j\alpha} f_{j\alpha} - 1)}$$

We now do a Hubbard-Stratonovich transformation to decouple the four-fermion interaction,

$$\begin{aligned} e^{-\int_0^\beta d\tau \left(\sum_{j\alpha} \bar{f}_{j\alpha} \partial_\tau f_{j\alpha} - \frac{J}{2} \sum_{\langle ij \rangle} \left[(f_{i\alpha}^\dagger f_{j\alpha}) (f_{j\beta}^\dagger f_{i\beta}) \right] \right)} &\propto e^{-\int_0^\beta d\tau \left(\sum_{j\alpha} \bar{f}_{j\alpha} \partial_\tau f_{j\alpha} - \frac{J}{2} \sum_{\langle ij \rangle} \left[(f_{i\alpha}^\dagger f_{j\alpha}) (f_{j\beta}^\dagger f_{i\beta}) \right] \right)} \\ &\int \mathcal{D}\chi_{ij} \mathcal{D}\chi_{ij}^* e^{-\int_0^\beta d\tau \frac{J}{2} \sum_{\langle ij \rangle} (\chi_{ij}^* \sum_{\alpha} \bar{f}_{i\alpha} f_{j\alpha}) (\chi_{ij} \sum_{\beta} \bar{f}_{j\beta} f_{i\beta})} \\ &= \int \mathcal{D}\chi_{ij} \mathcal{D}\chi_{ij}^* e^{-\int_0^\beta d\tau \frac{J}{2} \sum_{\langle ij \rangle} [|\chi_{ij}|^2 - \sum_{\alpha} (\chi_{ij} \bar{f}_{i\alpha} f_{j\alpha} + \chi_{ij}^* \bar{f}_{j\alpha} f_{i\alpha})]} \end{aligned}$$

Therefore, the partition function of the theory is given by the functional integral over all these fields,

$$Z = \int \mathcal{D}\bar{f} \mathcal{D}f \mathcal{D}\lambda \mathcal{D}\chi^* \mathcal{D}\chi e^{-S[\bar{f}, f, \lambda, \chi^*, \chi]}$$

with the action,

$$\begin{aligned} S[\bar{f}, f, \lambda, \chi^*, \chi] &= \int_0^\beta d\tau \left\{ \sum_{j\alpha} \bar{f}_{j\alpha} \partial_\tau f_{j\alpha} - i \sum_j \lambda_j(\tau) \left(\sum_{\alpha} \bar{f}_{j\alpha} f_{j\alpha} - 1 \right) \right. \\ &\quad \left. - \frac{J}{2} \sum_{\langle ij \rangle} \left(\chi_{ij} \bar{f}_{i\alpha} f_{j\alpha} + \chi_{ij}^* \bar{f}_{j\alpha} f_{i\alpha} \right) \right. \\ &\quad \left. + \frac{J}{2} \sum_{\langle ij \rangle} |\chi_{ij}|^2 \right\} \end{aligned}$$

Note that whereas the original spins that we represented using fermions live on the sites of a lattice, the new fields χ, χ^* live on the links of the lattice. Furthermore, by virtue of $\chi_{ij} = -\chi_{ji}$ we note that χ_{ij} is a directed field and therefore an orientation must be chosen at every link.

Saddle Point Solution (MFT)

Up to this point, the representation still leaves the partition function exactly but its calculation requires approximations to be made. The “zeroth level” mean-field solution amounts to taking the saddle point solution of the action S , in which the newly introduced fields uniformly condense to their static saddle point solutions,

$$\chi_{ij}(\tau) \rightarrow \chi_{ij} \quad i\lambda_j(\tau) \rightarrow \lambda$$

These are in general hard to justify but there is at least one limit, namely the large- N limit of the field theory in which this is exact. For the condensation of $\lambda(\tau)$, a way to think about it is as an effective chemical potential of the f -fermions which is generally complex in the field-theory but analytically continued to real values. The mean field Hamiltonian is then,

$$H_{MF} = -\frac{J}{2} \sum_{\langle ij \rangle} \left[f_{j\alpha}^\dagger f_{i\alpha} \chi_{ij}^* + \text{h.c.} \right] + \frac{J}{2} \sum_{\langle ij \rangle} |\chi_{ij}|^2$$

and thus an effective model of spinons hopping in the background of a static $U(1)$ gauge field.

The mean-field energy, E_{MF} , is now easy to compute as a function of χ_{ij} and λ , and we know that at the saddle point, their values should minimise it,

$$\frac{\partial E_{MF}}{\partial |\chi_{ij}|^2} = 0$$

Note that the ground state, $|\psi_{MF}\rangle$, is not even a variational ansatz state since it contains unphysical states. Indeed, the constraint is only fulfilled on average,

$$\langle \psi_{MF} | \hat{Q} | \psi_{MF} \rangle = \langle \psi_{MF} | \sum_{\alpha} f_{j\alpha}^\dagger f_{j\alpha} - 1 | \psi_{MF} \rangle = 0 \implies \sum_{j=1}^N \langle \psi_{MF} | \sum_{\alpha} f_{j\alpha}^\dagger f_{j\alpha} | \psi_{MF} \rangle = N$$

The mean-field theory can be improved as follows:

1. By including fluctuations of the gauge-field into account. This is similar to the case of, for example, superfluidity where phase fluctuations gives emergent phonons at low energies. This would lead to a gauge theory $U(1)$ if no spinon pairing of the form $\Delta_{ij} = \langle f_{i\uparrow}^\dagger f_{j\downarrow}^\dagger - f_{i\downarrow}^\dagger f_{j\uparrow}^\dagger \rangle$ is included, or $\mathbb{Z}_2$ if it is included (with hidden $SU(2)$ symmetry).
2. By introducing the Gutzwiller projection operator $\hat{P}_G$ that keeps only the physical states, so that work only with projected wavefunctions,

$$|\psi_{MF}\rangle \rightarrow \hat{P}_G |\psi_{MF}\rangle$$

which can be combined with Variational Monte Carlo methods.

Let us demonstrate how the second approach works, by taking a $1D$ spin-1/2 chain as an example, with PBC. Let,

$$f_{j\alpha} = \frac{1}{\sqrt{N}} \sum_k f_{k\alpha} e^{ikj}, \quad k = 0, \pm \frac{2\pi}{N}, \dots, \pi$$

so that the Hamiltonian takes the form,

$$H_{MF} = \sum_{k\alpha} (J\chi \cos(k) - \lambda) + \lambda N + J|\chi|^2 N$$

Consider now the constraint,

$$\langle \sum_{k\alpha} f_{k\alpha}^\dagger f_{k\alpha} \rangle = N$$

The mean-field ground state is a “spinon metal” with a Fermi surface state, with only one of $\pm\pi/2$ included, so that $k_F = \pi/2$,

$$\begin{aligned} |\psi_{MF}\rangle &= \prod_{-\frac{\pi}{2} \leq k < \frac{\pi}{2}} \prod_{\alpha=\uparrow, \downarrow} f_{k\alpha}^\dagger |0\rangle \\ &= \prod_{\alpha=\uparrow, \downarrow} \left(f_{-k_F, \alpha}^\dagger f_{-k_F + \frac{2\pi}{N}, \alpha}^\dagger \cdots f_{-k_F + \frac{2\pi}{N}(\frac{N}{2}-1), \alpha}^\dagger \right) |0\rangle \end{aligned}$$

and we recognise the term in the $(\dots)$ as a Slater determinant in real space. Explicitly,

$$\begin{aligned} f_{-k_F, \uparrow}^\dagger f_{-k_F + \frac{2\pi}{N}, \uparrow}^\dagger \cdots f_{-k_F + \frac{2\pi}{N}(\frac{N}{2}-1), \uparrow}^\dagger &= \left(\frac{1}{\sqrt{N}} \right)^{N/2} \sum_{x_1=1}^N \cdots \sum_{x_{N/2}=1}^N f_{x_1, \uparrow}^\dagger \cdots f_{x_{N/2}, \uparrow}^\dagger \times \\ &\quad e^{i(-k_F)x_1} \cdots e^{i(-k_F + \frac{2\pi}{N}(\frac{N}{2}-1))x_{N/2}} \\ &\propto \sum_{\substack{1 \leq x_1 < \cdots < \\ x_{N/2} \leq N}} \left[\sum_{p \in S_{N/2}} e^{i(-k_F)p(x_1)} \cdots e^{i(-k_F + \frac{2\pi}{N}(\frac{N}{2}-1))p(x_{N/2})} \right] \\ &\quad \times f_{x_1, \uparrow}^\dagger \cdots f_{x_{N/2}, \uparrow}^\dagger \\ &= \sum_{\substack{1 \leq x_1 < \cdots < \\ x_{N/2} \leq N}} \phi(x_1, x_2, \dots, x_{N/2}) f_{x_1, \uparrow}^\dagger \cdots f_{x_{N/2}, \uparrow}^\dagger \end{aligned}$$

where,

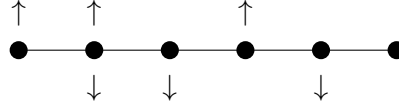
$$\begin{aligned} \phi(x_1, x_2, \dots, x_{N/2}) &= \det \begin{pmatrix} e^{i(-k_F)x_1} & e^{i(-k_F)x_2} & \cdots & e^{i(-k_F)x_{N/2}} \\ e^{i(-k_F + \frac{2\pi}{N})x_1} & e^{i(-k_F + \frac{2\pi}{N})x_2} & \cdots & e^{i(-k_F + \frac{2\pi}{N})x_{N/2}} \\ \vdots & \vdots & \ddots & \vdots \\ e^{i[-k_F + \frac{2\pi}{N}(N/2-1)]x_1} & e^{i[-k_F + \frac{2\pi}{N}(N/2-1)]x_2} & \cdots & e^{i[-k_F + \frac{2\pi}{N}(N/2-1)]x_{N/2}} \end{pmatrix} \\ &= e^{-ik_F(x_1+x_2+\cdots+x_{N/2})} \det \begin{pmatrix} 1 & 1 & \cdots & 1 \\ e^{-i\frac{2\pi}{N}x_1} & e^{-i\frac{2\pi}{N}x_2} & \cdots & e^{-i\frac{2\pi}{N}x_{N/2}} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-i\frac{2\pi}{N}(N/2-1)x_1} & e^{-i\frac{2\pi}{N}(N/2-1)x_2} & \cdots & e^{-i\frac{2\pi}{N}(N/2-1)x_{N/2}} \end{pmatrix} \\ &\propto e^{-\frac{\pi}{2}(x_1+x_2+\cdots+x_{N/2})} \prod_{1 \leq i < j \leq N/2} \left(e^{i\frac{\pi}{N}(x_i+x_j)} 2i \sin \left[\frac{\pi}{N}(x_i - x_j) \right] \right) \\ &\propto e^{-\frac{\pi}{2}(x_1+x_2+\cdots+x_{N/2})} \prod_{i=1}^{N/2} e^{i\frac{\pi}{N}x_i(N/2-1)} \prod_{1 \leq i < j \leq N/2} \sin \left[\frac{\pi}{N}(x_i - x_j) \right] \\ &= \left(\prod_{i=1}^{N/2} e^{-i\frac{\pi}{N}x_i} \right) \cdot \left(\prod_{1 \leq j < l \leq N/2} \sin \left[\frac{\pi}{N}(x_j - x_l) \right] \right) \end{aligned}$$

where from the third step, upon using that in 1D $k_F = \pi/2$, the rest is a Vandermonde determinant that can be simplified.

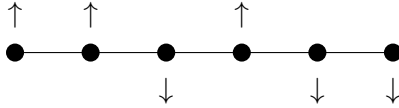
Altogether, the mean field ground state becomes,

$$|\psi_{MF}\rangle = \sum_{1 \leq x_1 < \dots < x_{N/2} \leq N} \sum_{1 \leq y_1 < \dots < y_{N/2} \leq N} \phi(\{x\})\phi(\{y\}) \times f_{x_1, \uparrow}^\dagger \dots f_{x_{N/2}, \uparrow}^\dagger f_{y_1, \downarrow}^\dagger \dots f_{y_{N/2}, \downarrow}^\dagger$$

The idea now is that, if the constraint is enforced only on average, half-filling is essentially independent of $\uparrow$ and $\downarrow$ spins. To see this, consider a chain of $N = 6$ spinons which can be unphysical,



but the following is physical,



To project the unphysical states out, we let $n_{j\alpha} = f_{j\alpha}^\dagger f_{j\alpha}$ and the “Gutzwiller projector operator”,

$$\hat{P}_G = \prod_{j=1}^N (n_{j\uparrow} - n_{j\downarrow})^2$$

whose action on the unphysical states is (take a single site j),

$$\begin{aligned} (n_{j\uparrow} - n_{j\downarrow})^2 |0\rangle_j &= 0 \\ (n_{j\uparrow} - n_{j\downarrow})^2 f_{j\uparrow}^\dagger f_{j\downarrow}^\dagger |0\rangle_j &= 0 \end{aligned}$$

while on the physical states,

$$\begin{aligned} (n_{j\uparrow} - n_{j\downarrow})^2 f_{j\uparrow}^\dagger |0\rangle_j &= f_{j\uparrow}^\dagger |0\rangle_j \\ (n_{j\uparrow} - n_{j\downarrow})^2 f_{j\downarrow}^\dagger |0\rangle_j &= f_{j\downarrow}^\dagger |0\rangle_j \end{aligned}$$

The Gutzwiller wavefunction,

$$|\psi\rangle = \hat{P}_G \prod_{-k_F \leq k < k_F} \prod_{\alpha=\uparrow, \downarrow} f_{k\alpha}^\dagger |0\rangle$$

therefore contains no unphysical configurations, and it can be used as a variational ansatz for the spin-1/2 chain. This would look like,

$$\begin{aligned} |\psi\rangle &= \hat{P}_G \sum_{1 \leq x_1 < \dots < x_{N/2} \leq N} \sum_{1 \leq y_1 < \dots < y_{N/2} \leq N} \phi(\{x\})\phi(\{y\}) f_{x_1, \uparrow}^\dagger \dots f_{x_{N/2}, \uparrow}^\dagger f_{y_1, \downarrow}^\dagger \dots f_{y_{N/2}, \downarrow}^\dagger |0\rangle \\ &= \sum_{\substack{\{x\}, \{y\} \\ \{x\} \neq \{y\}}} \phi(\{x\})\phi(\{y\}) f_{x_1, \uparrow}^\dagger \dots f_{x_{N/2}, \uparrow}^\dagger f_{y_1, \downarrow}^\dagger \dots f_{y_{N/2}, \downarrow}^\dagger |0\rangle \\ &= \sum_{\substack{\{x\}, \{y\} \\ \{x\} \neq \{y\}}} \text{sgn}(x_1, \dots, x_{N/2}, y_1, \dots, y_{N/2}) \phi(\{x\})\phi(\{y\}) \left| \uparrow_{x_1}, \uparrow_{x_2} \dots \uparrow_{x_{N/2}} \downarrow_{y_1}, \downarrow_{y_2} \dots \downarrow_{y_{N/2}} \right\rangle \end{aligned}$$

where on the last step we rearranged the order of the spinon operators to obtain a spin-state.

More generally, the variational energy,

$$E_{\text{var}} = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$$

can be calculated using the (Metropolis) Monte Carlo method. The special case of the 1D chain considered here allows for analytical results to be obtained. For example, the exact ground state energy per site, in the thermodynamic limit, of the spin-1/2 Heisenberg chain is,

$$E_{\text{exact}} = \frac{1}{4} - \ln 2 \approx -0.443147$$

Using the Gutzwiller wavefunction as a variational ansatz we get,

$$E_{\text{var}} \approx -0.442117$$

which is remarkably close to the exact value!

For completeness, let us note that the Gutzwiller wavefunction in 1D has a parent Hamiltonian known as the Haldane-Shastry model with inverse square interaction,

$$\begin{aligned} H_{HS} &= \sum_{1 \leq i < j \leq N} \frac{\vec{S}_i \cdot \vec{S}_j}{\left(\frac{N}{\pi}\right)^2 \sin^2 \left[\frac{\pi}{N}(i-j)\right]} \\ &\rightarrow \sum_{1 \leq i < j \leq N} \frac{\vec{S}_i \cdot \vec{S}_j}{|i-j|^2} \quad \text{as } N \rightarrow \infty \\ &= \sum_{j=1}^N \left(\vec{S}_j \cdot \vec{S}_{j+1} + \frac{1}{4} \vec{S}_j \cdot \vec{S}_{j+2} + \frac{1}{9} \vec{S}_j \cdot \vec{S}_{j+3} + \dots \right) \end{aligned}$$

For more details see F. D. M. Haldane, PRL 60, 635 (1988) and B.S. Shastry PRL 60, 639 (1988).

20 Lecture 20: Kitaev's Honeycomb Model

Let us use here the definition of a Quantum Spin Liquid (QSL) as the state of matter in which the entropy of spins is released in the absence of symmetry breaking down to the lowest temperatures, by forming a quantum spin state with long-range entanglement. The Kitaev Honeycomb Model is by now the most prominent example of a QSL which hosts quasiparticles with both Abelian and non-Abelian statistics. The classical Kitaev Honeycomb Model has a highly degenerate manifold of geometrically frustrated ground states. In the quantum version (which we consider here) the ground state is a highly entangled QSL which tunnels through the degenerate classical manifold and supports fractionalised excitations.

We consider a system of quantum spins with $S = \frac{1}{2}$ on the vertices of a honeycomb lattice. The inequivalent links of the honeycomb lattice are labeled by $x - y - z$ and the spins interact via NN interactions. The Hamiltonian is thus,

$$H = -J_x \sum_{x\text{-links}} \hat{\sigma}_i^x \hat{\sigma}_j^x - J_y \sum_{y\text{-links}} \hat{\sigma}_i^y \hat{\sigma}_j^y - J_z \sum_{z\text{-links}} \hat{\sigma}_i^z \hat{\sigma}_j^z \equiv - \sum_{\langle ij \rangle_\gamma} J_\gamma \sigma_i^\gamma \sigma_j^\gamma$$

and the spins satisfy the (anti)commutation relations,

$$\begin{aligned} \{\hat{\sigma}_i^\alpha, \hat{\sigma}_i^\beta\} &= 2\delta_{\alpha\beta} & (\text{same site}) \\ [\hat{\sigma}_i^\alpha, \hat{\sigma}_j^\beta] &= 0 & \text{for } i \neq j \end{aligned}$$

and using these we can show,

$$\begin{aligned} [\hat{\sigma}_i^\alpha \hat{\sigma}_j^\alpha, \hat{\sigma}_i^\beta \hat{\sigma}_j^\beta] &= 0 & \forall i \neq j \\ [\hat{\sigma}_i^\alpha \hat{\sigma}_j^\beta, \hat{\sigma}_i^\gamma \hat{\sigma}_j^\gamma] &= 0 & \text{for } \alpha, \beta \neq \gamma \end{aligned}$$

20.1 Parton Construction

There is parton-construction that allowed Kitaev to analytically solve the model. At each site there is two-dimensional Hilbert space for the spin-1/2 dof. Kitaev introduced four Majorana fermions (majoranas) $\{b^x, b^y, b^z, c\}$ which can be represented via,

$$\sigma^\alpha = ib^\alpha c, \quad \alpha = x, y, z$$

Note however that this enlarged the Hilbert space. A way to see this is by recalling that a canonical fermion representation of a spin,

$$|\uparrow\rangle = f_\uparrow^\dagger |\emptyset\rangle, \quad |\downarrow\rangle = f_\downarrow^\dagger |\emptyset\rangle$$

already enlarges the Hilbert space from dimension two to dimension four (the other two states being $|\emptyset\rangle, f_\uparrow^\dagger f_\downarrow^\dagger |\emptyset\rangle$). Now the majoranas can be constructed by these canonical (complex) fermions by,

$$f_\uparrow = b^x - ib^y \quad f_\downarrow = b^z - ic$$

and their fermion parity is,

$$\begin{aligned} (-1)^{f_\uparrow^\dagger f_\uparrow} &= 1 - 2f_\uparrow^\dagger f_\uparrow = ib^x b^y \\ (-1)^{f_\downarrow^\dagger f_\downarrow} &= 1 - 2f_\downarrow^\dagger f_\downarrow = ib^z c \end{aligned}$$

Equipped with the above we may conclude that there is a hidden constraint that excludes the unphysical dof, since in the end we should only work with two physical dof per site. We consider the product,

$$\sigma^x \sigma^y \sigma^z = i\mathbb{I}_{2 \times 2}$$

which in the majorana representation becomes,

$$b^x b^y b^z c = \mathbb{I}$$

This means that all physical states of the original Hamiltonian, written in the majorana basis, upon acted the above operator should give eigenvalue +1. The unphysical ones give -1. To understand this, note that the constraint can be written as,

$$(ib^x b^y)(ib^z c) = -1 = (-1)^{f_\uparrow^\dagger f_\uparrow} (-1)^{f_\downarrow^\dagger f_\downarrow}$$

so that the two complex fermions have opposite parity.

Consider now σ^z ,

$$\begin{aligned} \sigma^z &= f_\uparrow^\dagger f_\uparrow - f_\downarrow^\dagger f_\downarrow = \frac{1}{2} (1 - ib^x b^y) - \frac{1}{2} (1 - ib^z c) \\ &= -\frac{i}{2} b^x b^y + \frac{i}{2} b^z c \\ &= \frac{i}{2} b^z c + \frac{i}{2} b^z c = ib^z c \end{aligned}$$

where the last line only holds in the physical subspace, since to get it we used the constraint. Similarly we find the expressions for σ^x, σ^y , so that the Hamiltonian takes the form,

$$H = \sum_{\gamma \in x, y, z} J_\gamma \sum_{\langle j, l \rangle_\gamma} \underbrace{(ib_j^\gamma b_l^\gamma)}_{\equiv \hat{u}_{jl}} (ic_j c_l)$$

where $\hat{u}_{jl}$ is a fermion parity operator on a bond with two different sites, $\hat{u}_{jl} ib_j^\gamma b_l^\gamma$. These bond (or link) operators satisfy $\hat{u}_{ij} = -\hat{u}_{ji}$, $\hat{u}_{ij}^2 = \mathbb{I}$ and $\hat{u}_{ij}^\dagger = \hat{u}_{ij}$, and they further provide a set of local conserved quantities since,

$$[H, \hat{u}_{ij}] = 0$$

$\hat{u}_{ij}^2 = \mathbb{I}$ means that they have eigenvalues ± 1 and we thus recognise it as a static $\mathbb{Z}_2$ gauge field.

A generic eigenstate of H is thus written as follows,

$$|\tilde{\psi}\rangle = |\phi\rangle_c \otimes |\{u\}\rangle$$

where $|\phi\rangle_c$ is an eigenstate of the $c_j c_l$ part, and $|\{u\}\rangle$ is a state corresponding to the set of the $\hat{u}$ variables. Note that not all of these states are physical since the constraint is not taken into account so far. To obtain the physical states, we proceed with a Gutzwiller projection,

$$|\psi\rangle = \hat{P}_G |\tilde{\psi}\rangle = \hat{P}_G |\phi\rangle_c \otimes |\{u\}\rangle$$

Let the parity operator $Q_j = -b_j^x b_j^y b_j^z c = -1$ for the physical states and construct $\hat{P}_G$,

$$\hat{P}_G = \prod_{j=1}^N \frac{1 - \hat{Q}_j}{2}$$

so that $\hat{P}_G = 0$ if $Q_j = 1$. What remains is to prove that $|\psi\rangle$ is indeed an eigenstate. (Prove)

Given the above, the strategy to find the solution for the Kitaev Model becomes,

1. Choose a configuration of the gauge fields $\{u\}$
2. Diagonalise $H_c(\{u\})$ in the given configuration
3. Check the fermion parity of $|\phi\rangle \otimes |\{u\}\rangle$

We are now in position to find the ground state energy. This is ultimately related to the question of which configuration $\{u\}$ has the lowest energy. The answer is provided by Lieb's theorem: with the convention that $j \in A$ and $l \in B$ sublattice, the ground state configuration is the $u_{jl} = +1$ for all j, l .

20.2 Physical Operators

We note that the operators $\hat{u}_{jl}$ is not gauge invariant, does not preserve fermion parity (since it acts on two sites) and thus connects physical to unphysical states. Recall also that local, non-gauge invariant observables do not provide any useful characterisation of the system, so we need to construct a gauge-invariant quantity or correlation function.

A way to do this is, as is the case for any lattice gauge theory, is to construct a closed loop on the lattice. With the above convention that $j \in A$ and $l \in B$ sublattice, the plaquette operator,

$$W_p = u_{12}u_{32}u_{34}u_{54}u_{56}u_{16}$$

which is gauge invariant and can also be written in spin language as,

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z$$

The plaquette operators satisfy $[H, W_p] = 0$ for all p and they also have eigenvalues ± 1 . We can construct larger plaquettes by taking any closed loop so that $W_{pp'} = W_p W_{p'}$, whose eigenvalue is also the product of the single plaquette eigenvalue. Therefore longer loops might just be products of smaller loops.

There is an exception to the above line of thought. Suppose that we impose PBC along two different directions by putting the system on a torus and construct two different types of non-contractible loops, W_x and W_y , along two inequivalent directions. We have $[H, W_x] = [H, W_y] = 0$
...

20.3 Key Results

There are two key results of the Kitaev Honeycomb model which we list here. Essentially, the model realises two distinct phases, the A - and B - phase.

1. A-phase: gapped phase which supports abelian anyons. It is realized in the regime,

$$|J_z| > |J_x| + |J_y| + \text{cyc.}$$

The system can be mapped to the toric code.

2. B-phase: gapless phase with Majorana cones (which can be gapped by a small magnetic field, or a three spin interaction). Occurs when no interaction is dominant, and has a Chern number of $C = \pm 1$. This phase supports non-abelian anyons.

20.4 Effective theory for the non-abelian phase

In the non-abelian gapped phase, the existence of a non-zero Chern number implies the existence of chiral edge modes.

Recall the $(1+1)D$ massless Majorana field theory from lecture 15,

$$\begin{aligned} H &= \int dx \left[-i \frac{v}{2} (\chi^R \partial_x \chi^R - \chi^L \partial_x \chi^L) \right] \\ &= \sum_{k>0} vk [(\chi_k^R)^\dagger (\chi_k^R) + (\chi_{-k}^L)^\dagger (\chi_{-k}^L)] + \text{const.} \end{aligned}$$

but now the different modes χ^L and χ^R are spatially separated.

The (majorana) edge modes are charge neutral but they can carry heat. The thermal current carried by χ^R , at a temperature T much lower than the bulk energy gap (so that such excitations are negligible), is

$$\begin{aligned} I_R &= \int_0^\infty \frac{dk}{2\pi} n_{FD}(\epsilon_k) \epsilon_k v_k \\ &= \int_0^\infty \frac{dk}{2\pi} \frac{1}{e^{\beta vk} + 1} vk \cdot v \\ &= \frac{\pi}{24} T^2 \end{aligned}$$

so that the thermal Hall conductivity is,

$$\kappa = \frac{\partial I_R}{\partial T} = \frac{\pi}{12} T$$

Restoring the physical units $\hbar, k_B$,

$$\kappa/T = \underbrace{\frac{1}{2}} \frac{\pi k_B^2}{6\hbar}$$

The factor of $1/2$ in front is the half-quantised thermal Hall conductivity, and provides experimental sharp evidence of chiral Majorana edge states. Relevant experimental efforts can be found in Nature 559 227 (2018) and Science 373 568 (2021).

20.5 But why are there non-abelian Ising anyons? [PRL 99, 196805 (2007)]

We can answer the question by referring to the Jackiw-Rebbi mechanism, which predicts the existence of a zero-energy solution when there is a sign change of mass at an interface. Consider a lattice cut open along the longitudinal direction so that a row of bonds is removed. The cutting introduces in-gap edge modes with the upper carrying a right-moving edge mode, and the lower carrying a left-moving edge mode, each described by a 1D Dirac Hamiltonian.

Suppose that we couple the two now with weaker vertical bonds so that the spectrum becomes gapped again, albeit with a smaller gap. This means that the Dirac fermions acquired mass. The Hamiltonian is,

$$H = \int dx \left[-i \frac{v}{2} (\chi^R \partial_x \chi^R - \chi^L \partial_x \chi^L) + im(x) \chi^R \chi^L \right]$$

where we allowed the mass to be x -dependent. To achieve an x -dependence, we consider flipping the sign of half of the introduced vertical bonds.

According to the Jackiw-Rebbi mechanism, this implies the existence of a zero-mode per defect. These majorana zero-modes are localised at the edges where the mass changes sign and are therefore trapped there. These are half-quantum vortices and it is known (PRL 86, 268 (2002)) that their statistics is that of non-abelian anyons.